

PRICE AND VOLATILITY CO-JUMPS*

Federico M. Bandi^{†*} and Roberto Renò^{**}

*Johns Hopkins University and Edhec-Risk and **Università di Siena

First draft: April 14, 2011 - This draft: February 6, 2014

Abstract

The dependence between the magnitudes of discontinuous changes in asset prices and contemporaneous discontinuous changes in volatility (co-jumps) is a fundamental aspect of the price process contributing, among other effects, to skewness in the return distribution. Yet, its nature has been reported by many as being - in terms of sign, magnitude, and statistical significance - largely elusive. Using a novel identification strategy for stochastic volatility modelling in continuous time relying on trade-level information for *spot variance* estimation, as well as *infinitesimal cross-moments*, this paper documents that a sizeable proportion of discontinuous changes in asset prices are associated with strongly anti-correlated, contemporaneous changes in volatility. Not only are the price jump sizes strongly negatively correlated with the volatility jump sizes, but the absolute values of their (negative) mean and dispersion appear to increase with the volatility level, an additional effect which should lead to care in the management of joint directional and volatility jump risk. Using a possibly non-monotonic pricing kernel, we illustrate the equilibrium impact of price and volatility co-jumps on both return and variance risk premia.

Keywords: Stochastic volatility, jumps in prices, jumps in volatility, co-jumps, infinitesimal cross-moments, return risk premia, variance risk premia.

*We thank seminar participants at the 2012 WFA Annual Meetings (Las Vegas, NV), 2013 ICEEE (Genova, Italy), Bauer College of Business (University of Houston), Brown University, Carey Business School (Johns Hopkins University), CASS Business School (City University London), Edhec Business School, Georgetown University, Singapore Management University, Toulouse School of Economics, the Triangle Econometric Seminar, Wisconsin School of Business (University of Wisconsin), and University of Florence for useful discussions. We are very grateful to two anonymous reviewers and Dobri Dobrev for their comments and suggestions. Roberto Renò acknowledges financial support from The Castilla and León Council of Education and Science, grant VA016A10-1.

[†]Corresponding author. The Johns Hopkins Carey Business School, 100 International Drive. Baltimore, MD 21202. Phone: (410) 234-9407 Mail: fbandi1@jhu.edu

1 Introduction

Understanding the dynamic properties of stock returns continues to be at the heart of research in finance. Strong consensus has emerged about the need to allow for discontinuities in asset prices, "return jumps", while incorporating stochastic time-variation in the standard deviation of the continuous shocks to returns, "stochastic volatility". Even though both features generate fat tails in the return distribution, they do not appear to be sufficient to yield the rapid increases in volatility which have been historically experienced.

Adding jumps in volatility provides a solution to this issue. As pointed out in Eraker et al. (2003), jumps in returns and jumps in volatility serve different, but complementary, purposes. The former generate large, infrequently-observed, sudden movements, such as the October 1987 crash. These movements would require an unreasonably high level of volatility to occur. The latter lead to fast changes in the level of volatility and, due to volatility persistence, a lasting effect on the distribution of stock returns. Along with daily moves as large as 22%, during the 1987 crash, among other times of distress, the level of volatility more than doubled before slowly mean-reverting to lower levels. The inability of continuous-time return models with stochastic volatility and jumps only in returns to fit return/option data, and the need to incorporate jumps in volatility, has been recognized as early as in the work of, e.g., Bakshi et al. (1997), Bates (2000), and Pan (2002). Duffie et al. (2000) and Eraker et al. (2003) provide a thorough analysis of the effectiveness of discontinuities in returns and volatility by explicitly incorporating both components in their "double-jump" specifications.

A related problem has to do with the likelihood of discontinuous movements occurring *jointly* and the nature of the correlation - or lack thereof - between the resulting jump sizes. Importantly, while the presence of co-jumps is broadly accepted (see, *inter alia*, Jacod and Todorov, 2010), the study of the sign and magnitude of their dependence has led to a fundamental disconnect between the strand of the literature that uses derivative information and the one that solely relies on price data.

The former is rather emphatic about the presence of *negatively-correlated* price and volatility large co-movements. Eraker (2004) finds a negative correlation between jump sizes *only when* augmenting return data with option data, the negative correlation between co-jump sizes being identified off of the implied volatility smirk.¹ Using high-frequency observations on

¹This finding is coherent with the calibration results in Duffie et al. (2000) who, again, provide evidence

the VIX and the S&P500 index along with sample statistics developed in Jacod and Todorov (2009) and Todorov and Tauchen (2010), Todorov and Tauchen (2011) find evidence for a specification with a high likelihood of contemporaneous arrivals along with a lower likelihood of disjoint arrivals. Their reported sizes of the large price and volatility co-movements are strongly anti-correlated.

The literature which does not use derivative information is - on the other hand - drastically less clear-cut about the nature of the co-jump correlation, a strongly negative dependence not being a documented empirical feature in this case. Using tests developed in Jacod et al. (2012a), Jacod et al. (2012b) cannot reject the null of no correlation, if not in a single case. Even in this single case, removal of the Flash Crash leads to insignificant dependence between discontinuous price and volatility changes. The state of this strand of the literature is summarized effectively by Broadie et al. (2007), who write: "*Intuitively, ρ [the co-jump correlation] should be negative, as larger jumps in prices tend to occur with larger jumps in volatility, at least as we think of events such as the crash of 1987. Eraker, Johannes, and Polson (2003) and Chernov, Gallant, Ghysels and Tauchen (2003) find negative but insignificant estimates of ρ . Eraker (2004), on the other hand, finds a slightly positive but insignificant estimate. This parameter is extremely difficult to estimate, even with 15 or 20 years worth of data, because jumps are very rare events (...) Due to the difficulty in estimating this parameter precisely and for parsimony, we assume the sizes of the price jumps are independent of the sizes of the jumps in volatility.*"

Since pricing and risk management depends crucially on the nature of the dependence between discontinuous price and volatility changes, it is important to understand whether an economically-meaningful negative dependence is - indeed - a stylized feature of sensible price formation mechanisms. If this were the case, the resulting *joint* directional and volatility jump risk would have to be properly managed. In this sense, the disconnect between findings with and without derivative information is a concern. Is this disconnect mostly due to the impact of risk premia? In the case of the VIX, for instance, sudden implied volatility increases could be due to market fear (and increased risk premia), particularly in times of negative directional shocks (i.e., negative price jumps). This simple effect alone can give rise to a negative correlation between the volatility proxy (VIX) and market returns even in a situation where sudden, discontinuous changes in true volatility and market prices hardly

of anti-correlated price and volatility co-jump sizes by fitting the typically-found, pronounced smirk in option prices. Calibrations are also used in the more recent work of Cont and Kokholm (2011).

co-vary with each other. Or, is the disconnect - to a first degree - due to the way in which volatility is filtered, option information providing the resolution that traditional volatility filters may otherwise lack? We provide evidence for this latter thesis.

The analysis of a flexible stochastic volatility model with co-jumps, as well as independent jumps, is the subject of this paper. Volatility is, of course, a latent variable. We argue that its accurate identification is important to evaluate fine-grain dynamics such as those reflected in volatility jumps and price and volatility co-jumps. In essence, the mixed findings in the literature may be the result of volatility filters which are effective in identifying diffusive dynamics but may not be granular enough as to shed light on rare co-variations. There are two key steps to the procedure we propose. The first step is the estimation of *spot volatility*. Spot volatility, shown in Section 5 to be crucial for the unbiased identification of both co-jump intensity and co-jump correlation, is estimated nonparametrically for every hour in our sample using intra-daily price data. In the second step, volatility is utilized, along with price data, in the context of a novel moment-based procedure (dubbed Nonparametric Infinitesimal Method of Moments or NIMM) hinging on the notion of *infinitesimal cross-moment*. In essence, we evaluate the daily dynamics of returns and volatility by making use of the substantial informational content of intra-daily price data for the purpose of spot volatility filtering.

We emphasize that the use of a spot, or local, notion of volatility is not immaterial. We show that volatility filters which use too large a bandwidth for identification (as is the case for daily realized variance or daily variances from likelihood procedures) may cause attenuation to zero in the estimation of the cross-moments. Being cross-moment information implicitly or explicitly, as in our case, used by any co-jump identification procedure, including likelihood methods, the mixed findings in the existing literature - as reported above - have the potential to be the outcome of volatility filters which lack the granularity needed to identify rare co-movements and their dependence. When employing these filters, the use of derivatives in the identification of rare co-jumps may, therefore, be viewed as supplying the resolution which low-frequency price data, and the resulting filtered volatilities, could not provide. The presence of co-jumps and their negative dependence should, however, be revealed by price data only without potential contaminations imputable to measure changes and risk premia. Using high-frequency price data for the purpose of local volatility filtering, and our proposed modelling and identification approach, we show that the co-jumps, along with negatively correlated co-jump sizes, are an important stylized feature of the dynamic

evolution of daily market prices and volatility series, one that continuous-time models of price formation should embed. Price data *alone* forcefully uncover the negative dependence between large discontinuities in prices and volatilities, with no need for a cross-section of options.

The paper employs S&P 500 futures high-frequency returns from April 21, 1982, to February 5, 2009. Many of the large co-jumps are associated with well-known events like Black Monday (October 19, 1987), the Asian crisis (October 27, 1997), the Russian crisis (August 31, 1998), and the Lehman-Brothers default (September 29, 2008) during the recent financial crisis. The fine-grain dynamics of the return and volatility process are such that some of the discontinuous components are fully idiosyncratic while some are common to the return/volatility processes. The number of estimated yearly co-jumps (8/9) is slightly larger than the number of idiosyncratic discontinuities in the price process (6/7, per annum). The estimated yearly number of idiosyncratic volatility jumps is about 13.

We find that the size distribution of the price co-jumps depends on the underlying volatility level. The dispersion and negative mean of this distribution increase with volatility, thereby leading to more negative, and more erratic, price co-jumps in times of high volatility.

As stressed, the occurrence of co-jumps is also generally associated with price/volatility changes (i.e., sizes of the jumps) of opposite sign. We estimate a functional correlation between the size of the price jumps and the size of the volatility jumps very close to -1 . This effect is important. Among other issues, it introduces a second, less conventional, source of leverage in the model. Not only are the small Brownian shocks affecting prices and volatility negatively-correlated in the model (a traditional leverage effect), the infrequently large discontinuities are also strongly negatively-correlated (a leverage effect induced by discontinuous co-movements). The two combined effects lead to a form of "generalized leverage." Hence, there are two channels through which negative shocks to prices can increase variance and, therefore, induce skewness in the conditional (on volatility) distribution of stock returns. Similarly, there are two channels through which the price and volatility dynamics will lead to downward-sloping implied volatility curves across strikes for short and medium maturity options (the ubiquitous "implied volatility smirk").

Using a possibly non-monotonic pricing kernel, i.e., a pricing kernel decreasing in prices and increasing in variance, as discussed by Christoffersen et al. (2013), we derive equilibrium

implications of generalized leverage and, therefore, of the presence of negatively-correlated co-jumps. We show that generalized leverage delivers both a component of return risk premia and a component of variance risk premia. In other words, not only do the co-jumps lead to an additional source of skewness in stock returns, they also modify - in equilibrium - the mean of the return distribution while contributing to a negative variance risk premium. Should the stochastic discount factor not depend on variance, and therefore be monotonic in prices, then generalized leverage, in general, and the negative correlation between price and volatility co-jumps, in particular, would not matter to generate return risk premia. They would, however, still matter to determine variance risk premia.

We proceed as follows. Section 2 provides economic evidence of large co-jumps. We show how the most sizeable jumps in returns in our sample are associated with contemporaneous jumps in volatility of opposite sign. In section 3 we present a flexible stochastic volatility model with idiosyncratic and common jumps. We show how the structure of the model we propose generalizes to a nonparametric framework successful parametric specifications in the literature. Section 4 presents the notion of infinitesimal cross-moment and discusses its usefulness for co-jump identification. Section 5 discusses a strategy for (cross-) moment-based identification of the model. Section 6 provides nonparametric empirical findings about the joint price and volatility dynamics. Section 7 turns to a parametric evaluation. In Section 8 we discuss equilibrium pricing implications of co-jump presence both in terms of return risk premia and in terms of variance risk premia. The robustness of the model specification is tested in Section 9. Section 10 concludes. The asymptotic and finite sample properties of the proposed methods are discussed in two technical Appendices.

2 Large co-jumps: descriptive evidence

We go through our sample to identify unusually large price movements. In order to do so, we divide daily close-to-close price changes by the corresponding standard deviation over the previous day.² Should the changes be induced by continuous Brownian shocks, in light of

²The econometric analysis of this paper, described in detail in Section 5 and Appendix B, employs hourly (spot) variance estimates. To obtain the daily estimates used in this section, we scale up the hourly estimates to a daily value and average them for every 6-hour trading day. We then apply an exponential smoother with a 40-day lag to reduce measurement error. The results would be qualitatively very similar if the previous days' variances, which are displayed in Figure 6, were used without applying the smoother.

the approximate normality of Brownian motion over small time increments, they would be larger than 2 in absolute value with approximate probability equal to 5%. Unusually large, standardized, price changes would therefore signal jumps in the price process.³

Column 3 in Table 1 reports the 30 largest standardized (by volatility) daily price changes in our sample. Some of these are associated with well-known events, like Black Monday (October 19, 1987), the Asian crisis (October 27, 1997), the Russian crisis (August 31, 1998), 9/11 (September 17, 2001)⁴ and the more recent Lehman-Brothers default (September 29, 2008) during the 2008/2009 financial crisis. Importantly, the vast majority of the largest, abnormal, price changes are negative (24 out of 30). In other words, when large price discontinuities occur, they tend to be the result of negative shocks.

Column 4 in Table 1 reports the corresponding variance changes. In agreement with our subsequent analysis, we report changes in logarithmic variance, rather than in actual variance. Consistent with the price changes, the variance changes are close-to-close. They are obtained by virtue of spot (hourly) volatility estimates making use of intra-daily price data in the last hour of the trading day before being scaled up to a daily value (see Section 5 and Appendix B for details). As earlier in the case of prices, we divide the changes in logarithmic variance by the corresponding standard deviation (volatility of log-variance)⁵ to rule out changes which may be consistent with smooth Brownian dynamics. Two effects are striking. First, the log-variance changes associated with the largest standardized price changes have corresponding t-statistics which are, with very few exceptions, large and very statistically significant. Second, the volatility changes are, in general, of opposite sign with respect to the price changes. In other words, abnormally large price changes are associated with abnormally large volatility changes (co-jumps).

Figure 1 reports the estimated probability density functions for both the co-jumps in price and the co-jumps in volatility. It is interesting to observe the thickness of the tails of the distribution of the price co-jumps, signaling the need for a model specification with rich dynamics for the variance of the price co-jump sizes. We will return to this issue.

³This same logic is formally employed by Lee and Mykland (2008) to test for jumps using intra-daily return data.

⁴The actual discontinuity is not on 9/11 since markets were closed for five days after 9/11.

⁵Computation of the volatility of log-variance requires a full-blown dynamic model for the price dynamics. In particular, for internal consistency, the model should account for diffusive as well as discontinuous shocks. Here, we employ the volatility of log-variance identified from our proposed stochastic volatility model in Section 7.

Table 1: Descriptive evidence on co-jumps. The figures are for daily returns in percentage terms.

Largest price jumps and volatility jumps					
Day	Return (%)	t-stat	Log-variance change	t-stat	Events
03-Aug-1984	3.63	4.28	1.26	8.8	
18-Dec-1984	3.09	3.72	0.34	2.4	
08-Jan-1986	-3.30	-4.56	0.78	5.5	
11-Sep-1986	-5.19	-5.77	1.51	10.6	
16-Oct-1987	-7.35	-6.66	1.06	7.4	
19-Oct-1987	-30.01	-24.26	2.96	20.7	Black Monday
14-Apr-1988	-4.68	-4.27	1.86	13.0	Dollar plunge
17-Mar-1989	-2.75	-4.02	0.98	6.9	
13-Oct-1989	-6.85	-10.74	0.67	4.7	Friday 13th
12-Jan-1990	-3.43	-4.37	1.81	12.6	
22-Jan-1990	-3.47	-3.95	1.42	10.0	
17-Jan-1991	4.43	4.89	0.91	6.3	
21-Aug-1991	2.74	3.99	0.03	0.2	
15-Nov-1991	-4.08	-6.57	1.30	9.1	
16-Feb-1993	-2.52	-4.78	1.38	9.7	
04-Feb-1994	-2.33	-5.76	1.62	11.3	
08-Mar-1996	-3.94	-4.92	1.39	9.7	
05-Jul-1996	-2.36	-3.69	0.56	3.9	
27-Oct-1997	-7.80	-7.46	0.64	4.5	Asian Crisis
28-Oct-1997	5.68	5.09	0.75	5.3	
09-Jan-1998	-3.88	-4.08	1.02	7.2	
04-Aug-1998	-3.60	-3.76	1.21	8.5	
31-Aug-1998	-7.30	-5.41	0.47	3.3	Russian crisis
04-Jan-2000	-3.52	-3.99	-0.20	-1.4	
14-Apr-2000	-8.11	-4.90	1.11	7.7	Dot.com crash
03-Jan-2001	5.18	3.90	0.50	3.5	
17-Sep-2001	-5.02	-4.22	0.58	4.0	9/11
20-Jan-2006	-1.93	-3.64	0.67	4.7	
27-Feb-2007	-3.23	-7.07	2.58	18.0	Chinese Correction
29-Sep-2008	-6.93	-4.09	1.76	12.3	Lehman-Brothers default

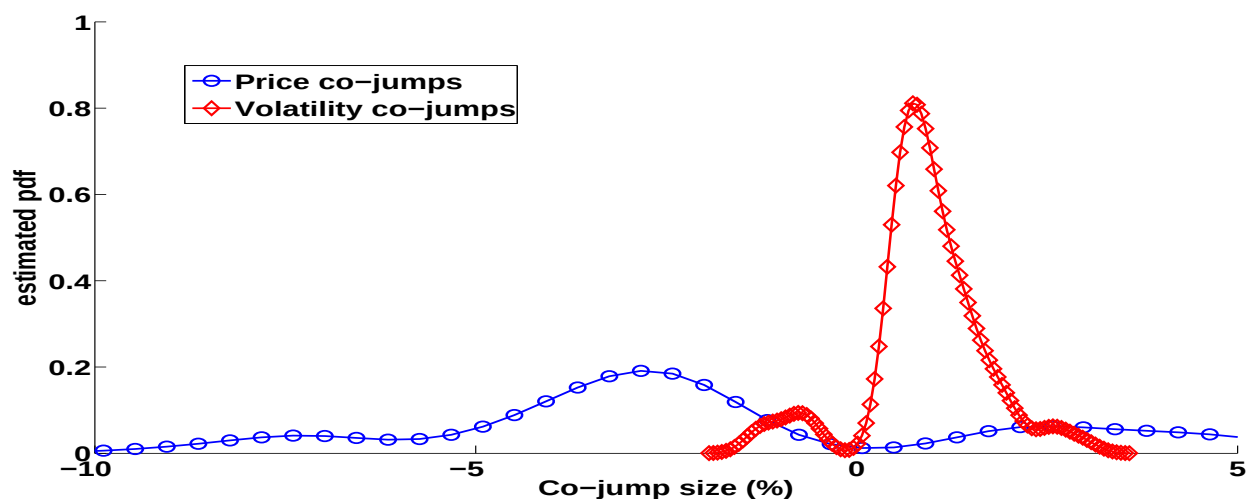


Figure 1: Estimated probability density function of the price and volatility co-jumps.

Because the sizes of the co-jumps may not have a zero mean, in order to gauge the extent of the correlation between price and volatility jump sizes, we report a scatter plot of the log-variance jumps versus the price jumps. Instead of reporting the co-jumps associated with the 30 largest price changes, as implied by Table 1, we now report the co-jumps associated with price and log-variance t-ratios jointly implying significance at the 0.5% level. The choice of this level is meant to highlight large price and variance discontinuities while, at the same time, preventing the insurgence of a sizeable number of false positives, something which is typical of jump tests. This procedure identifies 70 large daily jumps over 28 years which are plotted in Figure 2. For these jumps, the mean and standard deviation of the price discontinuities are estimated as being equal to -2.27% and 4.87% , whereas the estimated values of the mean and standard deviation of the log-variance jumps are 0.9 and 0.77 . Thus, when contemporaneous jumps in price and volatility occur, the volatility jumps tend to be positive whereas the price jumps tend to be negative. In addition, the scatter plot provides compelling visual evidence about a strongly negative correlation between co-jump sizes (measured at a significant value of about -0.5). In other words, price jumps below their (negative) mean generally occur alongside volatility spikes above their (positive) mean, thereby resulting in co-jumps with strongly negatively-correlated jump sizes.⁶

⁶There is one main exception in our sample. During the aftermath of the 1987 crash, on October 21st 1987, a percentage change of 16.11 in logarithmic volatility was accompanied by a percentage (positive) change of 15.5 in log-prices. The corresponding daily volatility was roughly 4.6% daily or 73% on a yearly basis. While we do not report this glaring outlier in the figure, the corresponding data point is used in all of our subsequent estimates.

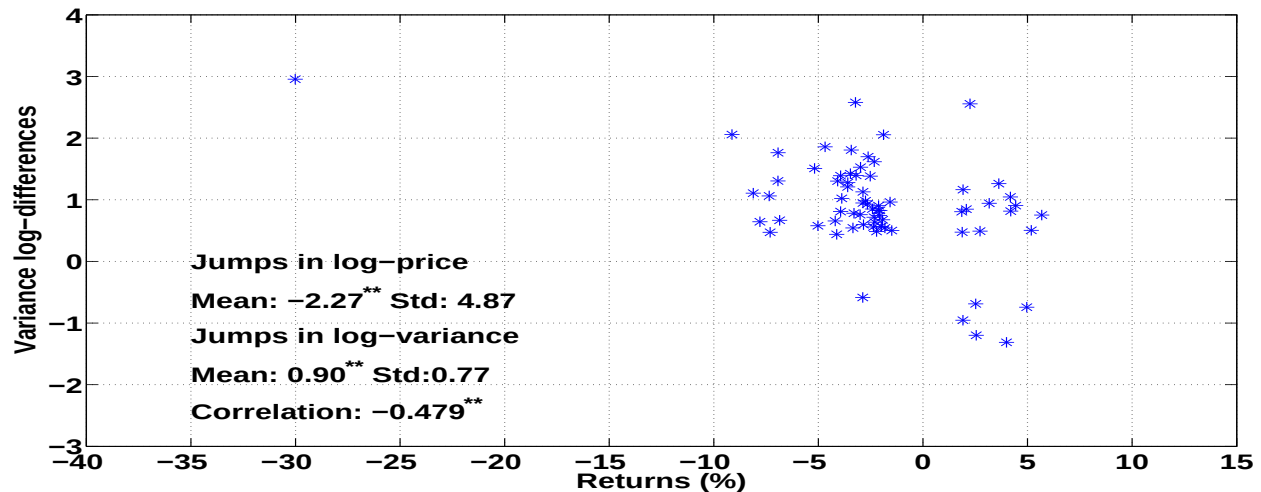


Figure 2: Scatter plot of the significant co-jumps in price and volatility.

By sorting with respect to the large price (and volatility) discontinuities, because we only detect the most sizeable jumps, this evidence should be viewed as providing a lower bound on the number of price and volatility co-jumps in our sample. In agreement with this observation, our point estimates (below) will imply 8/9 daily co-jumps per annum.

In Figure 3 we plot the large price discontinuities displayed in Figure 2 with respect to the volatility level prevailing at the time when the jumps occur. This figure suggests that both the mean and the standard deviation of the sizes of the price co-jumps vary substantially with the volatility level. Specifically, we find that larger volatility levels are associated with co-jumps that are more negative, in expectation, and more variable. This result points to the need for rich, possibly nonlinear, dynamics for the price co-jump sizes.

Next, we present a model featuring joint and idiosyncratic discontinuities in prices and volatility. Consistent with the evidence reported here, the model will allow for correlation between the jump sizes. By permitting the price and volatility jump sizes not to be mean zero, the model will also allow for a higher likelihood of negative price jumps and a higher likelihood of positive volatility jumps. Further, we will let the mean and standard deviation of price co-jumps vary with the volatility level. In the context of a flexible specification for the price and volatility dynamics, we will find estimates of the features of the co-jump sizes, including their correlation, which will be qualitatively and quantitatively similar to those reported in this preliminary descriptive analysis.

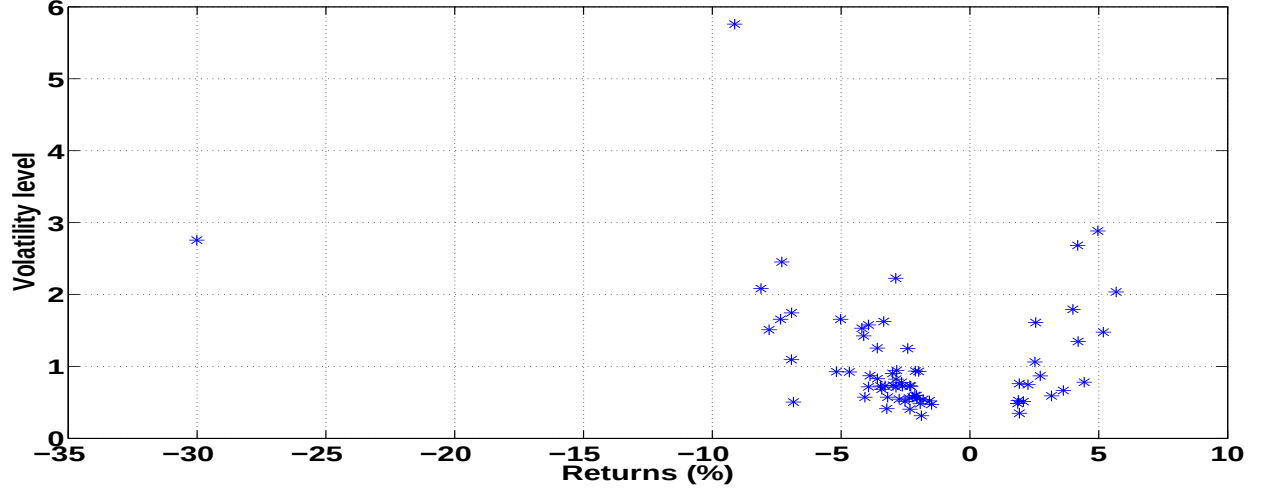


Figure 3: Scatter plot of the significant price co-jumps and the volatility level (% , daily) at which they occur.

3 A continuous-time stochastic volatility model with return/volatility co-jumps

We address the literature in continuous time⁷ by casting the problem in the context of a classical jump-diffusion specification with stochastic volatility. Differently from the existing literature, however, with the sole exception of conditions on the probability distribution of the jump sizes, we do not impose any parametric assumption on the functions driving returns and volatility dynamics. The return/variance system we consider is

$$\begin{aligned} d(\log p_t) &= \mu(\sigma_t)dt + \sigma_t \left\{ \rho(\sigma_t)dW_t^1 + \sqrt{1 - \rho^2(\sigma_t)}dW_t^2 \right\} + c_{r,t}^J dJ_r + c_{r,t}^{JJ} dJ_{r,\sigma} \\ d\xi(\sigma_t) &= m(\sigma_t)dt + \Lambda(\sigma_t)dW_t^1 + c_{\sigma,t}^J dJ_\sigma + c_{\sigma,t}^{JJ} dJ_{r,\sigma}, \end{aligned} \quad (1)$$

where $\xi(\cdot)$ is an increasingly monotonic function, $W = \{W^1, W^2\}$ is a bivariate standard Brownian motion vector, $J = \{J_r, J_\sigma, J_{r,\sigma}\}$ is an independent (of W) trivariate vector of mutually independent Poisson processes with intensities $\lambda_r(\sigma_t)$, $\lambda_\sigma(\sigma_t)$, and $\lambda_{r,\sigma}(\sigma_t)$, respectively. The functions $\mu(\cdot)$, $m(\cdot)$, $\Lambda(\cdot)$, $\lambda_r(\cdot)$, $\lambda_\sigma(\cdot)$, $\lambda_{r,\sigma}(\cdot)$ and $\rho(\cdot)$ satisfy mild smoothness conditions and are solely such that a unique strong solution to the system exists. We do not require stationarity of the system, but only *recurrence* of the latent variable, i.e., volatility,

⁷Alizadeh et al. (2002); Andersen et al. (2002); Bakshi et al. (1997); Bates (2000); Chernov and Ghysels (2000); Chernov et al. (2003); Duffie et al. (2000); Eraker (2004); Eraker et al. (2003); Heston (1993); Jones (2003); Pan (2002); Broadie et al. (2007), among other influential contributions.

as in Bandi and Renò (2009). The model has several features:

1. (Independent jumps and co-jumps) As mentioned, we allow for independent jumps J_r, J_σ as well as for common jumps $J_{r,\sigma}$.
2. (State-dependent intensities) The intensities of the jumps may be nonlinear functions of the underlying variance level. We therefore complement and extend models with affine variance-dependent price jump intensities (Bates, 2000, and Pan, 2002) as well as models with affine variance-dependent volatility jump intensities (Eraker, 2004).
3. (State-dependent jumps) The moments of the jump sizes are permitted to be functions of the underlying variance level. Assume $\xi(\cdot) = \log(\cdot)$, as in a log-variance model, for instance. Let the sizes of the idiosyncratic jumps be normally distributed: $(c_{r,t}^J, c_{\sigma,t}^J) \sim \mathcal{N}(\mu^J, \Sigma^J)$ with $\mu^J = (\mu_{J,r}, \mu_{J,\sigma})^\top$ and

$$\Sigma^J = \begin{pmatrix} \sigma_{J,r}^2 & \blacklozenge \\ 0 & \sigma_{J,\sigma}^2 \end{pmatrix}. \quad (2)$$

Similarly, let the sizes of the co-jumps be normally distributed: $(c_{r,t}^{JJ}, c_{\sigma,t}^{JJ}) \sim \mathcal{N}(\mu^{JJ}, \Sigma^{JJ})$ with $\mu^{JJ} = (\mu_{JJ,r}, \mu_{JJ,\sigma})^\top$ and

$$\Sigma^{JJ} = \begin{pmatrix} \sigma_{JJ,r}^2 & \blacklozenge \\ \rho_{JJ} \sigma_{JJ,r} \sigma_{JJ,\sigma} & \sigma_{JJ,\sigma}^2 \end{pmatrix}. \quad (3)$$

We can allow all matrices to be a function of σ . This dependence may, for instance, lead to more (less) erratic price discontinuities in times of higher (lower) diffusive variance, an effect which, as shown, is coherent with the properties of the data.

4. (Time-varying leverage) The correlation between shocks to prices and shocks to variance ($-1 \leq \rho(\cdot) \leq 1$) is also a flexible function of the underlying spot variance level. In a stochastic volatility model with independent jumps only, Bandi and Renò (2012) show that leverage is a decreasing function of the spot variance level. In other words, leverage becomes more negative as the variance level increases. Here, we generalize their framework by employing the same flexible functional form for $\rho(\cdot)$ in the context of a stochastic volatility model with co-jumps (along with independent jumps). The

addition of co-jumps is of course not immaterial for the purposes of leverage identification. Because $\rho(\cdot)$ is usually identified off of the covariance between shocks to prices and shocks to variance, the potential presence of co-jumps, which we accommodate explicitly, would make the estimates of $\rho(\cdot)$ contaminated by the (conditional) covariance between the discontinuous jump components in prices and in volatility. It therefore seems important to distinguish between the component of the covariance between price changes and variance changes which is due to the continuous shocks to the system ($\rho(\cdot)$) and the component due to discontinuous shocks. Below we expand on this issue.

5. (Possibly non-linear structures) We do not express the price and volatility drifts, the variance-of-variance, and the intensities of the jumps as affine (or, more generally, parametric) functions of the underlying state. While the non-affine nature of our problem prevents us from obtaining closed-form, or near closed-form, derivative prices, here we opt for robustness to potential misspecifications in the form of the driving functions. This robustness property may be beneficial. For example, there is evidence that the variance of variance ($\Lambda^2(\cdot)$) should be a more flexible, possibly CEV, function of the underlying spot variance (e.g., Jones, 2003). Similarly, it may be important to allow the variance process to mean-revert at different speeds depending on its level, as implied by a non-linear mean-reverting variance drift. Eraker (2004), for instance, emphasizes that such a specification could lead to the sharp declines in option implied volatilities following the high levels associated with the 1987 crash. In what follows, guided by the nonparametric estimates, we are explicit about the types of nonlinearities which appear to be in the data.

Two observations are in order. First, some recent work has emphasized the potential importance of a multifactor volatility specification (see, *inter alia*, Chernov, Gallant, Ghysels and Tauchen, 2003). In our context, such a model could be expressed as

$$\begin{aligned} d(\log p_t) &= \mu(\sigma_t)dt + \sigma_t dW_{r,t} + c_{r,t}^J dJ_r + c_{r,t}^{JJ} dJ_{r,\sigma} \\ d\xi(\sigma_t^2) &= dx_t + c_{\sigma,t}^J dJ_\sigma + c_{\sigma,t}^{JJ} dJ_{r,\sigma}, \end{aligned} \tag{4}$$

where x_t is a sum of continuous Ito semimartingales potentially correlated with $W_{r,t}$. While estimation of the full system in this case would require simulation methods beyond our scopes, independent jump and co-jump identification would proceed just like in the present

paper. As we will make clear below, our proposed identification hinges on the use of higher-order infinitesimal cross-moments and these cross-moments are not impacted, theoretically, by the diffusive dynamics. To this extent, in what follows, we will also just report co-jump estimates obtained by employing higher-order moments. These estimates will be statistically and economically very similar to those derived from the model in Eq. (1) and, importantly, will attest to the robustness of the reported co-jump dynamics to a potentially higher number of volatility factors.

Second, the only parametric assumption we make is on the probability distribution of the jump sizes. We emphasize that any parametric distribution with a finite number of parameters could be employed. In addition, the explicit dependence of the size parameters on the level of volatility (as in 3. above) provides unconditional features which go beyond the assumed distributional features of the jump sizes. For instance, when working with Gaussian sizes, as we do below and support empirically, these time-varying parameters will readily deliver unconditional skewness and tail fatness, thereby capturing the effects described in Figure 1. A comparison of Figure 1 with Figure 9 (below) illustrates these ideas.

We now turn to the identification of the system by virtue of an infinitesimal moment-based procedure.

4 Infinitesimal cross-moments

Let prices and volatilities evolve dynamically as in Eq. (1). The jump sizes are expressed as in Eq. (2) and Eq. (3) with μ^J , μ^{JJ} , $\Sigma^J(\cdot)$, and $\Sigma^{JJ}(\cdot)$. Specifically, we work with a logarithmic variance specification ($\xi(\sigma_t^2) = \log(\sigma_t^2)$) and Gaussian jumps. Section 9 provides a specification analysis which supports a logarithmic transformation for variance.

The key element of the identification method we propose is the generic *infinitesimal cross-moment* of order p_1, p_2 with $p_1 \geq p_2 \geq 0$, namely

$$\vartheta_{p_1, p_2}(\sigma) = \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \mathbf{E} \left[(\log p_{t+\Delta} - \log p_t)^{p_1} (\log(\sigma_{t+\Delta}^2) - \log(\sigma_t^2))^{p_2} \mid \sigma_t = \sigma \right]. \quad (5)$$

By varying the orders p_1 and p_2 , the cross-moments have useful interpretations in terms of the underlying functions of the system. Below, we provide the general expressions and a few

illustrative examples. For notational convenience, we drop the dependence on σ and write ϑ_{p_1, p_2} instead of $\vartheta_{p_1, p_2}(\sigma)$. All moments and functions, however, should be interpreted as being dependent on the underlying spot volatility process.

We begin with the volatility moments. Write

$$\vartheta_{0,1} = m + \vartheta_{0,1}^{Jump}, \quad (6)$$

$$\vartheta_{0,2} = \Lambda^2 + \vartheta_{0,2}^{Jump}, \quad (7)$$

$$\vartheta_{0,p_2} = \vartheta_{0,p_2}^{Jump} \quad p_2 \geq 3 \quad (8)$$

with

$$\vartheta_{0,p_2}^{Jump} = \lambda_{r,\sigma} \sum_{j=0}^{p_2} \binom{p_2}{j} G_{0,j} (\sigma_{JJ,\sigma})^j (\mu_{JJ,\sigma})^{p_2-j} + \lambda_{\sigma} \sum_{j=0}^{p_2} \binom{p_2}{j} G_{0,j} (\sigma_{J,\sigma})^j (\mu_{J,\sigma})^{p_2-j},$$

where $G_{0,0} = 1$ and, for $g, g_1, g_2 \geq 1$,

$$G_{0,2g} = (2g-1)!!,$$

$$G_{0,2g-1} = 0,$$

$$G_{g_1, g_2} = (g_1 + g_2 - 1) \rho_J G_{g_1-1, g_2-1} + (g_1 - 1)(g_2 - 1)(1 - \rho_J^2) G_{g_1-2, g_2-2}.$$

The expressions imply that

$$\vartheta_{0,1} = m + \lambda_{r,\sigma} \mu_{JJ,\sigma} + \lambda_{\sigma} \mu_{J,\sigma},$$

$$\vartheta_{0,2} = \Lambda^2 + \lambda_{r,\sigma} ((\mu_{JJ,\sigma})^2 + (\sigma_{JJ,\sigma})^2) + \lambda_{\sigma} ((\mu_{J,\sigma})^2 + (\sigma_{J,\sigma})^2),$$

$$\vartheta_{0,3} = \lambda_{r,\sigma} ((\mu_{JJ,\sigma})^3 + 3(\mu_{JJ,\sigma})(\sigma_{JJ,\sigma})^2) + \lambda_{\sigma} ((\mu_{J,\sigma})^3 + 3(\mu_{J,\sigma})(\sigma_{J,\sigma})^2),$$

and so forth. In other words, *all* infinitesimal variance moments depend on the corresponding moments of the idiosyncratic and common jumps. The first and the second infinitesimal moments also depend on the drift m and the variance of variance Λ^2 , respectively.

The price moments $\vartheta_{p_1, 0}$ can be defined similarly. We now turn to the genuine cross-moments:

$$\vartheta_{1,1} = \rho \Lambda \sigma + \vartheta_{1,1}^{Jump} \quad (9)$$

and

$$\vartheta_{1+p_1,1+p_2} = \vartheta_{1+p_1,1+p_2}^{Jump} \quad p_1 > 1 \text{ or } p_2 > 1 \quad (10)$$

with

$$\vartheta_{p_1,p_2}^{Jump} = \lambda_{r,\sigma} \sum_{j_1=0}^{p_1} \sum_{j_2=0}^{p_2} \binom{p_1}{j_1} \binom{p_2}{j_2} G_{j_1,j_2} (\sigma_{JJ,r})^{j_1} (\sigma_{JJ,\sigma})^{j_2} (\mu_{JJ,r})^{p_1-j_1} (\mu_{JJ,\sigma})^{p_2-j_2}. \quad (11)$$

The cross-moment expressions imply, for instance, that

$$\vartheta_{1,1} = \rho \Lambda \sigma + \lambda_{r,\sigma} (\rho_J \sigma_{JJ,r} \sigma_{JJ,\sigma} + \mu_{JJ,r} \mu_{JJ,\sigma}),$$

and

$$\begin{aligned} \vartheta_{2,2} = & \lambda_{r,\sigma} \{ (\mu_{JJ,\sigma})^2 (\mu_{JJ,r})^2 + (\sigma_{JJ,\sigma})^2 (\mu_{JJ,r})^2 + (\mu_{JJ,\sigma})^2 (\sigma_{JJ,r})^2 \\ & + (1 + 2\rho_J^2) (\sigma_{JJ,r})^2 (\sigma_{JJ,\sigma})^2 + 4\rho_J \mu_{JJ,r} \mu_{JJ,\sigma} \sigma_{JJ,r} \sigma_{JJ,\sigma} \}. \end{aligned}$$

Thus, the moment of order 1, 1 depends on the covariance between the continuous (Brownian) shocks to the system as well as on the conditional moment of order 1, 1 of the common jump sizes. This effect generates an additional source of leverage in the model. The higher (than 1, 1) cross-moments only depend on the corresponding cross-moments of the common jumps.

In essence, by setting $p_2 = 0$ and varying p_1 one can identify features specific to the price process. Conversely, by setting $p_1 = 0$ and varying p_2 one can identify features specific to the variance process. The genuine cross-moments with $p_1 \geq p_2 \geq 1$ are, however, required to identify anything that is common to the price and the variance process: Brownian leverage ρ , the intensity of the common jumps $\lambda_{r,\sigma}$, and the correlation between the jump sizes ρ_J , for instance.

Interestingly, subcases of the more general specification we consider may readily lead to closed-form formulations of the functions of interest as "functions" of the infinitesimal cross-moments. Assume, for example, $c_{r,t}^J = c_{r,t}^{JJ}$, $c_{\sigma,t}^J = c_{\sigma,t}^{JJ}$ and $\mu^{JJ} = 0$ (same sizes across idiosyncratic and common jumps and zero means for all of the jump sizes). In this case, the moments $\vartheta_{p_1,0}$ with $p_1 = 1, 2, 4, 6$, say, identify μ , $\sigma_{J,r}^2$ and $\lambda_r + \lambda_{r,\sigma}$, the moments ϑ_{0,p_2} with $p_2 = 1, 2, 4, 6$, say, identify m , Λ , $\sigma_{J,\sigma}^2$ and $\lambda_\sigma + \lambda_{r,\sigma}$, and the moments $\vartheta_{1,1}$, $\vartheta_{2,2}$, and $\vartheta_{1,3}$, among other genuine cross-moments, identify ρ , ρ_J , and $\lambda_{r,\sigma}$. In fact, this is one of many,

possible, identification schemes resulting in closed-form dependencies between the model's functions and the employed infinitesimal cross-moments. For any volatility level, the model's functions would simply be the solution to a system of equations, those defining the stated infinitesimal cross-moments.

Based on the preliminary descriptive evidence provided earlier, it is important for us to differentiate between the sizes of the independent jumps and those of the co-jumps. It is also important to allow for jumps with non-zero means. Hence, we estimate the more general model presented above and do not work with closed-form expressions. The moments, however, lend themselves to (a form of) moment estimation. In particular, given nonparametric estimates $\hat{\vartheta}_{p_1, p_2}$, we may select an appropriate number of moment conditions, by varying p_1 and p_2 , and identify all functions for every volatility level. This procedure, dubbed Nonparametric Infinitesimal Method of Moment (NIMM), is applied to any chosen level in the volatility range and results in nonparametric estimates. Similarly, by imposing a parametric structure on the relevant functions, we may use the estimated moments $\hat{\vartheta}_{p_1, p_2}$ to identify parameters for the full system. In what follows, we use both strategies. In particular, we employ the unrestricted nonparametric estimates as useful descriptive tools. Directed by the functional estimates, we then turn to a parametric evaluation of the full system's dynamics.

5 Continuous-time identification

The cross-moments are (infinitesimal) conditional expectations. We use sample analogues to conditional expectations based on classical kernel estimators to identify them. Consider a sample of T days and N intradaily equispaced times, defined as "knots", within each day. Assume availability of closing logarithmic prices ($\log p_{t,i}$) and spot volatility estimates ($\log \hat{\sigma}_{t,i}^2$) over each day $t = 1, \dots, T$ and each knot $i = 1, \dots, N$. The generic cross-moment estimator $\hat{\vartheta}_{p_1, p_2}$ is defined as

$$\hat{\vartheta}_{p_1, p_2}(\sigma) = \frac{\sum_{t=1}^{T-1} \sum_{i=1}^N \mathbf{K} \left(\frac{\hat{\sigma}_{t,i} - \sigma}{h} \right) (\log p_{t+1,i} - \log p_{t,i})^{p_1} (\log \hat{\sigma}_{t+1,i}^2 - \log \hat{\sigma}_{t,i}^2)^{p_2}}{\Delta \sum_{t=1}^T \sum_{i=1}^N \mathbf{K} \left(\frac{\hat{\sigma}_{t,i} - \sigma}{h} \right)}, \quad (12)$$

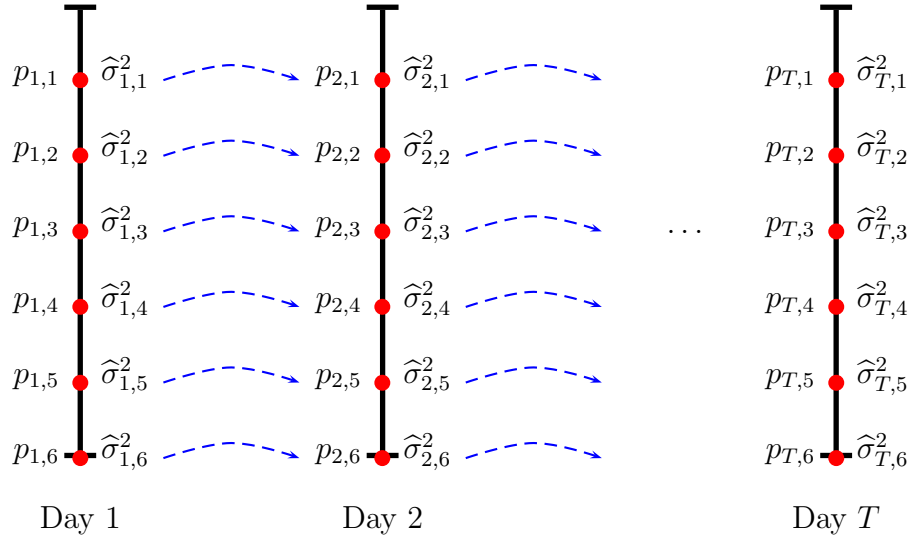


Figure 4: Pictorial representation of the way in which we employ data in Eq. (12). The estimates $\hat{\sigma}_{t,i}^2$ are computed with intraday data between prices $p_{t,i-1}$ and $p_{t,i}$. The dashed arrows indicate the direction along which we compute log-price and log-variance differences. These differences are always daily. Sampling volatility multiple times over each day, instead of only once per day, provides superior performance in estimating the cross-moments.

where $\mathbf{K}(\cdot)$ is a kernel function, with properties listed in Appendix A, whose role is to guarantee proper conditioning at σ for a small bandwidth h . In essence, then, the procedure is a two-step procedure. In the first step, spot volatility is estimated. In the second step, kernel estimates of the cross-moments are put to work to identify the full system's dynamics.

It is important to note that, in Eq. (12), we never "differentiate" volatility within the same day but always across subsequent days. Figure 4 illustrates our sampling. In this sense, all of our moment estimates are daily ($\Delta = 1$).

In the following subsection, in addition to describing the spot volatility estimation procedure, we explain why it is beneficial to use N daily spot variance differences instead of a single difference ($N = 1$) of integrated daily variances.

5.1 *Spot versus daily variance estimation*

In the implementation of the estimator in Eq. (12), we use the full trading day (from 9:30am to 4pm) and employ $N = 6$ knots in the interval 10.35am - 4.00pm, each separated by one hour and 5 minutes. Define one-minute logarithmic returns $r_{t,i,k} = \log p_{t,i,k} - \log p_{t,i,k-1}$, for $k = 1, \dots, 65$, over each hour and 5 minutes before a generic knot i .⁸ The spot variance estimates, for each day t and each knot i , are obtained by applying the jump-robust threshold bipower variation estimator

$$\hat{\sigma}_{t,i}^2 = \frac{65}{64 - n_j} \varsigma_1^{-2} \sum_{k=2}^{65} |r_{t,i,k}| |r_{t,i,k-1}| I_{\{|r_{t,i,k}| \leq \theta_{t,i,k}\}} I_{\{|r_{t,i,k-1}| \leq \theta_{t,i,k-1}\}}, \quad (13)$$

where $\varsigma_1 \simeq 0.7979$, $\theta_{t,i,k}$ is a suitable threshold, and n_j is the number of returns whose absolute value is greater than the threshold $\theta_{t,i,k}$ (Corsi et al., 2010). Even though, in order to span the trading day, the 6 knots are separated by 65 (rather than 60) minutes, we call this variance estimates "hourly" for conciseness. Alternative spot variance estimators may be used. However, any chosen estimator should identify spot (diffusive) variance and, hence, be robust to jumps in the price process. The estimator we employ eliminates jumps in two ways. Similarly to the classical bipower variation of Barndorff-Nielsen and Shephard (2004), discontinuities are annihilated asymptotically by the adjacent diffusive component. In addition, the jumps are discarded in finite samples, and asymptotically, when above a (vanishing, asymptotically) threshold $\theta_{t,i,k}$ (as in the threshold realized variance of Mancini, 2009).

Why local variance? The model and the moment estimates in Eq. (12) are written for a genuinely spot variance process. Even though traditional, the use of integrated daily estimates would likely induce attenuation effects in the estimation of the genuine cross-moments very much akin those determined by asynchronous effects in covariance estimation. Since, as explained earlier, the cross-moments are crucial to identify all aspects of the model which are common to the price and volatility series (the co-jumps, *in primis*), we opt for a more local notion of variance than in the existing literature, one that is nicely afforded by our use of high-frequency data for identification. This said, the relative size of the

⁸The one minute returns are constructed after pre-averaging all observed transaction prices within each minute. The pre-averaging is designed to reduce the impact of market microstructure noise (Jacod et al., 2009).

measurement error in variance estimation increases with the degree of locality. In other words, more integrated measures are expected to be relatively less noisy than less integrated ones. When considered jointly, these observations point to a probable bias-variance trade-off in the identification of the cross-moments, and the final nonparametric estimates, as a function of the choice of locality. More local variance estimates should lead to smaller biases but less precise estimates.

The simulation evidence in Figure 5 supports this intuition. We simulate 1,000 replications of the following model with co-jumps:

$$\begin{aligned} dX_t &= \sigma_t dW_t^1 + c_X dJ_t, \\ d\log \sigma_t^2 &= \eta dW_t^2 + c_\sigma dJ_t, \end{aligned} \tag{14}$$

where W^1 and W^2 are independent Brownian motions, $\eta = 0.6$, the intensity of the co-jumps λ is $50/252$, c_X and c_σ are normally distributed with zero mean and standard deviations 4 and 4η respectively, and the co-jump size correlation ρ_J is -0.9 . Each replication consists of 1,000 days. During each day, the price is sampled 360 times, mimicking one-minute returns over a 6-hour trading day. Then, as a function of the number of intradaily knots, we back out the intensity of the co-jumps and the correlation of their jump sizes using the infinitesimal cross-moments of order 2, 2 (for the intensity) and 3, 1 (for ρ_J). In both cases, in order for us to simply focus on the impact of "frequency" on the estimates, the parameters other than the intensity (when using the 2, 2 cross-moment) or the correlation (when using the 3, 1 cross-moment) are assumed to be fixed at their simulated value rather than being estimated. Estimation of all parameters would not modify the logic of the argument in this subsection.⁹

In Figure 5, the circles represent the average intensity values (left panel) and co-jump correlations (right panel) obtained by virtue of the true integrated variances between knots (the higher the number of knots, the more local the spot variance values). The stars represent average values yielded by the estimated counterparts of the previous quantities using thresholds (again, the higher the number of knots, the more local the spot variance estimates). The vertical bars are simply their 95% bounds across simulations. Finally, the crosses represent average values, across simulations, delivered by the true spot volatilities. In this case, the

⁹The simulation set-up in this section is designed to illustrate the impact of alternative volatility estimates on the identification of the cross-moments and the model's functions. For this reason, it is purposely simple. When evaluating the accuracy of all of the model's functions (see Appendix), we simulate from our more general parametric specification in Section 7.

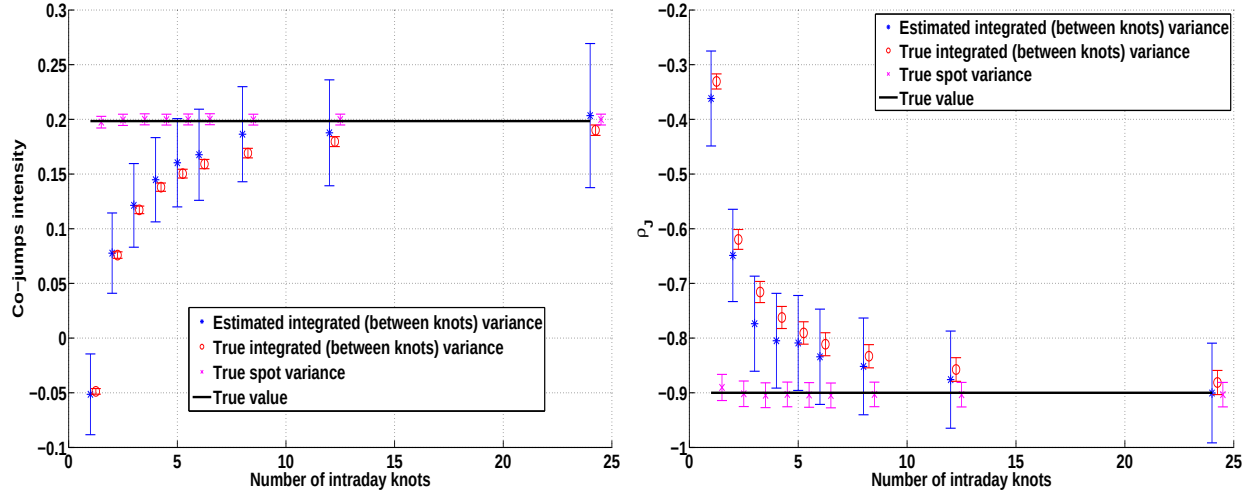


Figure 5: Impact of the choice of the number of intraday knots on the estimates of the co-jump intensity (left panel) and of the co-jump correlation (right panel).

choice of the number of knots only translates into more, or fewer, terms being averaged in Eq. (12). It does not affect the degree of locality of variance since, at each set of knots, the true spot variances simulated from the model are used. The feasible case is the second case (corresponding to the stars). In practise, coherently with the representation in Fig. 4, increasing the number of knots N translates into more local, than daily, variance estimates and a larger number of differences in the definition of the estimator in Eq. (12).

The use of daily ($N = 1$) integrated variances, whether true or estimated, causes substantial attenuation in the estimated likelihood of co-jumps and their correlation. The use of more local estimates results in smaller biases but larger noise. Our choice of hourly estimates ($N = 6$) is meant to balance the resulting finite-sample MSEs in a sensible way. Existing estimation methods in the stochastic volatility literature, like likelihood methods, use the information in the cross-moments implicitly, something which we do explicitly. In addition, existing volatility filters are daily in nature. Coherently with this simulation evidence, it is the loss of granularity due to the use of integrated (daily) variance estimates which may have led to a body of ambiguous evidence about the presence of co-jumps, and the magnitude of their size correlation, when using price data alone. In this experiment, we estimate a negative co-jump intensity (left panel) and a correlation of -0.35 (right panel) when the true value is -0.9 . In essence, overly integrated estimates may drastically reduce the likelihood of co-jumps as well as the size of the estimated correlation between co-jump sizes, thereby

justifying the uncorrelatedness usually found when employing price data recorded at daily frequencies (e.g., Eraker et al., 2003, and Eraker, 2004). The use of more local variance estimates, afforded here by the availability and use of high-frequency price observations, appears to provide needed precision in the identification of infrequent, large co-variations.

Two observations are in order. In a finite sample, the use of variance estimates which are more local in nature than daily integrated variance may be subject to intra-daily U-shaped effects. To this extent, we recall that the estimator in Eq. (12) averages (scaled) differences of hourly variance measures across days (t is a daily index and, again, Δ is equal to 1 day). Since, as is typically found, the U-shape structure of intra-daily variance effects is virtually unchanged between consecutive days in our sample, our final (daily) functional estimates are robust to this type of finite-sample effect. In addition, the spot variance estimates are scaled up to daily values so as to take into account the overnight returns. This is a typical adjustment to achieve robustness to biases due to the overnights.

We conclude this section by summarizing the logic behind our mixed frequency approach (minute-by-minute returns aggregated into 65-minute spot volatility estimates in order to model daily dynamics).

Estimating the model entirely at high frequency would require the introduction, in the specification of the volatility dynamics, of intraday effects superimposed on the assumed jump-diffusion structure. In order to fully capture high-frequency effects in the data, one would also have to carefully design specifications for market microstructure noise. By estimating the model at the one-day frequency after reducing the impact of noise, we may dispense with the need to model intradaily dynamics while still relying on accurate high-frequency estimates of spot volatility. As a result, we focus on an horizon which is relevant for typical financial applications and provide estimates which can be directly compared to those in the existing stochastic volatility literature (for which the one-day frequency is typical).

The use of one-minute returns is needed to exploit the power of realized estimates in the evaluation of spot volatility. We begin, in fact, with tick-by-tick data: consistent with pre-averaging, we aggregate returns over a one-minute interval solely to soften the impact of market microstructure noise.

The use of 65-minute volatilities is motivated by a bias-variance trade-off, illustrated above. The lower the frequency, the higher the bias in estimating the quantities of interest (co-

jumps intensity and correlation). The higher the frequency, the higher the variance. An intermediate choice was made to balance these two effects. While, theoretically, the optimal choice is necessarily a function of the unknown volatility dynamics, our simulations show that splitting a 6.5 trading day into 6 intervals (or 65 minutes, of course) represents an effective compromise between parsimony of the procedure and accuracy.

5.2 Cross-moment based identification: NIMM

The use of spot volatility estimates, as in Eq. (13), paired with the functional moment estimates in Eq. (12), will be used for pointwise identification of all of the system's functions. Our approach relates to two existing approaches in the literature. Bandi and Renò (2009) first introduce nonparametric identification of discontinuous stochastic volatility models under the assumption of independent jumps. Bandi and Renò (2011) focus explicitly on nonparametric leverage estimation and, in agreement with Bandi and Renò (2009), only allow for independent jumps. Differently from the previous work on the subject, this paper allows for co-jumps and focuses on an empirical treatment which highlights the statistical and economic relevance of discontinuous joint changes in prices and volatility. In particular, the addition of co-jumps with non-zero jump sizes, which represents the substantive core of our treatment, leads to two substantial innovations for the purpose of identification. First, we recognize the crucial role played by the *infinitesimal cross-moments* for co-jump identification. With the exception of the cross-moment of order 1, 1, which is also affected by diffusive leverage, the cross-moments are zero when the discontinuities are idiosyncratic, as in our previous work, and, therefore, play no role in their identification. Second, when allowing for the more complex jump dynamics in this paper, closed-form estimation of the system's functions in terms of the estimated moments, as conducted in Bandi and Renò (2009, 2011) is, in general, not feasible. Importantly, however, the infinitesimal cross-moments introduced here lend themselves to an estimation method akin to pointwise GMM (Hansen, 1982), something which we discuss next. The use of infinitesimal cross-moments to evaluate complex multivariate systems with discontinuities is, in our view, of general interest. We apply here to the evaluation of co-movements in prices and volatilities. It could be employed to study a host of alternative problems, contagion, for instance.

Denote by $g_1(\sigma), \dots, g_K(\sigma)$ the K functions driving the dynamics of the system. These are the estimation targets. Consider a set of N cross-moments $\hat{\vartheta}_{p_1, p_2}(\sigma)$ with $N \geq K$

for identification. The theoretical cross-moments $\vartheta_{p_1,p_2}(\sigma) = f_{p_1,p_2}(g_1(\sigma), \dots, g_K(\sigma))$ are a mapping f_{p_1,p_2} from the functions $g_1(\sigma), \dots, g_K(\sigma)$ to $\mathbb{R}$. For every value σ in the spot volatility range, the K vector of estimates $(\hat{g}_1(\sigma), \dots, \hat{g}_K(\sigma))$ is defined as:

$$(\hat{g}_1(\sigma), \dots, \hat{g}_K(\sigma)) = \arg \min_{(g_1(\sigma), \dots, g_K(\sigma))} (\hat{\vartheta}_{p_1,p_2}(\sigma) - \vartheta_{p_1,p_2}(\sigma))^\top W(\sigma) (\hat{\vartheta}_{p_1,p_2}(\sigma) - \vartheta_{p_1,p_2}(\sigma)),$$

where $W(\sigma)$ is an $N \times N$ symmetrical and positive definite weighting matrix. We implement a two-stage procedure to optimize the choice of $W(\sigma)$. In the first stage, $W(\sigma)$ is the identity matrix. The first-stage estimates are used to compute V , the variance-covariance matrix of the cross-moments, i.e., the matrix with generic element $V_{(p_1,p_2),(p_3,p_4)} = \frac{f_{p_1+p_3,p_2+p_4}(\hat{g}_1(\sigma), \dots, \hat{g}_K(\sigma))}{h(\sigma)\hat{L}_T(\sigma)}$, where $h(\sigma)$ is a volatility-specific bandwidth and $\hat{L}_T(\sigma)$ is an estimate of the volatility's occupation density. The estimation is then repeated with $\widehat{W}(\sigma) = V^{-1}$. The same procedure is implemented for different values of σ leading to the nonparametric estimates presented in the next section.

The functional estimates may provide useful guidance about parametric specification. Given a parametrization of the system's functions, the procedure we propose can be adapted to also yield parametric estimates. Denote by η a vector of M parameters. Select a number G of knots $\sigma_1, \dots, \sigma_G$, so that $N \times G \geq M$ for identification. Denote by $\hat{\vartheta}_{p_1,p_2}$ the $N \times G$ -vector of available estimated moments computed at the knots σ_i with $i = 1, \dots, G$ and by $\vartheta_{p_1,p_2}(\eta)$ the corresponding $N \times G$ -vector of theoretical moments. The parametric estimates are now given by:

$$\hat{\eta} = \arg \min_{\eta} (\hat{\vartheta}_{p_1,p_2} - \vartheta_{p_1,p_2}(\eta))^\top W(\sigma) (\hat{\vartheta}_{p_1,p_2} - \vartheta_{p_1,p_2}(\eta))$$

where $W(\sigma)$ is an $(N \times G) \times (N \times G)$ symmetrical and positive definite weighting matrix. To optimize the choice of $W(\sigma)$ and underweight relatively noisier moments, we evaluate the variance-covariance matrix between the N moments at the G knots (V) via simulation. We then employ $W = V^{-1}$.

Appendix A derives the consistency and asymptotic normality properties of $\hat{\vartheta}_{p_1,p_2}$, for a known σ , under a suitable asymptotic sampling scheme. Appendix A also provides conditions on $\theta_{t,i,k}$, among other choice variables, so that these properties are not affected by the estimation error necessarily introduced by the spot variance estimates obtained in Eq. (13) above. Appendix B discusses finite sample issues and issues of implementation.

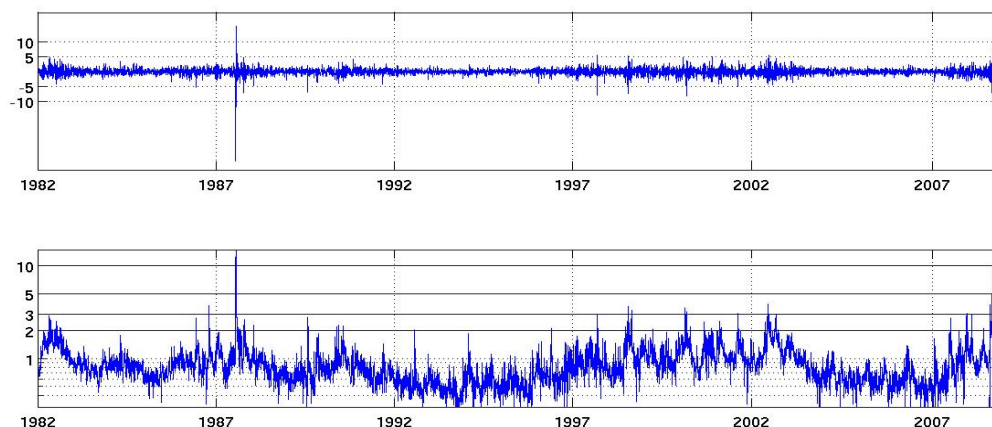


Figure 6: Top: Daily returns (%). Bottom: Filtered daily volatilities (in log scale) obtained by averaging hourly spot volatilities within each day (%).

6 Price and variance dynamics: functional estimates

As mentioned, we employ all transactions on S&P 500 futures from April 21, 1982, to February 5, 2009, for a total of 6,748 trading days. The daily return data and daily *filtered* (by virtue of high-frequency price data) logarithmic volatility series are reported in Figure 6.

Figure 7 provides the estimated moments of order p_1, p_2 with $p_1 + p_2 \leq 4$, plus the moments of order 2,3 and 3,2. These moments, which we employ for estimation, are sufficient for identification and are expected to be less noisy than their higher-order counterparts. The functions driving the price and volatility dynamics in Eq. (1) are reported in Figure 8. All functions are plotted with respect to daily volatility expressed as a percentage. In terms of interpretation, a value of 1 on the horizontal axis corresponds to a yearly volatility of about $\sqrt{252}\% = 15.87\%$.

We begin by commenting on the price process. The price drift μ is insignificantly different from zero across volatility states. The inability to detect risk-return trade-offs using daily return data is a notorious phenomenon which we confirm here.¹⁰ The intensity of the idiosyncratic jumps in prices λ_r is fairly stable, across volatility levels, and implies between

¹⁰Andersen et al. (2002), Pan (2002) and Eraker et al. (2004) also find a statistically insignificant dependence of the price drift on spot variance.

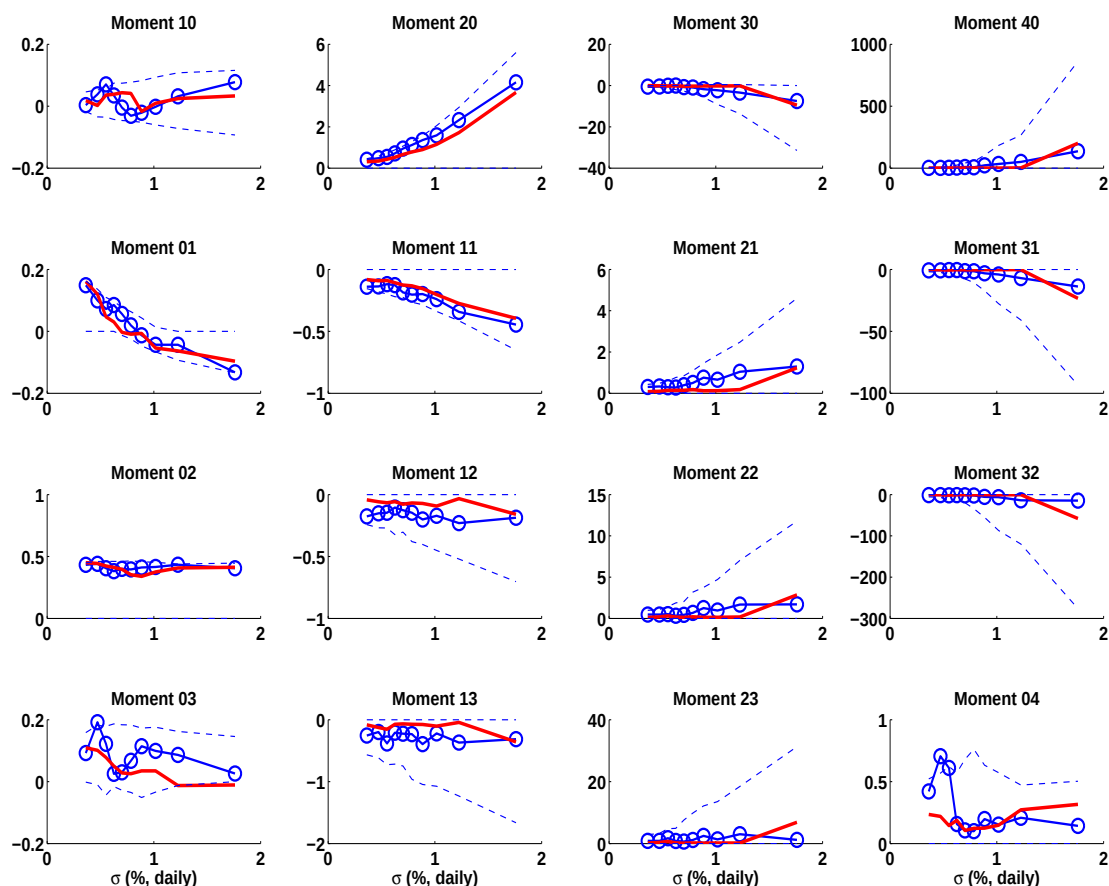


Figure 7: Estimated infinitesimal cross-moments (solid circled lines) with 95% confidence bands (dashed lines). The solid lines are implied *fitted* moments using functions, displayed in Figure 8, obtained via pointwise GMM.

roughly 5 and 15 jumps per year in terms of point estimates. As expected in this type of problems, the level of statistical uncertainty is substantial, particularly for relatively larger, seldom visited, volatility levels. The parametric estimates will refine this analysis. The mean of the idiosyncratic price jumps $\mu_{J,r}$ is indistinguishable from zero. The corresponding standard deviation $\sigma_{J,r}$ has a very slight tendency to increase with the volatility level, thereby suggesting the possibility of increasing jump sizes in times of higher uncertainty. This finding will be more evident in the case of the price co-jumps.

We now turn to stochastic variance. As expected, the variance drift m implies mean-reversion in log variance. The variance of variance Λ is relatively flat across volatility states. For

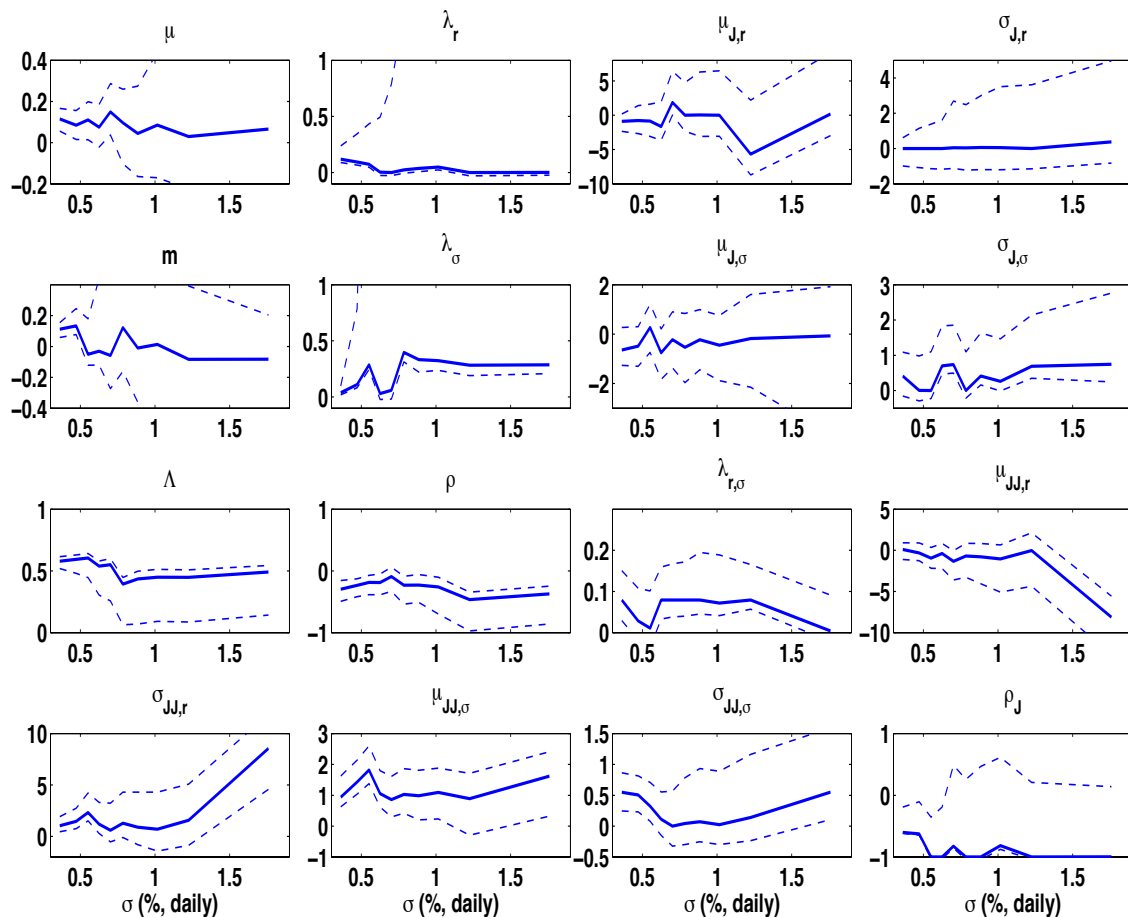


Figure 8: Estimated functions (solid lines) with 95% confidence bands (dashed lines).

instance, it does not display the CEV shape that has been emphasized as being important in some existing work (see, e.g., Chacko and Viceira, 2003, and Jones, 2003). This is, however, an unsurprising outcome of the logarithmic transformation of variance which we adopt in the present paper and justify in the subsequent specification analysis. The nonparametric inference presented here yields point estimates for the intensity of the variance jumps λ_σ which vary considerably across volatility states while also carrying a substantial degree of statistical uncertainty. The functional λ_σ estimates suggest a larger number of independent volatility jumps than independent price jumps, something which our parametric estimates in the next section will confirm. The mean of the idiosyncratic variance jumps is close to zero. The size of the independent variance jumps is relatively stable across volatility levels.

We now focus on the joint dynamics. The number of co-jumps implied by $\lambda_{r,\sigma}$ is larger than

zero and fairly stable across volatility levels, mainly around the bulk of the data ($\sigma = 0.8\%$). When jumping jointly with prices, volatility has a higher probability to increase than to decrease as indicated by a mean of the corresponding jump size distribution $\mu_{JJ,\sigma}$ which is significant and positive. In contrast, the mean of the common jumps in prices $\mu_{JJ,r}$ is negative. Also, more negative price co-jumps are associated with higher volatility levels. As in the idiosyncratic jump case, the standard deviation of the sizes of the common volatility jumps is relatively stable across volatility levels. The standard deviation of the sizes of the common price jumps $\sigma_{JJ,r}$ is, instead, strongly increasing with the volatility level. In sum, the estimated volatility co-jump size distribution is fairly invariant to the underlying volatility level. In contrast, the mean and the variance of the size distribution of the price co-jumps become more negative and increase, respectively, as volatility increases. Importantly, the correlation between discontinuous price changes and discontinuous volatility changes ρ_J is negative and very close to -1 . As emphasized, the magnitude and sign of this correlation have been rather elusive in work that uses - like we do - price data alone (see, e.g., Broadie et al., 2007). The economic and statistical strength of our reported finding, instead, clarifies that the negative co-jump correlation detected in earlier work employing option data (Eraker, 2004, and Todorov and Tauchen, 2011) cannot solely depend on the negative covariance between price jumps and market risk premia. It appears to be a genuine feature of the market price process, one that suitable derivative pricing and risk management should account for. We stress the importance of this additional source of return skewness in a specification which, differently from the bulk of the applied literature, explicitly separates the jump components into idiosyncratic and common factors.¹¹

We now turn to the most traditional source of skewness, Brownian leverage. As for other quantities, Figure 8 provides estimates of leverage as a function of spot volatility. This dependence has been emphasized as being important in a specification with independent jumps (Bandi and Renò, 2011). In agreement with Bandi and Renò (2011), but in the context of a more flexible model specification allowing for co-jumps, we find a decrease in the leverage estimates, i.e., values which are more negative associated with higher volatility levels, particularly around the center of the observed volatility range. Thus, the time-varying (with volatility) structure of ρ emphasized by Bandi and Renò (2011) appears to be robust

¹¹This separation is deemed important by Duffie et al. (2000) and defines their more general stochastic volatility model. The more general model is however not estimated. Eraker et al. (2003) and Eraker (2004) estimate models in which the jumps are either independent or perfectly dependent in terms of arrival times. In their framework, no allowance is made for independent jumps and common jumps.

to the presence of an additional source of skewness (further discussed below) induced by the joint occurrence of price and volatility jumps.

To summarize, the descriptive, nonparametric, analysis presented here suggests that logarithmic prices and logarithmic variances are likely to feature both independent and common jumps. The independent jumps are roughly mean zero. The common jumps have positive means, if in volatility, and negative means, if in prices. In the joint case, the price size distribution expands and re-centers around more negative values as volatility increases. The size distribution of the volatility co-jumps is, instead, fairly stable. The sizes of the price and volatility co-jumps are strongly anti-correlated.

7 Price and variance dynamics: parametric estimates

We now turn to a parametric assessment. The parametric model is motivated by the evidence in Section 2 and the previous nonparametric analysis. Its specification is:

$$\begin{aligned}
d \log p_t &= \mu_r dt + \sigma_t \left\{ \rho_t dW_t^1 + \sqrt{1 - \rho_t^2} dW_t^2 \right\} + c_{r,t}^J dJ_r + c_{r,t}^{JJ} dJ_{r,\sigma} \\
d \log(\sigma_t^2) &= (m_0 + m_1 \log(\sigma_t^2)) dt + \Lambda dW_t^1 + c_{\sigma,t}^J dJ_\sigma + c_{\sigma,t}^{JJ} dJ_{r,\sigma}, \\
\rho_t &= \max(\min(\rho_0 + \rho_1 \sigma_t, 1), -1), \\
\{J_r, J_\sigma, J_{r,\sigma}\} &\sim \text{Poisson}(\lambda_r, \lambda_\sigma, \lambda_{r,\sigma}) \\
c_{r,t}^J &\sim \mathcal{N}(\mu_{J,r}, \sigma_{J,r}^2) \\
c_{\sigma,t}^J &\sim \mathcal{N}(\mu_{J,\sigma}, \sigma_{J,\sigma}^2) \\
\begin{pmatrix} c_{r,t}^{JJ} \\ c_{\sigma,t}^{JJ} \end{pmatrix} &\sim \mathcal{N} \left(\begin{pmatrix} \mu_{JJ,r,0} + \mu_{JJ,r,0} \sigma_t \\ \mu_{JJ,\sigma} \end{pmatrix}, \begin{pmatrix} (\sigma_{JJ,r,0} + \sigma_{JJ,r,1} \sigma_t^{\sigma_{JJ,r,2}})^2 & \rho_J (\sigma_{JJ,r,0} + \sigma_{JJ,r,1} \sigma_t^{\sigma_{JJ,r,2}}) \sigma_{JJ,\sigma} \\ \sigma_{JJ,\sigma}^2 \end{pmatrix} \right). \tag{15}
\end{aligned}$$

The system, which is readily viewed as a special case of the model in Eq. (1), has 21 parameters whose estimates are reported in Column 5 of Table 2.¹² We estimate three relevant restricted models as well. The first model sets $J_{r,\sigma} = 0$ and has 12 parameters whose estimated values are reported under the title “no co-jumps” in Column 3 of Table

¹²The confidence bands are obtained by simulation. The estimated parametric model in Table 2, Column 5 is used to generate simulated samples as large as the original sample. The procedure is then applied to every simulated sample.

2. The second specification sets $J_r = J_\sigma = 0$ and has 15 parameters whose estimates are reported under the title “no independent jumps” in Column 4 of Table 2. The third model is without idiosyncratic jumps ($J_r = J_\sigma = 0$) and only uses higher-order moments ($p_1 + p_2 \geq 3$) for identification. For this latter model, which focuses on the co-jump part only and is theoretically robust to the specification of the diffusive part in Eq. (4), we report estimates in Column 2 of Table 2. The statistical and economic similarity between these results and the results for the full model attest to the robustness of the co-jump dynamics to alternative (possibly multivariate) specifications of the continuous, diffusive component of the market price process.

We begin the discussion, again, with the price equation. Consistent with the functional evidence reported earlier, preliminary estimation of a linear function for the price drift μ , namely $\mu_{r,0} + \mu_{r,1}\sigma$, resulted in an estimated slope coefficient $\mu_{r,1}$ very close to 0, numerically and statistically. Hence, we only report estimates for a constant drift. The extremely weak compensation for variance risk, when evaluated at the daily level, has been confirmed by several empirical studies, including Eraker et al. (2003). Idiosyncratic jumps in prices do not play a relevant role. Even though, when inferred from an estimated constant value for λ_r , the number of idiosyncratic price jumps is found to be around 6/7 per year (0.0252×252), these jumps are fairly small. The mean of the idiosyncratic price jumps is positive (1.39%, in terms of point estimate) but insignificantly different from zero. Their standard deviation is only 0.68%.

Before discussing co-jumps, we turn to the volatility equation. We fit a linear drift, namely, $m = m_0 + m_1 \log(\sigma^2)$, and find very accurately estimated linear mean-reversion in logarithmic volatility. In particular, the parameter m_1 is equal to -0.0597 with a 95% band equal to $(-0.0721, -0.0348)$. The volatility of volatility Λ is precisely estimated at a constant value equal to about 0.56. The corresponding 95% confidence interval is $(0.42, 0.59)$. The number of independent jumps in volatility is estimated at an annual value of about 13 (0.0528×252). The mean of the independent variance jumps is, again, statistically zero (point estimate equal to -0.45 with a 95% confidence band equal to $(-1.03, 0.20)$). The corresponding standard deviation is about 0.7.

The estimated number of co-jumps per year is 8.54 (0.0339×252) with a 95% confidence interval of $(4.79, 22.65)$ per annum. This number is consistent with the empirical evidence provided in Section 2. Further statistical evidence on the presence of significant co-jumps

Table 2: In Column 5 we display parametric estimates of the model in Eq. (15) with the corresponding 95% confidence intervals. Estimates of the same model with the restriction $J_{r,\sigma} = 0$ (no co-jumps) and $J_r = J_\sigma = 0$ (no independent jumps) are in Column 3 and 4, respectively. In Column 2 we display parametric co-jump estimates obtained by using only higher-order moments for identification. The parameters are relative to daily data, and imply returns expressed in percentage form.

Parametric estimates

parameter	only co-jumps	no co-jumps	no independent jumps	full model
μ_r	—	0.0423 (0.0000, 0.4848)	0.0631 (0.0530, 0.1158)	0.0306 (0.0000, 0.0949)
ρ_0	—	-0.2280 (-0.3627, -0.1891)	-0.0977 (-0.1500, 0.0222)	-0.0988 (-0.1979, 0.0322)
ρ_1	—	-0.0874 (-0.1148, 0.0203)	-0.1225 (-0.1921, -0.0560)	-0.1617 (-0.2801, -0.0814)
m_0	—	-0.0232 (-0.0399, -0.0051)	-0.0397 (-0.0865, -0.0241)	-0.0380 (-0.0961, 0.1382)
m_1	—	-0.0704 (-0.0824, -0.0539)	-0.0576 (-0.0662, -0.0321)	-0.0597 (-0.0721, -0.0348)
Λ	—	0.6048 (0.5752, 0.6151)	0.5950 (0.5674, 0.6045)	0.5583 (0.4234, 0.5902)
$\mu_{J,r}$	—	-0.1137 (-0.4279, 0.0759)	—	1.3948 (-0.5394, 3.1128)
$\mu_{JJ,r,0}$	-1.1333 (-2.5000, 1.5944)	—	0.5210 (-0.4511, 1.3991)	-0.0544 (-0.9658, 1.1614)
$\mu_{JJ,r,1}$	-0.7022 (-4.0000, 0.2566)	—	-1.8976 (-3.8597, -0.3171)	-1.0072 (-3.2868, 0.3475)
$\sigma_{J,r}$	—	1.2715 (0.1196, 1.9439)	—	0.6818 (0.0000, 1.7624)
$\sigma_{JJ,r,0}$	0.7652 (0.0000, 2.5921)	—	1.7428 (0.0000, 2.0801)	0.6246 (0.0000, 1.6966)
$\sigma_{JJ,r,1}$	0.8232 (0.0000, 6.8987)	—	0.1718 (0.0000, 3.1924)	2.2469 (0.6919, 4.8941)
$\sigma_{JJ,r,2}$	3.4350 (0.0633, 6.0732)	—	1.8828 (0.0000, 4.8503)	1.0863 (0.4230, 1.9144)
$\mu_{J,\sigma}$	—	0.3498 (-0.1498, 0.7563)	—	-0.4497 (-1.0275, 0.2038)
$\mu_{JJ,\sigma}$	1.2230 (0.2932, 2.0986)	—	0.7816 (0.5484, 0.8936)	1.4428 (0.9343, 1.5253)
$\sigma_{J,\sigma}$	—	1.2575 (0.8060, 1.3825)	—	0.7002 (0.0000, 1.0615)
$\sigma_{JJ,\sigma}$	0.4479 (0.0476, 1.2496)	—	0.4901 (0.1657, 0.5299)	0.1084 (0.0000, 0.4276)
ρ_J	-1.0000 (-1.0000, -0.3869)	—	-0.6416 (-1.0000, -0.7373)	-1.0000 (-1.0000, -0.1378)
λ_r	—	0.1033 (0.0329, 4.2099)	—	0.0252 (0.0001, 0.1591)
λ_σ	—	0.0279 (0.0043, 0.0480)	—	0.0528 (0.0003, 0.5293)
$\lambda_{r,\sigma}$	0.0297 (0.0000, 0.0803)	—	0.0489 (0.0401, 0.1081)	0.0339 (0.0190, 0.0899)

in the data is provided in Section 9. When jumping jointly with volatility, the price process displays a tendency to jump downwards. This tendency increases with the volatility level as implied by the point estimates of $\mu_{JJ,r,0}$ (-0.05%) and $\mu_{JJ,r,1}$ (-1.01). Guided by the nonparametric findings in the previous section and the preliminary analysis in Section 2, we fit a nonlinear structure to the standard deviation of the common price jumps, namely $\sigma_{J,r,0} + \sigma_{J,r,1}\sigma^{J,r,2}$. All parameters are positive and, with the exception of the constant term $\sigma_{J,r,0}$, statistically significant, thereby suggesting an increase in the size of the common price jumps along with increases in volatility. Variance has a tendency to jump upward. The mean of the common jumps in variance is estimated at a value equal to 1.44. The 95% confidence band for this parameter is (0.93, 1.52). The volatility of the common jumps in variance is estimated at a value of 0.1084 with a 95% confidence interval of (0, 0.43). To visualize the impact of these numbers, the size distributions of the price and volatility co-jumps are plotted in Figure 9. The distribution of the size of the price co-jumps is plotted as a function of three representative volatility levels (namely, 0.5%, 1%, and 2%). As volatility increases, this distribution expands and re-centers around more negative values. The implied unconditional distribution, also displayed in the figure, shows substantial kurtosis and some skewness, thereby strongly departing from normality. As discussed, these effects are generated by means and variances that are functions of spot volatility. The effects are in line with the preliminary descriptive evidence in Figure 1. Finally, importantly, the correlation between the common price jumps and the common volatility jumps is estimated at a value which is very clearly negative. The point estimate is at the lower boundary (-1) and the upper limit of the 95% confidence interval is -0.1378 .¹³

In sum, the estimates suggest that the price process is more likely than not to jump jointly with the volatility process (the number of price co-jumps is larger than the number of independent price jumps). When moving jointly in a discontinuous fashion, volatility tends to shift upward, whereas prices have a tendency to move downward. This said, the model estimates imply a substantial likelihood of positive price co-jumps, both at high and at low volatility levels (see Figure 9). This is, again, consistent with the preliminary evidence in

¹³This negative correlation is larger, in absolute value, than that reported in the descriptive analysis in Section 2. This is due to the fact the preliminary descriptive evidence in Section 2 is based on variance measures which are, by definition, estimates of the true variances and, therefore, affected by some measurement error. This measurement error will induce attenuation effects when evaluating the correlation between price and volatility jump sizes. Thus, on the one hand, we should view the reported correlation in Section 2 as being conservative and, likely, too small in absolute value. On the other hand, the finite sample adjustments leading to the estimates in this section are designed to lead to accurate identification. Appendix B provides details on these adjustments and confirms their effectiveness by simulation.

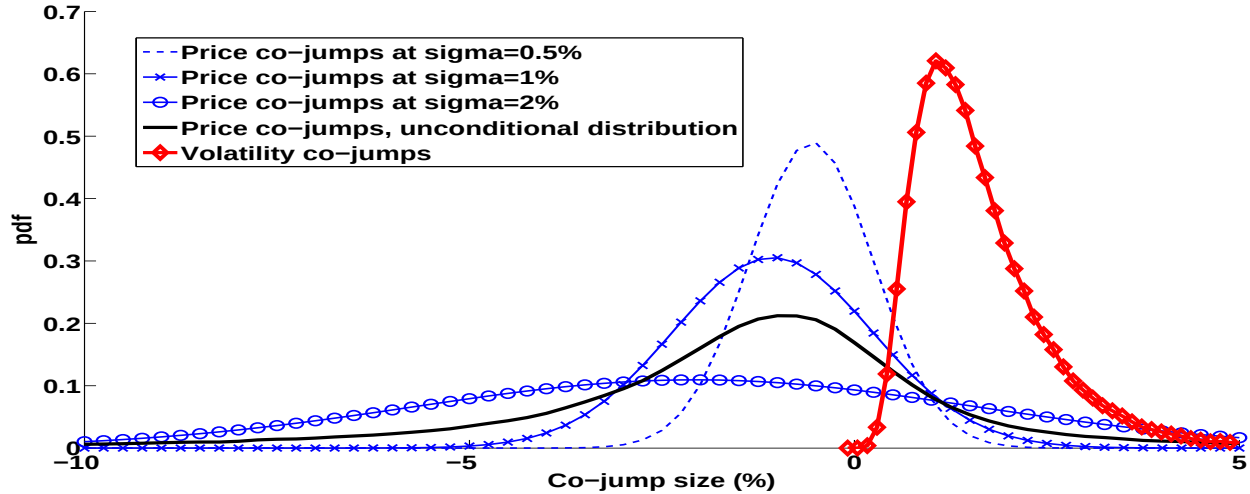


Figure 9: The unconditional and conditional distributions of the price and volatility co-jumps implied by the parametric estimates in Table 2, Column 5.

Section 2, Figure 3. Large positive price co-jumps are less frequent than negative price co-jumps, but they are in the data. Their mean and variability also increase with the underlying volatility level, something which the parametric model readily delivers. Hence, the size distribution of the price co-jumps is fat-tailed and negatively skewed. The size distribution of the volatility co-jumps is, instead, positively skewed. The asymmetry is, in this case, the result of normally distributed jumps in $\log \sigma^2$. Finally, when the volatility jumps are above their positive mean, the price jumps tend to be below their negative mean, as implied by a strongly negative correlation between the common jump sizes.

Motivated, again, by the nonparametric estimates, we fit a linear function to leverage, namely $\rho_0 + \rho_1 \sigma$. We find that leverage decreases with the volatility level ($\hat{\rho}_0 = -0.10$ and $\hat{\rho}_1 = -0.16$). For volatilities between 0.5% and 2%, estimated leverage varies between -0.18 and -0.42 . These values are lower, in absolute value, than those reported in the literature. Eraker et al. (2003), for instance, find a leverage value equal to -0.4838 for their specification with contemporaneously-arriving, but empirically uncorrelated in their framework, jumps. Similarly, in a model with the same logarithmic specification used here, but with no jumps in variance (neither idiosyncratic nor contemporaneous), Andersen et al. (2002) estimate a leverage parameter equal to -0.61 . This outcome is, however, unsurprising. The presence of *anti-correlated* common jumps is expected to reduce the size of classical leverage. Leverage is generally identified off of the covariance between price changes and volatility changes. In

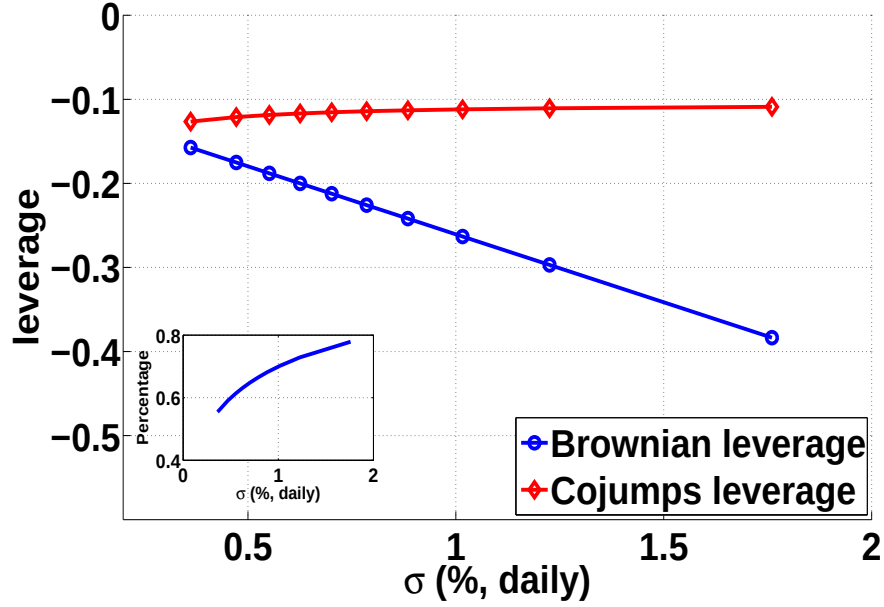


Figure 10: The leverage effect, as a function of volatility, decomposed into its Brownian part and the part due to co-jumps, as implied by the parametric estimates in Table 2, Column 5. In the inset we plot the percentage of ρ_{total} explained by Brownian correlation.

the model we study, this covariance is not just driven by Brownian correlation but also by the correlation between discontinuous shocks to the system. In fact

$$\rho_{total}(\sigma) = \frac{\vartheta_{1,1}}{\sigma\Lambda} = \rho + \frac{\lambda_{r,\sigma} (\rho_J \sigma_{JJ,r} \sigma_{JJ,\sigma} + \mu_{JJ,r} \mu_{JJ,\sigma})}{\sigma\Lambda} = \rho + \rho_{co-jumps}. \quad (16)$$

The standardization is given by spot volatility times the volatility of volatility and is designed to highlight ρ . Hence, $\rho_{co-jumps}$ is not, contrary to ρ , an actual correlation. Even though the number of yearly co-jumps $\lambda_{r,\sigma}$ may not be extremely high, the standardized covariance between the jump sizes is negative and large in absolute value. The means of the price/variance jump sizes are also large and of opposite sign. For a volatility level of $\sigma = 1\%$, for instance, we have $\rho = -0.26$, $\rho_{co-jumps} = -0.11$ and $\rho_{total} = -0.37$. Thus, the contribution of the jumps is found to be substantial. The parametric estimates also indicate (see Figure 10) that ρ is decreasing with the volatility level while $\rho_{co-jumps}$ is nearly constant. Figure 10 also plots the percentage of ρ_{total} attributable to the Brownian part computed using the parametric estimates in Table 2. The figure shows that this percentage is roughly 60/70% around the center of the volatility range (0.8%). The value varies somewhat with the volatility level, becoming increasingly important as spot volatility increases.

The size of Brownian leverage, and its implications for the skewness of stock returns, has drawn attention in the recent literature (see, e.g., Aït-Sahalia et al., 2013, and Bandi and Renò, 2011). Specifically, referring back to Eq. (16), compelling statistical arguments have been made about the need to account for noise in the estimation of the variance of variance Λ as well as downward biases in the estimation of the price and variance covariance $\vartheta_{1,1}$. Both effects have been shown to lead to attenuation effects in the estimation of leverage and, consequently, the belief that Brownian leverage may, in fact, be larger - i.e., more negative - than previously recognized. This paper addresses these important statistical concerns explicitly by relying on *spot* (hourly) volatility measures as well as by reducing the residual measurement error in the spot variance measures explicitly (see Appendix B). Monte Carlo simulations from the parametric model in Table 2 (Figure 14, Appendix B) show that our small sample adjustments are reliable in providing unbiased estimate of the infinitesimal cross-moments, including the moment $\vartheta_{1,1}$, which is crucial for the identification of leverage effects. Monte Carlo analysis on the estimated leverage parameters, and all other parameters of the systems, provides further, more direct, evidence. Following our small sample adjustments, this analysis provides evidence of minimal distortions in estimating both ρ_0 and ρ_1 (Figure 15, Appendix B, for a visual representation).

This paper, however, emphasizes an economic argument which may run counter to the idea that true Brownian leverage may be more negative than previously believed. Because a portion of the covariance between price and volatility changes is due to common jumps, the presence of anti-correlated co-jumps will, as was shown by virtue of the decomposition illustrated in Figure 10, lead to smaller Brownian correlations.

Having made these points, it is now worth asking what would happen to the model estimates should one impose the restrictions that (1) the intensity of the common jumps is equal to zero and (2) the intensity of the independent jumps is equal to zero (Table 2, Column 3 and Column 4, respectively). Both specifications, implying presence of independent or common jumps *only*, represent important models in the literature (see, e.g., Eraker et al., 2003), the mixed case discussed in this paper combining aspects of the two. In the first case, i.e., independent jumps only, all functions and parameters are qualitatively and quantitatively similar to the unrestricted model with two interesting exceptions. First, the correlation between Brownian shocks has now a more familiar magnitude. For volatilities between 0.5% and 2%, the estimated leverage values vary between -0.27 and -0.40 . The absence of co-jumps justifies an increased Brownian leverage. Second, some discontinuous changes in prices

and volatility are, in terms of their number, now attributed by the model to the independent jumps. In terms of point estimates, we find roughly 26 yearly price jumps (with small sizes) and 7 yearly volatility jumps. In the second case, i.e., perfectly correlated jump arrivals or co-jumps only, we expect not accounting for independent jumps to possibly attenuate the negative correlation between the common jump sizes. This would happen if some of the independent jumps were attributed to common discontinuous variation. Consistent with this observation, we find attenuation in the jump size correlation ($-.64$). This attenuation is, however, not accompanied by drastic changes in Brownian leverage since it is compensated by an increase in the likelihood of co-jumping.

8 The pricing of co-jumps: economic implications

We placed particular emphasis on the substantial negative dependence between price and volatility co-jumps. We now turn to some of its economic implications. Specifically, we show that not only negatively-correlated co-jumps add needed flexibility to the empirical dynamics of the price process, but they also represent a valuable economic channel leading to broader notions of return and volatility risk premia. We begin with the general nonparametric specification discussed in the paper. We then specialize the model to an alternative parametric structure in order to illustrate important effects in closed form.

8.1 The nonparametric case

We assume that the stock price follows the model in Eq. (1). We also assume that there exists a pricing kernel M_t taking the form

$$M_t = M_0 \left(\frac{p_t}{p_0} \right)^\phi e^{\int_0^t \delta(\sigma_s) ds + \psi(\xi(\sigma_t) - \xi(\sigma_0))}, \quad (17)$$

where ϕ and ψ are parameters controlling aversion to price and volatility risk, respectively, while $\delta(\cdot)$ is a function controlling time preferences. A similar pricing kernel has been described by the recent work of Christoffersen et al. (2013) as being able to reconcile a variety of empirical facts about option prices, from the implied volatility puzzle (leading to a negative price of variance risk and implied volatilities that are higher than the observed,

realized volatilities) to the difficulty in explaining out-of-the money option prices. The assumed pricing kernel is monotonically decreasing in prices when $\phi < 0$ (keeping volatility constant) and monotonically increasing in volatility when $\psi > 0$ (keeping prices constant). However, since high prices are often associated with high volatility, the pricing kernel may have a U shape when projected onto prices. We will later return to the nonmonotonicity of the pricing kernel. Below, we use the notation $E_{\sigma_t}[\cdot]$ to signify $E[\cdot|\sigma_t]$. The following proposition holds.

Proposition 8.1. *Denote by r the risk-free rate and write the return risk premium as*

$$\tilde{\mu}(\sigma_t) = \mu(\sigma_t) - r + \frac{1}{2}\sigma_t^2.$$

Given Eq. (17), absence of arbitrage implies

$$\begin{aligned} \delta(\sigma_t) = & -r(1 + \phi) - \phi \left(\tilde{\mu}(\sigma_t) - \frac{1}{2}\sigma_t^2 \right) - \psi m(\sigma_t) - \frac{1}{2} (\phi^2 \sigma_t^2 + \psi^2 \Lambda^2(\sigma_t) + 2\phi\psi\rho(\sigma_t)\sigma_t\Lambda(\sigma_t)) \\ & - \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}[e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1] - \lambda_r(\sigma_t)E_{\sigma_t}[e^{\phi c_{r,t}^J} - 1] - \lambda_\sigma(\sigma_t)E_{\sigma_t}[e^{\psi c_{\sigma,t}^J} - 1], \end{aligned} \quad (18)$$

and

$$\begin{aligned} \tilde{\mu}(\sigma_t) = & -\phi\sigma_t^2 - \psi\rho(\sigma_t)\sigma_t\Lambda(\sigma_t) \\ & - \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}[e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}}(e^{c_{r,t}^{JJ}} - 1)] - \lambda_r(\sigma_t)E_{\sigma_t}[e^{\phi c_{r,t}^J}(e^{c_{r,t}^J} - 1)]. \end{aligned} \quad (19)$$

Further, assume that the price process' risk-neutral dynamics are given by:

$$\begin{aligned} d(\log p_t) = & \left(r - \frac{1}{2}\sigma_t^2 - \lambda_{r,\sigma}^*(\sigma_t)E_{\sigma_t}[c_{r,t}^{JJ*}] - \lambda_r^*(\sigma_t)E_{\sigma_t}[c_{r,t}^{J*}] \right) dt \\ & + \sigma_t \left\{ \rho(\sigma_t)dW_t^{1*} + \sqrt{1 - \rho^2(\sigma_t)}dW_t^{2*} \right\} + c_{r,t}^{JJ*}dJ_{r,\sigma}^* + c_{r,t}^{J*}dJ_r^* \\ d\xi(\sigma_t) = & m^*(\sigma_t)dt + \Lambda(\sigma_t)dW_t^{1*} + c_{\sigma,t}^{JJ*}dJ_{r,\sigma}^* + c_{\sigma,t}^{J*}dJ_\sigma^*, \end{aligned}$$

where $J^* = \{J_r^*, J_\sigma^*, J_{r,\sigma}^*\}$ is an independent (of the bivariate vector $W^* = \{W_t^{1*}, W_t^{2*}\}$) trivariate vector of mutually independent Poisson jumps with risk-neutral intensities $\lambda_r^*(\sigma_t)$, $\lambda_\sigma^*(\sigma_t)$ and $\lambda_{r,\sigma}^*(\sigma_t)$, respectively. The risk-neutral jump sizes are $c_{r,t}^{J*}$, $c_{\sigma,t}^{J*}$, $c_{r,t}^{JJ*}$ and $c_{\sigma,t}^{JJ*}$. We have

$$m^*(\sigma_t) - m(\sigma_t) = \phi\rho(\sigma_t)\sigma_t\Lambda(\sigma_t) + \psi\Lambda^2(\sigma_t) \quad (20)$$

and, for all $u \in \mathbb{R}$ and all attainable σ_t values,

$$\lambda_{r,\sigma}^*(\sigma_t) = \lambda_{r,\sigma}(\sigma_t) E_{\sigma_t} \left[e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} \right], \quad E_{\sigma_t} \left[e^{iuc_{\sigma,t}^{JJ*}} \right] = \frac{E_{\sigma_t} \left[e^{iuc_{\sigma,t}^{JJ} \left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} \right)} \right]}{E_{\sigma_t} \left[e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} \right]}, \quad (21)$$

$$E_{\sigma_t} \left[e^{iuc_{r,t}^{JJ*}} \right] = \frac{E_{\sigma_t} \left[e^{iuc_{r,t}^{JJ} \left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} \right)} \right]}{E_{\sigma_t} \left[e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} \right]}, \quad (22)$$

$$\lambda_r^*(\sigma_t) = \lambda_r(\sigma_t) E_{\sigma_t} \left[e^{\phi c_{r,t}^J} \right], \quad E_{\sigma_t} \left[e^{iuc_{r,t}^{J*}} \right] = \frac{E_{\sigma_t} \left[e^{iuc_{r,t}^J \left(e^{\phi c_{r,t}^J} \right)} \right]}{E_{\sigma_t} \left[e^{\phi c_{r,t}^J} \right]}, \quad (23)$$

$$\lambda_{\sigma}^*(\sigma_t) = \lambda_{\sigma}(\sigma_t) E_{\sigma_t} \left[e^{\psi c_{\sigma,t}^J} \right], \quad E_{\sigma_t} \left[e^{iuc_{\sigma,t}^{J*}} \right] = \frac{E_{\sigma_t} \left[e^{iuc_{\sigma,t}^J \left(e^{\psi c_{\sigma,t}^J} \right)} \right]}{E_{\sigma_t} \left[e^{\psi c_{\sigma,t}^J} \right]}. \quad (24)$$

Proposition 8.1 outlines the restrictions on the coefficients $\delta(\cdot)$, ϕ and ψ induced by absence of arbitrage along with the risk-neutral transformations compatible with the pricing kernel in Eq. (17).

Eq. (18) describes time preferences. It is convenient to interpret it in light of the Black-Scholes model for which $\sigma_t = \sigma$ and $\mu(\sigma) = \bar{\mu} - \sigma^2/2$ are constants and there are no jumps. In this case, it is known that the pricing kernel takes the form

$$M_t^{BS} = M_0 \exp \left(- \left(r + \frac{\lambda^2}{2} \right) t - \lambda W_t \right)$$

with $\lambda = \frac{\bar{\mu} - r}{\sigma}$, thereby yielding

$$M_t^{BS} = M_0 \left(\frac{p_t}{p_0} \right)^{-\frac{\lambda}{\sigma}} \exp \left(-r \left(1 - \frac{\lambda}{\sigma} \right) + \frac{\lambda^2}{2} - \frac{\lambda \sigma}{2} \right) t.$$

Thus, in the Black-Scholes model, $\phi_{BS} = - \left(\frac{\bar{\mu} - r}{\sigma^2} \right)$, $\psi_{BS} = 0$, and

$$\delta_{BS} = -r(1 + \phi_{BS}) + \frac{1}{2} \phi_{BS}^2 \sigma^2 + \frac{1}{2} \phi_{BS} \sigma^2,$$

which represents a simplified version of Eq. (18) when taking into account that, in this case,

$$\tilde{\mu}(\sigma) = \bar{\mu} - r = -\phi_{BS}\sigma^2. \quad (25)$$

Our more general specification includes an additional adjustment for the stochastic volatility mean ($-\psi m(\sigma_t)$), extra components generated by the volatility of stochastic volatility and leverage effects ($\psi^2 \Lambda^2(\sigma_t) + 2\phi\psi\rho(\sigma_t)\sigma_t\Lambda(\sigma_t)$), as well as adjustments due to the means of the three jump components.¹⁴

Eq. (19) describes the return risk premium $\tilde{\mu}(\sigma_t)$. Note that

$$-\frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^c}{M_t}\frac{dp_t^c}{p_t}\right] = \underbrace{-\phi\sigma_t^2}_{-\frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^{c,p}}{M_t}\frac{dp_t^c}{p_t}\right]} + \underbrace{-\psi\rho(\sigma_t)\Lambda(\sigma_t)\sigma_t}_{-\frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^{c,\xi(\sigma)}}{M_t}\frac{dp_t^c}{p_t}\right]}, \quad (26)$$

where $\frac{dM_t^{c,p}}{M_t}$ denotes the price shock component of the evolution of the stochastic discount factor, whereas $\frac{dM_t^{c,\xi(\sigma)}}{M_t}$ denotes the corresponding volatility shock component. Since, given $\phi < 0$, positive price shocks are associated with lower stochastic discounting and negative price shocks are associated with higher stochastic discounting, the contribution of the first term to the risk premium ($-\phi\sigma^2$) is positive. This term is classical (c.f., Eq. (25)). Given $\psi > 0$, the term $-\psi\rho(\sigma_t)\Lambda(\sigma_t)\sigma_t$ also contributes positively to the return risk premium provided $\rho(\sigma_t) < 0$, a well-known stylized fact further illustrated above. The logic is as follows: positive price shocks are associated with low volatility (given $\rho(\sigma_t) < 0$) and low discounting (if $\psi > 0$). Similarly, negative price shocks lead to high volatility and high discounting. Christoffersen et al. (2013) discuss the importance of this term. Here, in agreement with the documented relation between leverage and volatility (Bandi and Renò, 2012), we let leverage be a function of the underlying spot volatility as well. Importantly, however, $\rho(\sigma_t)$ only captures one component of leverage, i.e., diffusive leverage, the second component being associated with co-jumps. We now turn to it, and the corresponding contribution to the return risk premium, as formalized in Eq. (19).

¹⁴The mapping between the Black-Scholes model and our specification is clear when taking into account that the drift in Eq. (1) can be expressed as follows: $\mu(\sigma_t) = \bar{\mu}(\sigma_t) - \frac{1}{2}\sigma_t^2$. This implies that $\tilde{\mu}(\sigma_t)$ is, as stated in Proposition 8.1, a return risk premium.

Writing $\mu^{JJ} = -\lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} \left[e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} (e^{c_{r,t}^{JJ}} - 1) \right]$, we have

$$\begin{aligned} \mu^{JJ} &= -\lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} \left[\left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} + 1 - 1 \right) (e^{c_{r,t}^{JJ}} - 1) \right] \\ &= -\lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} \left[\left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1 \right) (e^{c_{r,t}^{JJ}} - 1) \right] - \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} \left[e^{c_{r,t}^{JJ}} - 1 \right] \\ &= \underbrace{-\frac{1}{dt}E_{\sigma_t} \left[\frac{dM_t^{JJ}}{M_t} \frac{dp_t^{JJ}}{p_t} \right]}_{\text{co-jump risk premium}} - \underbrace{\frac{1}{dt}E_{\sigma_t} \left[\frac{dp_t^{JJ}}{p_t} \right]}_{\text{compensation}}, \end{aligned}$$

where M_t^{JJ} and p_t^{JJ} denote the co-jump part of the pricing kernel and the price process, respectively. The same considerations apply to the idiosyncratic jump part.

To gain intuition, let us approximate the risk premium due to co-jumps using a first-order expansion (for small jump sizes):

$$-\lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} \left[\left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1 \right) (e^{c_{r,t}^{JJ}} - 1) \right] \approx -\phi \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} \left[(c_{r,t}^{JJ})^2 \right] - \psi \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} \left[c_{r,t}^{JJ} c_{\sigma,t}^{JJ} \right],$$

where the first term represents the positive (since $\phi < 0$) contribution of the price co-jump variance and the second term represents the positive (since $\psi > 0$) contribution of the co-jump leverage (which was found to be negative), thereby leading to a *positive* contribution of the co-jumps to the overall return risk premium.

We note that, consistent with the leverage effect in Eq. (16), the risk premium due to co-jumps depends on $\lambda_{r,\sigma}(\sigma_t)E_{\sigma_t} [c_{r,t}^{JJ} c_{\sigma,t}^{JJ}]$ instead of the co-jump covariance. This is intuitive since deterministic jump sizes could also lead to leverage and a risk premium.

In the co-jump case, large negative price shocks are associated with high discounting through lower prices (since $\phi < 0$) and higher volatility (since $\psi > 0$). Hence, there are two channels through which the co-jumps enter return risk premia: prices, via ϕ , and variance, via ψ . The first effect is the co-jump analogue of $-\phi\sigma_t^2$, the second effect is the co-jump analogue of $-\psi\rho(\sigma_t)\Lambda(\sigma_t)\sigma_t$.

Remark 1 (The impact of co-jumps on the *return risk premium*). *In sum, generalized leverage, as given by diffusive leverage and leverage induced by the co-jumps, contributes to expected returns and return risk premia provided $\psi > 0$. Hence, not only do the co-jumps lead to an additional source of skewness in stock returns. They also modify, in equilibrium, the mean of the return distribution by virtue of their corresponding risk compensation.*

We now turn to the variance risk premium:

$$(m^*(\sigma_t) + \lambda_{r,\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ*}] + \lambda_{\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{J*}]) - (m(\sigma_t) + \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ}] + \lambda_{\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^J]) .$$

Eq. (20) through (24) provide its components. Write

$$\underbrace{\phi\rho(\sigma_t)\sigma_t\Lambda(\sigma_t) + \psi\Lambda^2(\sigma_t)}_{m^*(\sigma_t)-m(\sigma_t)} + \underbrace{\lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}\left[\left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1\right)c_{\sigma,t}^{JJ}\right]}_{\lambda_{r,\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ*}] - \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ}]} + \underbrace{\lambda_{\sigma}(\sigma_t)E_{\sigma_t}\left[\left(e^{\psi c_{\sigma,t}^J} - 1\right)c_{\sigma,t}^J\right]}_{\lambda_{\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{J*}] - \lambda_{\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^J]}$$

using extensively the properties of the measure change provided by M_t . In fact,

$$E_{\sigma_t}[dJ_{r,\sigma}^*] = E_{\sigma_t}\left[dJ_{r,\sigma}e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}}\right] = \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}\left[c_{\sigma,t}^{JJ}e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}}\right] \quad (27)$$

$\xRightarrow{\text{implies}}$

$$\lambda_{r,\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ*}] - \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ}] = \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}\left[c_{\sigma,t}^{JJ}\left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1\right)\right] \quad (28)$$

and

$$E_{\sigma_t}[dJ_{\sigma}^*] = E_{\sigma_t}\left[dJ_{\sigma}e^{\psi c_{\sigma,t}^J}\right] = \lambda_{\sigma}(\sigma_t)E_{\sigma_t}\left[c_{\sigma,t}^Je^{\psi c_{\sigma,t}^J}\right] \quad (29)$$

$\xRightarrow{\text{implies}}$

$$\lambda_{\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{J*}] - \lambda_{\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^J] = \lambda_{\sigma}(\sigma_t)E_{\sigma_t}\left[c_{\sigma,t}^J\left(e^{\psi c_{\sigma,t}^J} - 1\right)\right]. \quad (30)$$

The term $m^*(\sigma_t) - m(\sigma_t)$ can be interpreted similarly to Eq. (26) since $\frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^{c,\xi(\sigma)}}{M_t}d\xi(\sigma_t)\right] = \psi\Lambda^2(\sigma_t)$ and $\frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^{c,p}}{M_t}d\xi(\sigma_t)\right] = \phi\rho(\sigma_t)\sigma_t\Lambda(\sigma_t)$. In other words, price uncertainty enters the variance risk premium through leverage while variance uncertainty enters the variance risk premium through the variance of variance. Analogously,

$$\lambda_{r,\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ*}] - \lambda_{r,\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{JJ}] = \frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^{JJ}}{M_t}d\xi^{JJ}(\sigma_t)\right],$$

and

$$\lambda_{\sigma}^*(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^{J*}] - \lambda_{\sigma}(\sigma_t)E_{\sigma_t}[c_{\sigma,t}^J] = \frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^J}{M_t}d\xi^J(\sigma_t)\right].$$

With $\psi > 0$, large idiosyncratic volatility shocks require a *negative* volatility premium (since $\frac{1}{dt}E_{\sigma_t}\left[\frac{dM_t^J}{M_t}d\xi^J(\sigma_t)\right] > 0$).

We also expect the volatility co-jumps to lead to a negative variance premium if the correlation between the price co-jumps and the volatility co-jumps is negative. To gain intuition, we again use a first-order expansion (for small jump sizes):

$$E_{\sigma_t} \left[c_{\sigma,t}^{JJ} \left(e^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1 \right) \right] \approx \phi E_{\sigma_t} [c_{r,t}^{JJ} c_{\sigma,t}^{JJ}] + \psi E_{\sigma_t} \left[(c_{\sigma,t}^{JJ})^2 \right] > 0,$$

leading to a *positive* contribution of the co-jumps to the variance risk premium.

In this case, large positive variance shocks are associated with high discounting through higher variance (since $\psi > 0$) and lower prices (since $\phi < 0$). As in the case of the price equation, there are two channels through which the co-jumps enter variance risk premia: variances, via ψ , and prices, via ϕ . The first effect is the co-jump analogue of $\psi \Lambda^2(\sigma_t)$, the second effect is the co-jump analogue of $\phi \rho(\sigma_t) \sigma_t \Lambda(\sigma_t)$.

Eq. (21) through Eq. (24) imply that the risk-neutral intensities are expected to be larger than the corresponding intensities under the natural measure, thus determining a larger number of (idiosyncratic and common) jumps under the change of measure.

Finally, Eq. (21) through Eq. (24) also provide the risk-neutral characteristic functions of the sizes of all jumps. In order to study the relation between natural and risk-neutral size moments more structure ought to be imposed on the problem. We do so in what follows.

Remark 2 (The impact of co-jumps on the *variance risk premium*). *Generalized leverage, as given by diffusive leverage and leverage induced by the co-jumps, contributes to variance risk premia provided $\phi < 0$. Hence, the co-jumps lead to an additional source of skewness in stock returns and modify - in equilibrium - the mean of the return distribution by virtue of their corresponding risk compensation (see Remark 1 above) while also inducing a variance risk premium.*

Remark 3 (Co-jumps and nonmonotonic pricing kernels). *Should the stochastic discount factor not depend on variance, and therefore be monotonic in prices, the negative correlation between price and volatility co-jumps would not matter to determine return risk premia. It would, however, still matter to determine variance risk premia.*

To better understand the risk compensations (in returns and variance) due to co-jumps, we now employ a parametric model.

8.2 A parametric example: affine

Assume that the price process conforms with the specification

$$\begin{aligned} d \log p_t &= \left(r + \mu_1 + \left(\mu_2 - \frac{1}{2} \right) \sigma_t^2 \right) dt + \sigma_t \left[\rho dW_t^1 + \sqrt{(1 - \rho^2)} dW_t^2 \right] + c_r dJ_t \\ d\sigma_t^2 &= \alpha(\beta - \sigma_t^2)dt + \Lambda \sqrt{\sigma_t^2} dW_t^1 + c_\sigma dJ_t, \end{aligned} \quad (31)$$

where r is the constant risk-free rate, $\mu_1, \mu_2, \rho, \alpha, \beta$ and Λ are constant parameters and J_t is a Poisson process with constant intensity λ regulating the likelihood of co-jumps. The co-jump sizes are given by

$$c_r \sim \mathcal{N}(\mu_r + \rho_J c_\sigma, \sigma_r^2)$$

and

$$c_\sigma \sim \exp(\mu_\sigma),$$

with $\mu_r < 0, \rho_J < 0, \sigma_r$ and $\mu_\sigma > 0$ being additional constants whose sign restrictions are coherent with data. The return specification is consistent with Duffie et al. (2000). We exclusively include co-jumps, and dispense with idiosyncratic jumps, in order to focus on the main features of our treatment.

We further assume, as in Christoffersen et al. (2013), that the pricing kernel takes the form

$$M_t = M_0 \left(\frac{p_t}{p_0} \right)^\phi e^{\delta t + \eta \int_0^t \sigma_s^2 ds + \psi(\sigma_t^2 - \sigma_0^2)}, \quad (32)$$

where $\delta > 0, \eta > 0, \phi < 0$ and $\psi > 0$ are, again, constants. Their signs were discussed previously.

The nonparametric findings in the previous subsection can be specialized to the current set-up. The following proposition follows from Proposition 8.1., Eq. (31), and Eq. (32).

Proposition 8.2. *Assume Eq. (31) and Eq. (32) hold. Also, assume $\mu_\sigma < \frac{1}{\psi + \phi \rho_J}$. Absence of arbitrage implies*

$$\delta = -r(1 + \phi) - \phi \mu_1 - \psi \alpha \beta - \lambda \left(\frac{e^{\phi \mu_r + \frac{1}{2} \phi^2 \sigma_r^2}}{1 - \mu_\sigma(\psi + \phi \rho_J)} - 1 \right), \quad (33)$$

$$\eta = - \left(\mu_2 - \frac{1}{2} \right) + \psi \alpha - \frac{1}{2} (\phi^2 + \psi^2 \Lambda^2 + 2\phi\psi\rho\Lambda), \quad (34)$$

$$\mu_1 = -\lambda \left[\frac{e^{(\phi+1)\mu_r + \frac{1}{2}(\phi+1)^2\sigma_r^2}}{1 - \mu_\sigma(\psi + (\phi+1)\rho_J)} - \frac{e^{\phi\mu_r + \frac{1}{2}\phi^2\sigma_r^2}}{1 - \mu_\sigma(\psi + \phi\rho_J)} \right] > 0, \quad (35)$$

$$\mu_2 = -\phi - \psi\rho\Lambda > 0. \quad (36)$$

Furthermore, assume that the price process' risk-neutral dynamics are given by

$$\begin{aligned} \frac{dp_t}{p_t} &= (r - \lambda^* E[e^{c_r^*} - 1])dt + \sigma_t \left[\rho dW_t^{1*} + \sqrt{(1 - \rho^2)} dW_t^{2*} \right] + (e^{c_r^*} - 1) dJ_t^* \\ d\sigma_t^2 &= \underbrace{a(\beta - \sigma_t^2)}_{m(\sigma_t)} + \chi\sigma_t^2 dt + \Lambda\sqrt{\sigma_t^2} dW_t^{1*} + c_\sigma^* dJ_t^*, \end{aligned}$$

where J^* is a risk-neutral co-jump process with constant intensity λ^* independent of the bivariate vector $W^* = \{W_t^{1*}, W_t^{2*}\}$, c_r^* and c_σ^* are the risk-neutral co-jump sizes, and χ is a constant given by

$$\chi = \phi\rho\Lambda + \psi\Lambda^2 > 0.$$

Finally, for all $u \in \mathbb{R}$,

$$\frac{\lambda^*}{\lambda} = \frac{e^{\phi\mu_r + \frac{1}{2}\phi^2\sigma_r^2}}{1 - \mu_\sigma(\psi + \phi\rho_J)} > 1, \quad (37)$$

$$E \left[e^{ic_r^* u} | c_\sigma \right] = e^{iu(\mu_r + \rho_J c_\sigma + \phi\sigma_r^2) - \frac{1}{2}u^2\sigma_r^2}, \quad (38)$$

$$E \left[e^{ic_\sigma^* u} \right] = \frac{1}{1 - iu \left(\frac{\mu_\sigma}{1 - \mu_\sigma(\psi + \phi\rho_J)} \right)}. \quad (39)$$

Proof. Proposition 8.1 and Lemma B.1 give the result.

In this case, Eqs. (35) and (36) can be solved to find ϕ and ψ which, in turn, can be substituted into Eqs. (33) and (34) to find δ and η , thereby determining the unique pricing kernel compatible with Eqs. (31) and (32).

We notice that

$$\begin{aligned} \tilde{\mu} &= \mu_1 + \mu_2\sigma^2 \\ &= -\phi\sigma^2 - \psi\rho\Lambda\sigma^2 - \lambda \left(\frac{e^{(\phi+1)\mu_r + \frac{1}{2}(\phi+1)^2\sigma_r^2}}{1 - \mu_\sigma(\psi + (\phi+1)\rho_J)} - \frac{e^{\phi\mu_r + \frac{1}{2}\phi^2\sigma_r^2}}{1 - \mu_\sigma(\psi + \phi\rho_J)} \right). \end{aligned} \quad (40)$$

Eq. (40) highlights the structural components of the return risk premium. In this specification, the correlation between price and volatility co-jumps plays a role in the risk-return relation (through the parameter ρ_J) even if $\psi = 0$. Coherently with our discussion in the previous section, however, this finding does *not* point to a risk compensation (in market returns) for the co-jump component of generalized leverage even when $\psi = 0$. The result is solely a by-product of the adopted specification for the co-jump sizes. In this case, in fact, the size of the jumps in volatility appears in the mean of the jumps in prices (and, thus, in the risk premium) even when $\sigma_r = 0$, namely when there is no covariance between the co-jump sizes.¹⁵

Using the results in the previous subsection, the variance risk premium becomes

$$\begin{aligned}\chi\sigma_t^2 + \lambda^* E[c_\sigma^*] - \lambda E[c_\sigma] &= \phi\rho\Lambda\sigma^2 + \psi\Lambda^2\sigma^2 + \lambda E[c_\sigma (e^{\phi c_r + \psi c_\sigma} - 1)] \\ &= \phi\rho\Lambda\sigma^2 + \psi\Lambda^2\sigma^2 + \mu_\sigma \left(\frac{e^{\phi\mu_r + \frac{1}{2}\phi^2\sigma_r^2}}{(1 - \mu_\sigma(\psi + \phi\rho_J))^2} - 1 \right) > 0\end{aligned}\quad (42)$$

given the assumptions on the parameters. This finding implies that the expected value of the variance process under the risk neutral measure is larger than the expected value of the variance process under the objective measure. This is consistent with a variance implied from option prices larger than realized variance and, when viewing risk premia as the difference between natural mean and risk-neutral mean, with a *negative* variance premium. Eq. (42)

¹⁵It is easy to consider an alternative parametric specification which clarifies this point further, i.e., the logarithmic case. Assume

$$d\log\sigma_t^2 = \alpha(\beta - \sigma_t^2)dt + \Lambda\sqrt{\sigma_t^2}dW_t^1 + c_\sigma dJ_t$$

with

$$\begin{pmatrix} c_r \\ c_\sigma \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_r \\ \mu_\sigma \end{pmatrix}, \begin{pmatrix} \sigma_r^2 & \rho_J\sigma_r\sigma_\sigma \\ \rho_J\sigma_r\sigma_\sigma & \sigma_\sigma^2 \end{pmatrix}\right)\quad (41)$$

and the pricing kernel

$$M_t = M_0 \left(\frac{p_t}{p_0}\right)^\phi \left(\frac{\sigma_t^2}{\sigma_0^2}\right)^\psi \exp^{\delta t + \eta \int_0^t \sigma_s^2 ds}.$$

We would then have

$$\begin{aligned}\mu_1 + \frac{1}{dt} E\left(\frac{dp_t^{JJ}}{p_t}\right) &= -\lambda_J E[(e^{\phi c_r + \psi c_\sigma} - 1)(e^{c_r} - 1)] \\ &= -\lambda_J \left(e^{(1+\phi)\mu_r + \psi\mu_\sigma + \frac{1}{2}((1+\phi)^2\sigma_r^2 + \psi^2\sigma_\sigma^2 + 2(1+\phi)\psi\rho_J\sigma_r\sigma_\sigma)} - e^{\phi\mu_r + \psi\mu_\sigma + \frac{1}{2}(\phi^2\sigma_r^2 + \psi^2\sigma_\sigma^2 + 2\phi\psi\rho_J\sigma_r\sigma_\sigma)} \right) \\ &\quad - \lambda_J (-e^{\mu_r + \frac{1}{2}\sigma_r^2} + 1)\end{aligned}$$

Thus, $\mu_1 + \frac{1}{dt} E\left(\frac{dp_t^{JJ}}{p_t}\right)$ would not depend on ρ_J if $\psi = 0$.

characterizes the components of the variance risk premium structurally.

Let us now turn to the risk-neutral features of the co-jump sizes. Eq. (39) is the characteristic function of an exponential random variable with parameter

$$E[c_\sigma^*] = \mu_\sigma^Q = \frac{\mu_\sigma}{1 - \mu_\sigma(\psi + \phi\rho_J)} > \mu_\sigma.$$

Thus, the expected value and variance of the risk-neutral jump sizes are *larger* than the corresponding natural moments. Eq. (38) is the characteristic function (conditionally on volatility jumps) of a normal random variable with mean and variance given by

$$\begin{aligned} E^Q(c_r|c_\sigma) &= E(c_r^*|c_\sigma) = \mu_r + \rho_J c_\sigma + \phi \sigma_r^2, \\ V^Q(c_r|c_\sigma) &= \sigma_r^2. \end{aligned}$$

We note that

$$E^Q(c_r) = E(c_r^*) = \mu_r + \rho_J \mu_\sigma + \phi \sigma_r^2 < \mu_r,$$

which implies that the risk-neutral first moment of the price jump sizes is *smaller* than the natural first moment (if $\rho_J < 0$, as in the data, and $\phi < 0$).

One important observation is in order. This subsection has exploited affinity solely as a theoretical means to further clarify the economics of co-jump pricing, as laid out in Subsection 8.1. In the context of a ubiquitous and tractable specification for which closed-form solutions can be obtained, we have derived sign restrictions on the co-jump premia. These restrictions confirm the intuition produced by the nonparametric model in Subsection 8.1. We discuss specification analysis, and the empirical accuracy of an affine variance process, in Section 9.

8.3 Nonmonotonic pricing kernels

While classical models postulate a pricing kernel that is monotonically decreasing with prices, some empirical work points to lack of monotonicity. In a recent paper, for example, Christoffersen et al. (2013) propose Eq. (32) as a way to explain important stylized facts in derivative pricing. They show that $\psi > 0$ implies a U-shaped structure in prices. The intuition goes as follows. Volatility is high in correspondence with low or high asset prices. When prices are low and volatility is high the discount factor is also high (given $\phi < 0$ and $\psi > 0$).

When prices are high and volatility is high, the discount factor is relatively higher than it would be if volatility were not in it. In fact, higher prices lower it (through ϕ), whereas higher volatility increases it (through ψ). It is the effect of volatility at high prices which induces nonmonotonicity. We can now build on this intuition to understand the impact of (1) the presence of co-jumps and (2) the presence of negatively-correlated co-jumps. Adding co-jumps gives the model a chance of relatively higher volatility at low prices and relatively lower volatility at high prices (due to leverage induced by the co-jumps), thereby leading to a rotation of the U-shaped stochastic discount factor. This rotation should be larger, the larger the correlation between the co-jump sizes.

We confirm this intuition using our estimated nonparametric model. To this extent, the price process is simulated using the estimated specification in Section 7 without independent jumps. The parameters are those in Table 2, Column 4. The only exception is the parameter ρ_J which is set equal to either 0 or -1 .

We employ the pricing kernel in Eq. (17) with δ constant, $\phi = -0.5$, and $\psi = 1.5$ for a maturity of one day. For every simulation (indexed by i), the pricing model delivers a value $M_t(p_{i,t}, \sigma_{i,t})$ at maturity. In Figure 11, we report the average value of the pricing kernel for each price level (and, in the insert, the ratios with respect to the case without co-jumps). This average is computed as a nonparametric moment. We report

$$\widehat{M}_t(p) = \frac{\sum_{i=1}^N \mathbf{K}\left(\frac{p_{i,t}-p}{\zeta}\right) M_t(p_{i,t}, \sigma_{i,t})}{\sum_{i=1}^N \mathbf{K}\left(\frac{p_{i,t}-p}{\zeta}\right)},$$

where ζ is Silverman's bandwidth.

As expected, the pricing kernel is U-shaped. The inclusion of co-jumps rotates it. The extent of the rotation depends on the magnitude of the correlation between the co-jump sizes, a higher correlation being associated with a stronger rotation.

This discussion is meant to illustrate the main effects but is necessarily qualitative in nature. A quantitative study of co-jump risk premia may lead to alternative parametrizations of the stochastic discount factor for the purpose of effective derivative pricing while, possibly, preserving nonmonotonicity and its empirical appeal. Such a study requires the use of derivative data. It is beyond the scopes of the present paper but is left for future work.

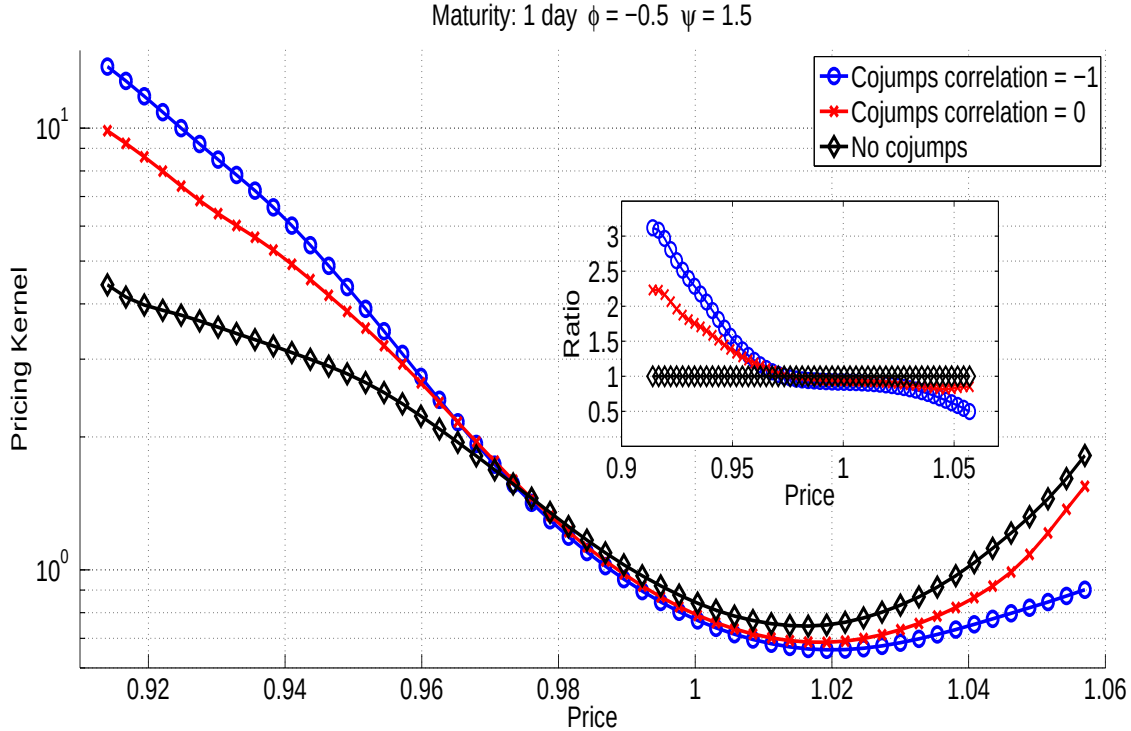


Figure 11: Pricing kernel as a function of the stock price. In the insert we report the ratios with respect to the case without co-jumps.

9 Specification analysis

Much emphasis has been placed on affine structures and their usefulness in deriving near closed-form prices for a wide array of securities (see, e.g., Piazzesi, 2010, for a review). More generally, emphasis has been placed on parametric specifications in the presence of models with latent variables, such a stochastic volatility. In these models, the estimation of the parameters and the filtering of the latent states are a joint problem, one that has been successfully undertaken in several recent papers (see, for a recent discussion, Andersen and Benzoni, 2011). By its very nature, however, this joint problem is bound to link the filtering of volatility to the identification of the system's parametric structure.

We dispense with this link and identify daily volatility, *before* inference on the dynamics begins, by virtue of intra-daily price data. In this sense, our filtered volatility series is especially useful to select a function $\xi(\cdot)$ and, hence, a variance transformation, which conforms with the assumed jump-diffusion structure and can therefore be modelled as a jump-diffusion

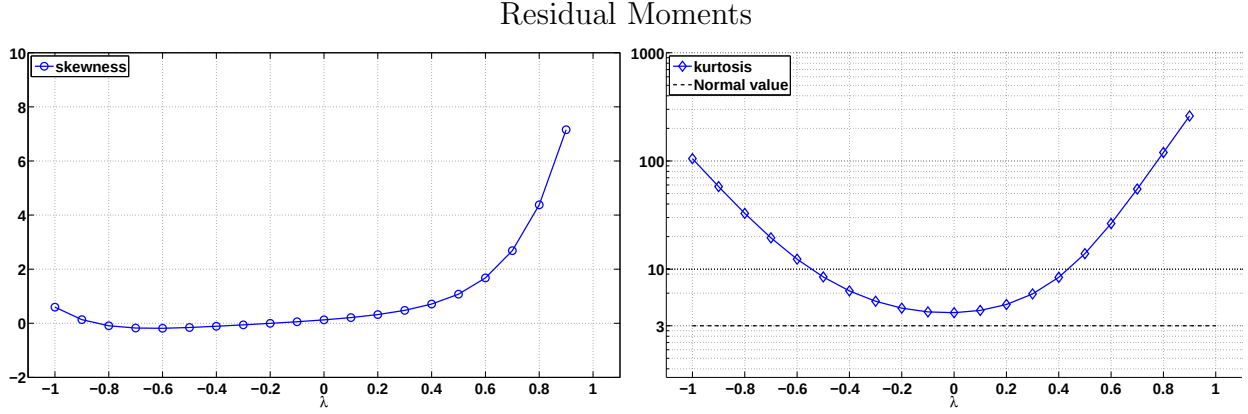


Figure 12: Box-Cox specification test.

process. Write

$$\varepsilon_{t,t+\Delta} = \frac{f_{\lambda}(\sigma_{t+\Delta}^2) - f_{\lambda}(\sigma_t^2) - m_{\lambda}(\sigma_t)\Delta}{\Lambda_{\lambda}(\sigma_t)\sqrt{\Delta}},$$

where $f_{\lambda}(\cdot)$ is a Box-Cox transformation, namely $f_{\lambda}(\cdot) = \frac{\cdot^{\lambda}-1}{\lambda}$ for $\lambda \neq 0$ and $f_{\lambda}(\cdot) = \log(\cdot)$ for $\lambda = 0$. The functions $m_{\lambda}(\sigma_t)$ and $\Lambda_{\lambda}(\sigma_t)$ are the volatility drift and the volatility of volatility function associated with the same transformation.

Consider a specification without jumps. Because the system's shocks are assumed to be Brownian shocks, the (standardized) residuals $\varepsilon_{t,t+\Delta}$ should be locally Gaussian. The most suitable choice of λ is, therefore, the choice which guarantees Gaussianity of the residuals. In a model with symmetric Poisson jumps, the residual should still have zero skewness and only inflated kurtosis. The excess kurtosis should be an increasing function of the number of jumps and the standard deviation of the jump size. To this extent, Figure 12 reports the skewness and kurtosis of the volatility residuals for different values of the parameter λ . A value of λ equal to 0 yields zero skewness and the lowest kurtosis level. Consistent with a model with some discontinuities, this level is slightly larger than the Gaussian one. A zero λ value coincides with the logarithmic transformation which we adopt in this paper.¹⁶ The second panel in Figure 6 shows the filtered logarithmic variances. It is visually clear that the daily logarithmic variances conform rather well with a model in which the Brownian

¹⁶Papers on stochastic volatility estimation in discrete time which employ a logarithmic transformation are Chib et al. (2002); Harvey and Shephard (1996); Jacquier et al. (1994, 2004), and Yu (2005, 2012) among others.

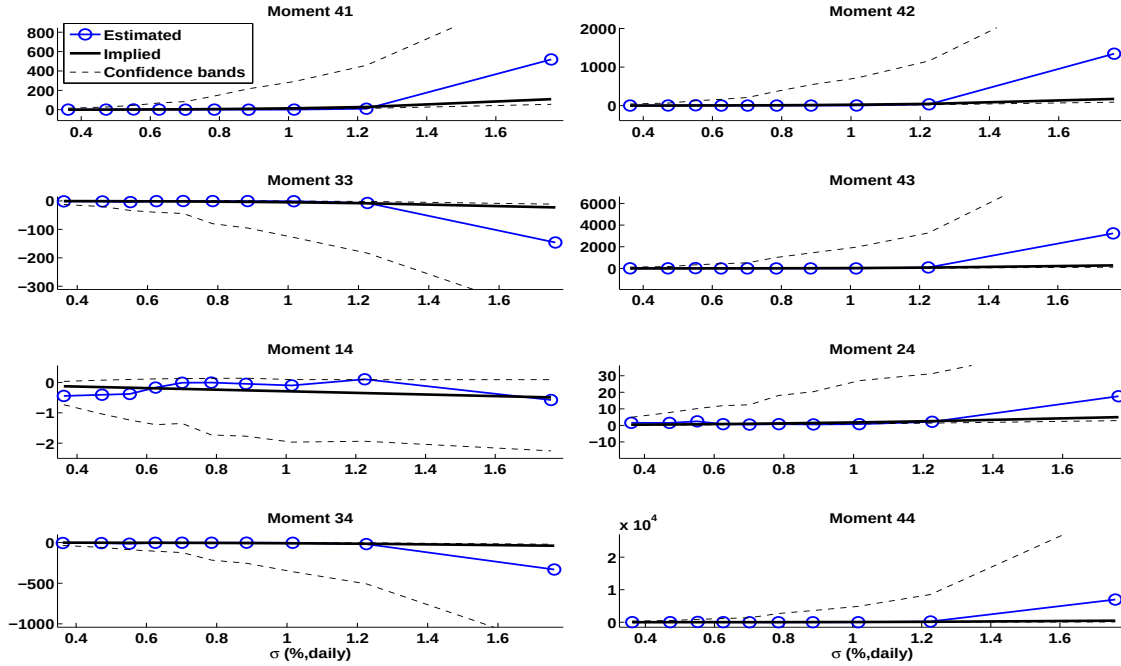


Figure 13: Overidentifying restrictions.

shocks occur along with discontinuous shocks. In addition, the Brownian shocks appear to have a rather constant variance $\Lambda^2(\cdot)$ in our sample. Similarly, the volatility jumps seem to have a rather small standard deviation $\sigma_{j,\sigma}^2$. Both observations are consistent with the nonparametric and parametric analysis in Sections 6 and 7.

We now turn to specification testing on the functional estimates. In particular, we focus on co-jumps. The use of moment conditions for estimation nicely lends itself to tests of over-identifying restrictions. We begin with a visual assessment. We employ infinitesimal cross-moments which were not used for estimation. We then compare, in Figure 13, the estimated parametric forms implied by the model to the estimated empirical moments. We find that the model-implied estimates are close to the "free" estimates, thereby providing visual evidence of satisfactory model specification.

We now turn to a more formal analysis. The (nonparametric and parametric) estimation of the model in Section 6 and 7 provides, as a straightforward by-product, a series of tests for the presence of co-jumps in our sample (the alternative hypothesis $\mathcal{H}_1 : \lambda_{r,\sigma} > 0$ being tested against the null of absence of co-jumps $\mathcal{H}_0 : \lambda_{r,\sigma} = 0$). We begin with $\mathcal{H}_0 : \langle \vartheta_{p_1,p_2} \rangle = 0$

Table 3: Tests for the presence of co-jumps and corresponding p-values.

	test	value	p-value
Co-jumps tests	J-test	239.4	0.20%
	$\langle \vartheta_{2,2} \rangle$	0.2862	0.10%
	$\langle \vartheta_{1,2} \rangle$	-0.0681	0.00%
	$\langle \vartheta_{2,1} \rangle$	0.2669	0.00%
	$\langle \vartheta_{1,3} \rangle$	-0.1091	0.00%
	$\langle \vartheta_{3,1} \rangle$	-0.9861	0.20%
	$\langle \vartheta_{2,3} \rangle$	0.4846	0.10%
	$\langle \vartheta_{3,2} \rangle$	-2.3185	0.10%
	$\langle \vartheta_{3,3} \rangle$	-6.2580	0.10%

($p_1 \geq 1$ and $p_2 \geq 1$ with $(p_1, p_2) \neq (1, 1)$), where $\langle \vartheta_{p_1, p_2} \rangle$ denotes the estimated moment of order (p_1, p_2) averaged over the estimated volatility density, tested against $\mathcal{H}_1 : \langle \vartheta_{p_1, p_2} \rangle \neq 0$. When p_1 and p_2 are both ≥ 1 , and at least one of them is strictly larger than 1, theory dictates that these higher-order cross-moments should all be directly proportional to $\lambda_{r, \sigma}$. Specifically, $\langle \vartheta_{p_1, p_2} \rangle = 0$ if $\lambda_{r, \sigma} = 0$. Conversely, a zero moment ($\langle \vartheta_{p_1, p_2} \rangle = 0$) implies $\lambda_{r, \sigma} = 0$, if $\rho_J \neq 0$, since $\sigma_{JJ, r} > 0$ and $\sigma_{JJ, \sigma} > 0$. Somewhat more explicitly, we also employ a GMM-type J -test directly on the restriction $\lambda_{r, \sigma} = 0$. The distribution, and resulting size, of all the proposed tests under the null is evaluated using simulations from the parametric model in Table 2 after setting $\lambda_{r, \sigma} = 0$. Table 3 provides the test values along with their simulated p-values. All tests yield a clear rejection of the null of absence of co-jumps at the 1% level.

One final observation is in order. While the use of infinitesimal cross-moments for co-jump identification is justified asymptotically in Appendix A, statistical uncertainty in the functional estimates, in the parametric estimates, and in the above tests is evaluated throughout this work by simulating the model using the estimated parameter values in Table 2 before running the described procedures for every simulated data set. In this sense, our inference is finite sample in nature, conservative, and unaffected by the likely overrated precision which would characterize asymptotic inference.

10 Final remarks

Market participants, as well as more casual observers, would believe that negative market shocks are associated with sudden, positive, changes in the level of volatility. This belief, confirmed by several recent financial crisis including the rather wild market gyrations of August 2011, originates from the frequent observation of the parallel, but opposite in sign, behavior of major stock indices and volatility indices, like the VIX. Negative price changes often occur along with positive spikes in the VIX. Just to mention a couple of recent events, between August 5th 2011 (a Friday) and August 8th 2011 (a Monday) the VIX rose from 32 to 48. The associated, negative, S&P 500 change on Monday, August 8th, was about 6%. Similarly, on August 17th the VIX rose from 31 to 42. The corresponding market price change was about -4.5% . Because it is an important barometer of market sentiment, the VIX is a complicated object capturing market fear, through risk-premia, as well as changes in fundamental values.

Interestingly, at daily frequencies, attempts to relate volatility measures unaffected by risk premia to sudden price changes have been largely unsuccessful. This outcome may be the result of pricing models which do not explicitly allow for independent price and volatility jumps along with co-jumps and, therefore, excessively constrain the jump dynamics. It may also be the result of volatility filtering methods relying on low-frequency return data, as well as on the parametric structure of the model, which may not be capable of yielding enough resolution, in terms of final estimates, so as to thoroughly identify the fine-grain dynamics of rare events, like volatility jumps and co-jumps.

Combining jump-robust, high-frequency, spot variance estimates with a novel identification procedure based on infinitesimal cross-moments, we provide empirical evidence about the likelihood of price and volatility jumps occurring jointly. When co-jumping, price and volatility are strongly negatively correlated. We emphasize that this result does not hinge on sudden changes in risk premia associated with market downturns (as possibly yielded by the VIX) or implied volatility smirks (as given by cross-sectional option prices). Said differently, the effect is solely revealed by the dynamic properties of stock prices, once volatility is filtered effectively and a sufficiently rich specification is adopted, without the need for the, arguably economically confounding (due to risk premia), information contained in traded or synthetic derivatives.

Market participants care about directional price changes as well as volatility changes. The latter may affect short-term investors needing to cash-in in turbulent markets. It also affects long-term investors in that higher short-term volatility will, in general, lead to a higher likelihood of long-term losses. An inverse association between large price changes and large volatility changes, which we document extensively, exacerbates the adverse implications of negative price moves. The corresponding positive volatility shocks will, in fact, lead to persistently higher volatility levels. These self-reinforcing price and volatility effects have important implications for the asset allocation decisions of both investors with a short-term outlook and investors with a long-term horizon. In terms of pricing, the hedging of contemporaneous directional and volatility changes appears to be particularly cumbersome. This is likely to translate into a substantial compensation for the risk associated with co-jump dynamics. When extracted from quoted derivatives, this compensation may represent a generally overlooked, first-order effect to be accounted for in the modeling of risk-neutral dynamics for the purpose of effective derivative pricing.

A procedure which identifies the nature of the price and volatility jump dependence without the need for the information in derivatives, like the one we have discussed, has also the potential to provide reliable information about risk compensation once derivative data are brought to bear. We leave this issue for future work.

Appendices

A Asymptotic properties

Consider a bivariate jump-diffusion process with compound Poisson jumps expressed as follows:

$$\begin{aligned} dX_t &= \mu_X(Y_t)dt + \sigma_X(Y_t) \left\{ \rho(Y_t)dW_t^1 + \sqrt{1-\rho^2(Y_t)}dW_t^2 \right\} + c_X dJ_t^X + d_X dJ_t, \\ dY_t &= \mu_Y(Y_t)dt + \sigma_Y(Y_t)dW_t^1 + c_Y dJ_t^Y + d_Y dJ_t, \end{aligned} \quad (43)$$

where $dW^X = \rho(Y_t)dW_t^1 + \sqrt{1-\rho^2(Y_t)}dW_t^2$, $dW^Y = dW^1$ are correlated diffusion processes with correlation coefficient $\rho(Y_t)$ and dJ^X, dJ^Y , and dJ are independent (of the Brownian motions W^1 and W^2 , as well as each other) Poisson processes with intensities $\lambda_X(Y_t), \lambda_Y(Y_t)$, and $\lambda_{XY}(Y_t)$, respectively. The functions are such that a strong solution to the system exists. The system does not have to be stationary. It is sufficient for the state variable $\{Y_t, t \geq 0\}$ to be Harris recurrent, something which we assume.

We observe the skeleton $\{X_{\Delta_{n,T}}, Y_{\Delta_{n,T}}\}, \{X_{2\Delta_{n,T}}, Y_{2\Delta_{n,T}}\}, \dots, \{X_{n\Delta_{n,T}}, Y_{n\Delta_{n,T}}\}$, i.e., n equally-spaced observations sampled at intervals $\Delta_{n,T} = T/n$. The asymptotic design is such that $n \rightarrow \infty$, $T \rightarrow \infty$, and $\Delta_{n,T} \rightarrow 0$. We estimate

$$\vartheta_{p_1, p_2}(y) = \lim_{\varepsilon \rightarrow 0} \frac{E[(X_{t+\varepsilon} - X_t)^{p_1} (Y_{t+\varepsilon} - Y_t)^{p_2} | Y_t = y]}{\varepsilon}, \quad (44)$$

by using classical Nadaraya-Watson kernel estimates, namely

$$\hat{\vartheta}_{p_1, p_2}(y) = \frac{\sum_{i=1}^{n-1} \mathbf{K}\left(\frac{Y_{iT/n} - y}{h_{n,T}}\right) (X_{(i+1)T/n} - X_{iT/n})^{p_1} (Y_{(i+1)T/n} - Y_{iT/n})^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}\left(\frac{Y_{iT/n} - y}{h_{n,T}}\right)}. \quad (45)$$

Define $\hat{L}_{n,T}(y) = \frac{\Delta_{n,T}}{h_{n,T}} \sum_{i=1}^n \mathbf{K}\left(\frac{Y_{iT/n} - y}{h_{n,T}}\right) = \frac{\Delta_{n,T}}{h_{n,T}} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)$, the empirical occupation density of the Y process.

The following proofs provide details which are specific to inference for *infinitesimal cross-moments*. For more details on the general method of proof in this type of models, we refer the interested reader to the treatment in Bandi and Renò (2009). The notation in this Appendix is purposely for a general system of observable processes $\{(X_t, Y_t), t \geq 0\}$. The next subsection specializes the analysis to the case $(X_t, Y_t) = (\log p_t, \hat{\sigma}_t^2)$ while providing limiting conditions for a vanishing measurement error in the variance estimates $\hat{\sigma}_t^2$. Finally, we note that, when the jump sizes depend on Y (or σ below), the notation $E[\cdot]$ should be interpreted as $E[\cdot|y]$ (or $E[\cdot|\sigma]$). We are concise here to avoid clutter but are explicit in the main text.

Assumption. The function $\mathbf{K}(x)$ is a nonnegative, bounded, continuous, and symmetric kernel defined on a compact set S satisfying $\int_S \mathbf{K}(s)ds = 1$, $\mathbf{K}_2 = \int_S \mathbf{K}^2(s)ds < \infty$, and $\mathbf{K}_1 = \int_S s^2 \mathbf{K}(s)ds < \infty$. The kernel's first and second derivative are absolutely integrable.

Theorem 1. (Consistency.) If $n, T \rightarrow \infty$ and $\Delta_{n,T} = T/n \rightarrow 0$ so that $h_{n,T} \widehat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$ and $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$, then

$$\widehat{\vartheta}_{p_1,0}(y) \xrightarrow{p} \begin{cases} \mu_X(y) + \lambda_X(y)E[c_X] + \lambda_{XY}(y)E[d_X] & p_1 = 1 \\ \sigma_X^2(y) + \lambda_X(y)E[c_X^2] + \lambda_{XY}(y)E[d_X^2] & p_1 = 2 \\ \lambda_X(y)E[c_X^{p_1}] + \lambda_{XY}(y)E[d_X^{p_1}] & p_1 \geq 3 \end{cases},$$

$$\widehat{\vartheta}_{1,1}(y) \xrightarrow{p} \rho(y)\sigma_X(y)\sigma_Y(y) + \lambda_{XY}(y)E[d_X d_Y],$$

and, without loss of generality, for $p_1 \geq p_2 \geq 1$ (with $p_1 > p_2$ if $p_2 = 1$),

$$\widehat{\vartheta}_{p_1,p_2}(y) \xrightarrow{p} \lambda_{XY}(y)E[d_X^{p_1} d_Y^{p_2}].$$

Theorem 2. (Weak convergence.) Let $n, T \rightarrow \infty$ and $\Delta_{n,T} = T/n \rightarrow 0$ so that $h_{n,T} \widehat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$ and $\frac{\Delta_{n,T} \sqrt{\widehat{L}_{n,T}(y)}}{h_{n,T}^{3/2}} \xrightarrow{a.s.} 0$. If

$$h_{n,T}^5 \widehat{L}_{n,T}(y) = O_{a.s.}(1),$$

then

$$\sqrt{h_{n,T} \widehat{L}_{n,T}(y)} \left\{ \widehat{\vartheta}_{p_1,p_2}(y) - \vartheta_{p_1,p_2}(y) - \Gamma_{\vartheta_{p_1,p_2}}(y) \right\} \Rightarrow \mathbf{N}(0, \mathbf{K}_2 \vartheta_{2p_1,2p_2}(y)),$$

with

$$\Gamma_{\vartheta_{p_1,p_2}} = h_{n,T}^2 \mathbf{K}_1 \left(\frac{\partial \vartheta_{p_1,p_2}(y)}{\partial y} \frac{\frac{\partial s(y)}{\partial y}}{s(y)} + \frac{1}{2} \frac{\partial^2 \vartheta_{p_1,p_2}(y)}{\partial^2 y} \right),$$

where $s(dx) = s(x)dx$ is the invariant measure of the Y process.

Lemma A.1. Given a Borel measurable function $g(y) : \mathcal{R} \rightarrow \mathcal{R}$, write

$$\chi_{n_1,n_2}(y) = \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{n_1} (Y_{s-} - Y_{iT/n})^{n_2} g(Y_s) ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)}$$

with $n_1 = 0, 1, \dots$ and $n_2 = 0, 1, \dots$. Assume $n, T \rightarrow \infty$ with $\Delta_{n,T} \rightarrow 0$. If $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$, then

$$\chi_{0,0}(y) \xrightarrow{p} g(y),$$

and, if $n_1 \neq 0$ or $n_2 \neq 0$,

$$\chi_{n_1,n_2}(y) \xrightarrow{p} 0.$$

Proof of Lemma A.1. Using the ratio-limit theorem (Revuz and Yor, 1994) as in Bandi and Renò

(2009), and handling discretization the way they do, we have

$$\chi_{0,0}(y) = \frac{\frac{\int_0^T \mathbf{K}_s(y) g(Y_s) ds}{\int_0^T \mathbf{K}_s(y) ds} + O_p\left(\frac{\Delta_{n,T}}{h_{n,T}^2}\right)}{\frac{\int_0^T \mathbf{K}_s(y) ds}{\int_0^T \mathbf{K}_s(y) ds} + O_p\left(\frac{\Delta_{n,T}}{h_{n,T}^2}\right)} \xrightarrow{p} g(y),$$

if $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$. And, immediately,

$$\chi_{n_1,n_2}(y) = O_p\left(\Delta_{n,T}^{1/2}\right)^{n_1+n_2} \chi_{0,0}(y) \xrightarrow{p} 0.$$

Lemma A.2. Given two Borel measurable functions $g_1(y) : \mathcal{R} \rightarrow \mathcal{R}$ and $g_2(y, z) : \mathcal{R}^2 \rightarrow \mathcal{R}$, write

$$\Psi_{n_1,n_2}(y) = \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{n_1} (Y_{s-} - Y_{iT/n})^{n_2} g_1(Y_s) dW_s}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)},$$

and

$$\Xi_{n_1,n_2}(y) = \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{n_1} (Y_{s-} - Y_{iT/n})^{n_2} \int_Z g_2(Y_s, z) \bar{\nu}(ds, dz)}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)},$$

with $n_1 = 0, 1, \dots$ and $n_2 = 0, 1, \dots$. Assume $n, T \rightarrow \infty$ with $\Delta_{n,T} \rightarrow 0$. If $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$ and $h_{n,T} \hat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$, then

$$\Psi_{n_1,n_2}(y) \xrightarrow{p} 0,$$

and

$$\Xi_{n_1,n_2}(y) \xrightarrow{p} 0.$$

Proof of Lemma A.2. Both $\Psi_{0,0}(y)$ and $\Xi_{0,0}(y)$ are appropriately re-scaled (by $\hat{L}_{n,T}(y)$) sums of martingale difference sequences. Following Bandi and Renò (2009), if $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$ and $h_{n,T} \hat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$,

$$\Psi_{0,0}(y) \xrightarrow{p} 0,$$

$$\Xi_{0,0}(y) \xrightarrow{p} 0.$$

Thus, immediately,

$$\Psi_{n_1,n_2}(y) = O_p\left(\Delta_{n,T}^{1/2}\right)^{n_1+n_2} \Psi_{0,0}(y) \xrightarrow{p} 0,$$

and

$$\Xi_{n_1,n_2}(y) = O_p\left(\Delta_{n,T}^{1/2}\right)^{n_1+n_2} \Xi_{0,0}(y) \xrightarrow{p} 0.$$

Lemma A.3. Consider Ψ_{n_1,n_2} and Ξ_{n_1,n_2} as defined in Lemma A.2. Let $n, T \rightarrow \infty$ so that

$\Delta_{n,T} = T/n \rightarrow 0$. If $h_{n,T} \widehat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$ and $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$, we have

$$\sqrt{h_{n,T} \widehat{L}_{n,T}(y)} \Psi_{0,0} \Rightarrow \mathbf{N}(0, \mathbf{K}_2 g_1^2(y)).$$

For $n_1 \geq 0$, if $h_{n,T} \widehat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$ and $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$,

$$\sqrt{\frac{h_{n,T} \widehat{L}_{n,T}(y)}{\Delta_{n,T}}} \Psi_{n_1,0} \Rightarrow \mathbf{N}\left(0, \frac{1}{2} \mathbf{K}_2 g_1^2(y) \vartheta_{2n_1,0}(y)\right).$$

For $n_1 \geq n_2 \geq 1$, if $h_{n,T} \widehat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$ and $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$,

$$\sqrt{\frac{h_{n,T} \widehat{L}_{n,T}(y)}{\Delta_{n,T}}} \Psi_{n_1,n_2} \Rightarrow \mathbf{N}\left(0, \frac{1}{2} \mathbf{K}_2 g_1^2(y) \vartheta_{2n_1,2n_2}(y)\right).$$

Similar expressions apply to Ξ_{n_1,n_2} with $E[g_2^2(y, z)]$ replacing $g_1^2(y)$.

Proof of Lemma A.3. Without loss of generality, write $n_1 \geq n_2 \geq 1$ and consider Ξ_{n_1,n_2} . Write

$$\begin{aligned} \Xi_{n_1,n_2}^{num}(y) &: = \frac{1}{\sqrt{h_{n,T} \Delta_{n,T}}} \sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{n_1} (Y_{s-} - Y_{iT/n})^{n_2} \int_Z g_2(Y_s) \bar{\nu}(ds, dz) \\ &: = \frac{1}{\sqrt{\Delta_{n,T}}} \sum_{i=1}^{n-1} u_i. \end{aligned}$$

We immediately have $\frac{1}{\sqrt{\Delta_{n,T}}} \sum_{i=1}^{n-1} E[u_i | \mathfrak{F}_{iT/n}] = 0$. Using arguments in Bandi and Renò (2009), we obtain

$$\begin{aligned} \frac{1}{\Delta_{n,T}} \sum_{i=1}^{n-1} E[u_i^2 | \mathfrak{F}_{iT/n}] &= \frac{1}{2} \int_0^T \mathbf{K}_s^2(y) E[g_2^2] \lambda_{XY}(Y_s) E[d_X^{2n_1} d_Y^{2n_2}] ds + \frac{\Delta_{n,T}}{h_{n,T}^2} O_p\left(\frac{1}{h_{n,T}} \int_0^T \mathbf{K}\left(\frac{Y_s - x}{h_{n,T}}\right) ds\right) \\ &= \mathbf{U}_{n,T}^2, \end{aligned}$$

and

$$\frac{1}{\Delta_{n,T}} \sum_{i=1}^{n-1} E[u_i^2 \mathbf{1}(|u_i| > \epsilon) | \mathfrak{F}_{iT/n}] \xrightarrow{p} 0.$$

Theorem VIII. 3.33 in Jacod and Shiryaev (2003) allows us to conclude that, uniformly in T and $h_{n,T}$ in $\mathcal{H}(\varepsilon) = \left\{ \frac{\Delta_{n,T}^{1/2}}{\varepsilon} < h_{n,T} < \varepsilon \right\}$ for a small $\varepsilon > 0$, $\Xi_{n_1,n_2}^{num}(y) \xrightarrow{\Delta_{n,T} \rightarrow 0} W_{\mathbf{U}^2}$. Hence,

$$\sqrt{\frac{h_{n,T} \widehat{L}_{n,T}(y)}{\Delta_{n,T}}} \Xi_{n_1,n_2}(y) = \frac{\Xi_{n_1,n_2}^{num}(y)}{\sqrt{\frac{\Delta_{n,T}}{h_{n,T}} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)}} \stackrel{d}{=} W \frac{\mathbf{U}_{n,T}^2}{\int_0^T K_s(y) ds},$$

if $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$. The ratio-limit theorem now yields

$$\frac{\mathbf{U}_{n,T}^2}{\int_0^T \mathbf{K}_s(y) ds} \xrightarrow[T \rightarrow \infty, h_{n,T} \rightarrow 0, \Delta_{n,T} \rightarrow 0]{p} \frac{1}{2} \mathbf{K}_2 \mathbf{E}[g_2^2(y)] \lambda_{X,Y}(y) \mathbf{E}[d_X^{2n_1} d_Y^{2n_2}].$$

Theorem 4.1 in van Zanten (2000) implies the desired result. The case $n_2 = 0$ can be treated in a similar way. Next, consider Ψ_{n_1, n_2} and write,

$$\Psi_{n_1, n_2}^{num}(y) = \frac{1}{\sqrt{h_{n,T} \Delta_{n,T}}} \sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{n_1} (Y_{s-} - Y_{iT/n})^{n_2} g_1(Y_s) dW_s.$$

Compute now its quadratic variation, i.e.,

$$[\Psi_{n_1, n_2}^{num}(y)] = \frac{1}{h_{n,T} \Delta_{n,T}} \sum_{i=1}^{n-1} (\mathbf{K}_{i,n,T}(y))^2 \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{2n_1} (Y_{s-} - Y_{iT/n})^{2n_2} g_1^2(Y_s) ds.$$

Using Ito's lemma, we obtain

$$\begin{aligned} & (X_{s-} - X_{iT/n})^{2n_1} (Y_{s-} - Y_{iT/n})^{2n_2} \\ &= \int_{iT/n}^s 2n_1 (X_{u-} - X_{iT/n})^{2n_1-1} (Y_{u-} - Y_{iT/n})^{2n_2} \mu_{X,s} du \\ &+ \int_{iT/n}^s 2n_2 (X_{u-} - X_{iT/n})^{2n_1} (Y_{u-} - Y_{iT/n})^{2n_2-1} \mu_{Y,s} du \\ &+ \int_{iT/n}^s 2n_1 (X_{u-} - X_{iT/n})^{2n_1-1} (Y_{u-} - Y_{iT/n})^{2n_2} \sigma_{X,s} dW_u^X \\ &+ \int_{iT/n}^s 2n_2 (X_{u-} - X_{iT/n})^{2n_1} (Y_{u-} - Y_{iT/n})^{2n_2-1} \sigma_{Y,s} dW_u^Y \\ &+ \int_{iT/n}^s \frac{1}{2} 2n_1(2n_1-1) (X_{u-} - X_{iT/n})^{2n_1-2} (Y_{u-} - Y_{iT/n})^{2n_2} \sigma_{X,s}^2 du \\ &+ \int_{iT/n}^s \frac{1}{2} 2n_2(2n_2-1) (X_{u-} - X_{iT/n})^{2n_1} (Y_{u-} - Y_{iT/n})^{2n_2-2} \sigma_{Y,s}^2 du \\ &+ \int_{iT/n}^s 2n_1 2n_2 (X_{u-} - X_{iT/n})^{2n_1-1} (Y_{u-} - Y_{iT/n})^{2n_2-1} \rho_s \sigma_{X,s} \sigma_{Y,s} du \\ &+ \sum_{\Delta X_u \neq 0 \text{ or } \Delta Y_u \neq 0} [(X_{u-} + \Delta X_u - X_{iT/n})^{2n_1} (Y_{u-} + \Delta Y_u - Y_{iT/n})^{2n_2} \\ &\quad - (X_{u-} - X_{iT/n})^{2n_1} (Y_{u-} - Y_{iT/n})^{2n_2}]. \end{aligned} \tag{46}$$

If $n_1 \geq n_2 \geq 1$, the higher-order term above is given by $\sum (\Delta X_u)^{2n_1} (\Delta Y_u)^{2n_2}$. Thus,

$$[\Psi_{n_1, n_2}^{num}(y)] \xrightarrow{a.s.} \frac{1}{2} \frac{1}{h_{n,T}} \int_0^T \mathbf{K}_s^2(y) g_1^2(Y_s) \lambda_{X,Y}(Y_s) \mathbf{E} \left(d_X^{2n_1}(z) d_Y^{2n_2}(z) \right) ds$$

Reasoning in the same way as for Ξ_{n_1, n_2} , we have

$$\sqrt{\frac{h_{n,T} \widehat{L}_{n,T}(y)}{\Delta_{n,T}}} \Psi_{n_1, n_2} \Rightarrow \mathbf{N} \left(0, \frac{1}{2} \mathbf{K}_2 g_1^2(y) \vartheta_{2n_1, 2n_2} \right),$$

if $h_{n,T} \widehat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$ and $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$. Clearly, for $n_1 \geq 0$,

$$\sqrt{\frac{h_{n,T} \widehat{L}_{n,T}(y)}{\Delta_{n,T}}} \Psi_{n_1, 0} \Rightarrow \mathbf{N} \left(0, \frac{1}{2} \mathbf{K}_2 g_1^2(y) \vartheta_{2n_1, 0} \right),$$

if $h_{n,T} \widehat{L}_{n,T}(y) \xrightarrow{a.s.} \infty$ and $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$.

Proof of Theorem 1. Consider the $p_2 = 0, p_1 \geq 3$ case. Ito's Lemma gives

$$\begin{aligned} (X_{(i+1)T/n} - X_{iT/n})^{p_1} &= p_1 \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{p_1-1} \mu_{X,s} ds \\ &\quad + p_1 \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{p_1-1} \sigma_{X,s} dW_s^X \\ &\quad + \frac{1}{2} p_1 (p_1 - 1) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{p_1-2} \sigma_{X,s}^2 ds \\ &\quad + \sum_{\Delta X_s \neq 0} [(X_{s-} + \Delta X_s - X_{iT/n})^{p_1} - (X_{s-} - X_{iT/n})^{p_1}]. \end{aligned} \quad (47)$$

Now, write

$$\begin{aligned} &\sum_{\Delta X_s \neq 0} [(X_{s-} + \Delta X_s - X_{iT/n})^{p_1} - (X_{s-} - X_{iT/n})^{p_1}] \\ &= \sum_{k=0}^{p_1-1} \binom{p_1}{k} \int_{iT/n}^{(i+1)T/n} \int_Z (X_{s-} - X_{iT/n})^k \left(c_X^{p_1-k}(z) \nu_X(ds, dz) + d_X^{p_1-k}(z) \nu_{XY}(ds, dz) \right). \end{aligned}$$

Compensating the random jump measures and denoting by $\bar{\nu}_X$ and $\bar{\nu}_{XY}$ the corresponding compensated random jump measures, we write $\bar{\nu}_X(ds, dy) = \nu_X(ds, dy) - \lambda_{X,s} E[c_X^{p_1-k}] ds$ and $\bar{\nu}_{XY}(ds, dy) = \nu_{XY}(ds, dy) - \lambda_{XY,s} E[d_X^{p_1-k}] ds$. Thus,

$$\begin{aligned} \widehat{\vartheta}_{p_1, 0}(y) &= \frac{\sum_{i=1}^{n-1} p_1 \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{p_1-1} \mu_{X,s} ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ &\quad + \frac{\sum_{i=1}^{n-1} p_1 \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{p_1-1} \sigma_{X,s} dW_s^X}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \end{aligned}$$

$$\begin{aligned}
& + \frac{\sum_{i=1}^{n-1} \frac{1}{2} p_1 (p_1 - 1) \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^{p_1-2} \sigma_{X,s}^2 ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\
& + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \sum_{k=0}^{p_1-1} \binom{p_1}{k} \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^k \int_Y \left(c_X^{p_1-k}(y) \bar{\nu}_X(ds, dy) + d_X^{p_1-k}(y) \bar{\nu}_{XY}(ds, dy) \right)}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\
& + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \sum_{k=0}^{p_1-1} \binom{p_1}{k} \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^k \left(\lambda_{X,s} \mathbf{E}[c_X^{p_1-k}] + \lambda_{XY,s} \mathbf{E}[d_X^{p_1-k}] \right) ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\
& = \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3 + \mathbf{R}_4 + \mathbf{A}_5.
\end{aligned}$$

The terms $\mathbf{R}_1$ through $\mathbf{R}_4$ converge to zero in probability given Lemma A.1. and Lemma A.2. Now write $\mathbf{A}_5$ as follows

$$\begin{aligned}
\mathbf{A}_5 & = \frac{\sum_{i=1}^{n-1} \int_{iT/n}^{(i+1)T/n} \mathbf{K}_{i,n,T}(y) \left(\lambda_{X,s} \mathbf{E}[c_X^{p_1}] + \lambda_{XY,s} \mathbf{E}[d_X^{p_1}] \right) ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\
& + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \sum_{k=1}^{p_1-1} \binom{p_1}{k} \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n})^k \left(\lambda_{X,s} \mathbf{E}[c_X^{p_1-k}] + \lambda_{XY,s} \mathbf{E}[d_X^{p_1-k}] \right) ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\
& = \mathbf{A}_5^1 + \mathbf{A}_5^2.
\end{aligned}$$

Lemma A.1. gives

$$\begin{aligned}
\mathbf{A}_5^1 & \xrightarrow{P} \lambda_X(y) \mathbf{E}[c_X^{p_1}] + \lambda_{XY}(y) \mathbf{E}[d_X^{p_1}], \\
\mathbf{A}_5^2 & \xrightarrow{P} 0,
\end{aligned}$$

which proves the stated result. Given Eq. (47) the cases $p_1 = 1$ and $p_1 = 2$ are obvious. Consider the case $p_2 > p_1 \geq 1$, without loss of generality. Use Ito's lemma as in Eq. (46) and rewrite the last term as

$$\begin{aligned}
& \sum_{\Delta X_s \neq 0 \text{ or } \Delta Y_s \neq 0} \left[(X_{s-} + \Delta X_s - X_{iT/n})^{p_1} (Y_{s-} + \Delta Y_s - Y_{iT/n})^{p_2} - (X_{s-} - X_{iT/n})^{p_1} (Y_{s-} - Y_{iT/n})^{p_2} \right] \\
& = \sum_{\Delta X_s \neq 0 \text{ or } \Delta Y_s \neq 0} \left(\sum_{k=0}^{p_1-1} \binom{p_1}{k} (X_{s-} - X_{iT/n})^k (\Delta X_s)^{p_1-k} \right) \left(\sum_{k=0}^{p_2-1} \binom{p_2}{k} (Y_{s-} - Y_{iT/n})^k (\Delta Y_s)^{p_2-k} \right) \\
& \quad - \sum_{\Delta X_s \neq 0 \text{ or } \Delta Y_s \neq 0} (X_{s-} - X_{iT/n})^{p_1} (Y_{s-} - Y_{iT/n})^{p_2} \\
& = \sum_{\Delta X_s \Delta Y_s \neq 0} (\Delta X_s)^{p_1} (\Delta Y_s)^{p_2} + \sum_{\Delta X_s \neq 0 \text{ or } \Delta Y_s \neq 0} \left((X_{s-} - X_{iT/n}) f_{1,i} + (Y_{s-} - Y_{iT/n}) f_{2,i} \right)
\end{aligned}$$

$$= \int_{iT/n}^{(i+1)T/n} \int_Y d_X^{p_1}(z) d_Y^{p_2}(z) \nu_{XY}(dz, ds) + \int_{iT/n}^{(i+1)T/n} (X_{s-} - X_{iT/n}) f_{1,s} + (Y_{s-} - Y_{iT/n}) f_{2,s} ds$$

for suitable $f_{1,s}, f_{2,s}$ functions. Now, write

$$\begin{aligned} \widehat{\vartheta}_{p_1, p_2}(y) = & \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \int_Y d_X^{p_1}(y) d_Y^{p_2}(y) \nu_{XY}(dy, ds)}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} p_1 (X_{s-} - X_{iT/n})^{p_1-1} (Y_{s-} - Y_{iT/n})^{p_2} \mu_{X,s} ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} p_2 (X_{s-} - X_{iT/n})^{p_1} (Y_{s-} - Y_{iT/n})^{p_2-1} \mu_{Y,s} ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} p_1 (X_{s-} - X_{iT/n})^{p_1-1} (Y_{s-} - Y_{iT/n})^{p_2} \sigma_{X,s} dW_s^X}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} p_2 (X_{s-} - X_{iT/n})^{p_1} (Y_{s-} - Y_{iT/n})^{p_2-1} \sigma_{Y,s} dW_s^Y}{\Delta_{n,T} \sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \frac{1}{2} p_1 (p_1 - 1) (X_{s-} - X_{iT/n})^{p_1-2} (Y_{s-} - Y_{iT/n})^{p_2} \sigma_{X,s}^2 ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \frac{1}{2} p_2 (p_2 - 1) (X_{s-} - X_{iT/n})^{p_1} (Y_{s-} - Y_{iT/n})^{p_2-2} \sigma_{Y,s}^2 ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} p_1 p_2 (X_{s-} - X_{iT/n})^{p_1-1} (Y_{s-} - Y_{iT/n})^{p_2-1} \rho_s \sigma_{X,s} \sigma_{Y,s} ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\ & + \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} ((X_{s-} - X_{iT/n}) f_{1,i} + (Y_{s-} - Y_{iT/n}) f_{2,i}) ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \end{aligned} \quad (48)$$

$$\begin{aligned}
& \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \int_Y d_X^{p_1}(y) d_Y^{p_2}(z) \nu_{XY}(dz, ds)}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \\
& + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3 + \mathbf{R}_4 + \mathbf{R}_5 + \mathbf{R}_6 + \mathbf{R}_7 + \mathbf{R}_8.
\end{aligned}$$

Finally, we compensate the random measure ν_{XY} and use Lemmas A.1. and A.2. to obtain

$$\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4, \mathbf{R}_5, \mathbf{R}_6, \mathbf{R}_7, \mathbf{R}_8 \xrightarrow{p} 0,$$

and

$$\frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \int_Y d_X^{p_1}(z) d_Y^{p_2}(z) \nu_{XY}(dz, ds)}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} \xrightarrow{p} \lambda_{XY}(y) \mathbb{E}[d_X^{p_1} d_Y^{p_2}].$$

Given Eq. (48), the case $p_2 = p_1 = 1$ is obvious since

$$\mathbf{R}_7 \xrightarrow{p} \rho(y) \sigma_X(y) \sigma_Y(y).$$

Proof of Theorem 2. Consider the case $p_1 > p_1 \geq 1$. Use Ito's lemma as in Eq. (46) and re-write the jump term as

$$\begin{aligned}
& \sum_{\Delta X_s \neq 0 \text{ or } \Delta Y_s \neq 0} [(X_{s-} + \Delta X_s - X_{iT/n})^{p_1} (Y_{s-} + \Delta Y_s - Y_{iT/n})^{p_2} - (X_{s-} - X_{iT/n})^{p_1} (Y_{s-} - Y_{iT/n})^{p_2}] \\
& = \int_{iT/n}^{(i+1)T/n} \int_Z d_X^{p_1}(z) d_Y^{p_2}(z) \nu_{XY}(ds, dz) + \int_{iT/n}^{(i+1)T/n} f_{3,i} ds,
\end{aligned}$$

for a suitable $f_{3,i} = o_p \left(\int_{iT/n}^{(i+1)T/n} \int_Z d_X^{p_1}(z) d_Y^{p_2}(z) \nu_{XY}(ds, dz) \right)$. Then, as in the consistency proof, after compensating the random measure ν_{XY} , write

$$\begin{aligned}
& \hat{\vartheta}_{p_1, p_2}(y) - \underbrace{\frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \int_Z \mathbb{E}(d_X^{p_1}(z) d_Y^{p_2}(z)) \lambda_{XY,s} ds dz}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)}}_{\mathbf{B}} \\
& : = \mathbf{A} + \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3 + \mathbf{R}_4 + \mathbf{R}_5 + \mathbf{R}_6 + \mathbf{R}_7 + \mathbf{R}_8',
\end{aligned} \tag{49}$$

where

$$\mathbf{A} = \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \int_Z d_X^{p_1}(z) d_Y^{p_2}(z) \bar{\nu}_{XY}(ds, dz)}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)},$$

and

$$\mathbf{R}'_8 = \frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} f_{3,i}(s) ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)}.$$

Now, using Lemma A.3., we have that

$$\sqrt{h_{n,T} \widehat{L}_{n,T}(y)} \mathbf{A} \Rightarrow \mathbf{N} \left(0, \mathbf{K}_2 \lambda_{XY}(y) \mathbf{E}[d_X^{2p_1} d_Y^{2p_2}] \right).$$

Naturally, $\mathbf{A}$ is the dominating term in the sum. Finally, note that the right-hand side of Eq. (49) can be expressed as

$$\widehat{\vartheta}_{p_1,p_2}(y) - \vartheta_{p_1,p_2}(y) - \underbrace{\left(\frac{\sum_{i=1}^{n-1} \mathbf{K}_{i,n,T}(y) \int_{iT/n}^{(i+1)T/n} \vartheta_{p_1,p_2}(Y_s) ds}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_{i,n,T}(y)} - \vartheta_{p_1,p_2}(y) \right)}_{\mathbf{B}'},$$

with

$$\mathbf{B}' = h_{n,T}^2 \left(\int_S s^2 \mathbf{K}(s) ds \right) \left(\frac{\partial \vartheta_{p_1,p_2}(y)}{\partial x_i} \frac{\partial s(y)}{\partial x_i} + \frac{1}{2} \frac{\partial^2 \vartheta_{p_1,p_2}(y)}{\partial y} \right) + O_p \left(\frac{\Delta_{n,T}}{h_{n,T}^2} \right).$$

This completes the proof.

A.1 Replacing spot variance with its estimated value

Rewrite the estimator in Eq. (13) as $\widehat{\sigma}_{t,i}^2 = \frac{TBPV_{t,i}}{\phi}$ with

$$TBPV_{t,i} = \varsigma_1^{-2} \sum_{j=2}^k |r_{t,i,j}| |r_{t,i,j-1}| I_{\{|r_{t,i,j}| \leq \theta_{t,i,j}\}} I_{\{|r_{t,i,k-1}| \leq \theta_{t,i,j-1}\}}.$$

In our case ϕ is equal to 1 hour and $k = 60$ minutes. We show that, under assumptions, the spot variance estimator is consistent for spot variance and the resulting measurement error can be made asymptotically negligible.

Theorem 3. *We let $\phi \rightarrow 0$ with $k \rightarrow \infty$. Assume $\theta_{t,i} = \xi_{t,i} \Theta(\frac{\phi}{k})$, where $\Theta(\frac{\phi}{k})$ is a real function satisfying $\Theta(\frac{\phi}{k}) \xrightarrow{\phi \rightarrow 0, k \rightarrow \infty} 0$ and $\frac{1}{\Theta(\frac{\phi}{k})} \left(\frac{\phi}{k} \log \left(\frac{\phi}{k} \right) \right) \xrightarrow{\phi \rightarrow 0, k \rightarrow \infty} 0$ and $\xi_{t,i}$ is an a.s. bounded process with a strictly positive lower bound. Write $\omega_{n,k,\phi} = \sqrt{\frac{\log(n)}{k}} + \sqrt{\phi}$. Consider $\widehat{\vartheta}_{p_1,p_2}(\cdot)$ with $(p_1, p_2) = (1, 0)$ or $(0, 1)$. If*

$$\frac{\omega_{n,k,\phi}}{\Delta_{n,T}} \rightarrow 0,$$

the consistency result in Theorem 1 holds when replacing $\sigma_{iT/n}^2$ with $\widehat{\sigma}_{iT/n}^2$. For any other combina-

tion of (p_1, p_2) , if

$$\frac{\omega_{n,k,\phi}}{\Delta_{n,T}^{1/2} h_{n,T}} \rightarrow 0,$$

the consistency result in Theorem 1 holds when replacing $\sigma_{iT/n}^2$ with $\hat{\sigma}_{iT/n}^2$. Assume $(p_1, p_2) = (1, 0)$ or $(0, 1)$, if

$$\sqrt{h_{n,T} \hat{L}_{n,T}(\sigma^2)} \frac{\omega_{n,k,\phi}}{\Delta_{n,T}} \rightarrow 0,$$

where $\hat{L}_{n,T}(\sigma^2)$ is the estimated occupation density of spot variance process, the weak convergence results in Theorem 2 holds when replacing $\sigma_{iT/n}^2$ with $\hat{\sigma}_{iT/n}^2$. For any other combination of (p_1, p_2) , if

$$\sqrt{h_{n,T} \hat{L}_{n,T}(\sigma^2)} \frac{\omega_{n,k,\phi}}{\Delta_{n,T}^{1/2} h_{n,T}} \rightarrow 0,$$

the weak convergence results in Theorem 2 holds when replacing $\sigma_{iT/n}^2$ with $\hat{\sigma}_{iT/n}^2$.

Proof of Theorem 3. In what follows, $X = \log(p)$ and $Y = \hat{\sigma}^2$. The transformation $\log(\hat{\sigma}^2)$ can be treated similarly. Given the results in Bandi and Renò (2009, 2011), we only have to consider the case in which $p_1 \geq 1$ and $p_2 \geq 1$, with at least one being strictly larger than 1. Write

$$\begin{aligned} \hat{\vartheta}_{p_1, p_2}(x) &= \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_i} \\ &= \frac{\sum_{i=1}^{n-1} \hat{\mathbf{K}}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} - \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} \\ &+ \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} - \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} \\ &+ \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} - \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_i} \\ &= (a) + (b) + (c). \end{aligned}$$

In the presence of jumps, we have

$$(\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} = O_p(\Delta_{n,T}^{1/2}),$$

and

$$(\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2} = O_p(\Delta_{n,T}^{1/2}).$$

In the presence of co-jumps, a simple application of Ito's lemma yields

$$(\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2} = O_p(\Delta_{n,T}^{1/2}).$$

Also write $\mathbf{M}_{n,k,T} = \max_{1 \leq i \leq n} |\hat{\sigma}_{iT/n}^2 - \sigma_{iT/n}^2| = O_p(\omega_{n,k,\phi})$ as shown in Bandi and Renò (2009, 2011). Thus

$$(a) = \frac{\sum_{i=1}^{n-1} (\hat{\mathbf{K}}_i - \mathbf{K}_i) (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\hat{\sigma}_{(i+1)T/n}^2 - \sigma_{(i+1)T/n}^2 + \sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2 + \sigma_{iT/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i}.$$

Now, for suitable constants $c_{i,j,k}$,

$$\begin{aligned} & (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\hat{\sigma}_{(i+1)T/n}^2 - \sigma_{(i+1)T/n}^2 + \sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2 + \sigma_{iT/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2} \\ &= \sum_{i+j+k=p_2} c_{i,j,k} (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{iT/n}^2 - \sigma_{iT/n}^2)^k (\hat{\sigma}_{(i+1)T/n}^2 - \sigma_{(i+1)T/n}^2)^i (\hat{\sigma}_{iT/n}^2 - \sigma_{iT/n}^2)^j \\ &= O_p(\Delta_{n,T}^{1/2}). \end{aligned}$$

Thus, using the mean-value theorem,

$$\begin{aligned} (a) &= O_p(\Delta_{n,T}^{1/2}) \frac{\max_{1 \leq i \leq n} \left| \frac{\hat{\sigma}_{i\Delta_{n,T}}^2 - \sigma_{i\Delta_{n,T}}^2}{h_{n,T}} \right| \sum_{i=1}^{n-1} \left| \mathbf{K}' \left(\frac{\sigma_{iT/n}^2 + O_p(\mathbf{M}_{n,k,T}) - x}{h_{n,T}} \right) \right|}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} \\ &= O_p \left(\frac{\mathbf{M}_{n,k,T} \Delta_{n,T}^{1/2}}{\Delta_{n,T} h_{n,T}} \right) \frac{\frac{1}{h_{n,T}} \int_0^T \left| \mathbf{K}' \left(\frac{\sigma_s^2 - x}{h_{n,T}} \right) \right| ds}{\frac{1}{h_{n,T}} \int_0^T \mathbf{K} \left(\frac{\sigma_s^2 - x}{h_{n,T}} \right) ds} + O_p \left(\frac{\Delta_{n,T}}{h_{n,T}^2} \right) \\ &= O_p \left(\frac{\mathbf{M}_{n,k,T}}{\Delta_{n,T}^{1/2} h_{n,T}} \right) \frac{1}{1 + O_p \left(\frac{\Delta_{n,T}}{h_{n,T}^2} \right) + O_p(g(n, T, k, \phi))} \\ &= O_p \left(\frac{\mathbf{M}_{n,k,T}}{\Delta_{n,T}^{1/2} h_{n,T}} \right) \end{aligned}$$

if $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$ and $g(n, T, k, \phi) \rightarrow 0$. We now turn to (b).

$$(b) = \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} \left[(\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2} - (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2} \right]}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i}.$$

We can prove that $\xi_i := (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} \left[(\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2} - (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2} \right]$ is $O_p(\mathbf{M}_{n,k,T} \Delta_{n,T}^{1/2})$. This is straightforward if $p_2 = 1$. If $p_2 > 1$, using $a^p - b^p = (a-b) \sum_{k=1}^p a^{p-k} b^{k-1}$, we have

$$\begin{aligned} & (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2} - (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2} \\ &= (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2 - \sigma_{(i+1)T/n}^2 + \sigma_{iT/n}^2) \sum_{k=1}^{p_2} (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2-k} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{k-1}. \end{aligned}$$

We can now decompose ξ_i into its terms, the first corresponding to $k = 1$:

$$\begin{aligned} & (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2 - \sigma_{(i+1)T/n}^2 + \sigma_{iT/n}^2)(\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2-1} \\ &= O_p(\mathbf{M}_{n,k,T}) (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\hat{\sigma}_{(i+1)T/n}^2 - \sigma_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2 + \sigma_{iT/n}^2 + \sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2-1} \\ &= o_p(\mathbf{M}_{n,k,T} \Delta_{n,T}^{1/2}). \end{aligned}$$

For the terms with $2 \leq k \leq p_2 - 1$, we may write:

$$\begin{aligned} & (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2 - \sigma_{(i+1)T/n}^2 + \sigma_{iT/n}^2)(\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2)^{p_2-k} \\ & \cdot \underbrace{(\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{k-1}}_{O_p(\Delta_{n,T}^{1/2})} \\ &= O_p(\mathbf{M}_{n,k,T} \Delta_{n,T}^{1/2})(\hat{\sigma}_{(i+1)T/n}^2 - \sigma_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2 + \sigma_{iT/n}^2 + \sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2-k} \\ &= o_p(\mathbf{M}_{n,k,T} \Delta_{n,T}^{1/2}). \end{aligned}$$

The term $k = p_2$ is the dominating one since

$$(\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2-1} (\hat{\sigma}_{(i+1)T/n}^2 - \hat{\sigma}_{iT/n}^2 - \sigma_{(i+1)T/n}^2 + \sigma_{iT/n}^2) = O_p(\mathbf{M}_{n,k,T} \Delta_{n,T}^{1/2}).$$

Thus,

$$(b) = O_p \left(\frac{\mathbf{M}_{n,k,T} \Delta_{n,T}^{1/2}}{\Delta_{n,T}} \right) \frac{\Delta_{n,T} \sum_{i=1}^{n-1} \mathbf{K}_i}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} = O_p \left(\frac{\mathbf{M}_{n,k,T}}{\Delta_{n,T}^{1/2}} \right),$$

if $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$ and $g(n, T, k, \phi) \rightarrow 0$. Finally,

$$\begin{aligned} (c) &= \frac{\sum_{i=1}^{n-1} \mathbf{K}_i (\log(p_{(i+1)T/n}) - \log(p_{iT/n}))^{p_1} (\sigma_{(i+1)T/n}^2 - \sigma_{iT/n}^2)^{p_2}}{\Delta_{n,T} \sum_{i=1}^n \mathbf{K}_i} \times \frac{\Delta_{n,T} \left(\sum_{i=1}^n \hat{\mathbf{K}}_i - \sum_{i=1}^n \mathbf{K}_i \right)}{\Delta_{n,T} \sum_{i=1}^n \hat{\mathbf{K}}_i} \\ &= O_p \left(\frac{\max_{1 \leq i \leq n} \left| \frac{\hat{\sigma}_{i\Delta_{n,T}}^2 - \sigma_{i\Delta_{n,T}}^2}{h_{n,T}} \right| \frac{\Delta_{n,T}}{h_{n,T}} \sum_{i=1}^n \left| \mathbf{K}' \left(\frac{\sigma_{i\Delta_{n,T}}^2 - \sigma_{i\Delta_{n,T}}^2}{h_{n,T}} \right) \right|}{\frac{\Delta_{n,T}}{h_{n,T}} \sum_{i=1}^n \hat{\mathbf{K}}_i} \right) \\ &= O_p \left(\frac{\mathbf{M}_{n,k,T}}{h_{n,T}} \right), \end{aligned}$$

if $\frac{\Delta_{n,T}}{h_{n,T}^2} \rightarrow 0$ and $g(n, T, k, \phi) \rightarrow 0$. Now, notice that (c) is of higher order than (b) since $\frac{\mathbf{M}_{n,k,T}}{h_{n,T}} =$

$\frac{\mathbf{M}_{n,k,T}}{\Delta_{n,T}^{1/2}} \frac{\Delta_{n,T}^{1/2}}{h_{n,T}} = \frac{\mathbf{M}_{n,k,T}}{\Delta_{n,T}^{1/2}} o(1)$. Thus

$$(a) + (b) + (c) = O_p \left(\frac{\mathbf{M}_{n,k,T}}{\Delta_{n,T}^{1/2} h_{n,T}} \right),$$

which completes the proof.

B Economic implications

Proof of Proposition 8.1. All results, and their implications, could be proved by using the properties of the change of measure, given by the assumed stochastic discount factor, as in the main text (see, e.g., Eq. 27). Here we follow a different, but equivalent, strategy. As in Christoffersen et al. (2013), we show the results by using pricing restrictions.

From Eq. (17), we have

$$\log M_t - \log M_0 = \phi(\log(p_t) - \log(p_0)) + \int_0^t \delta(\sigma_s) ds + \psi(\xi(\sigma_t) - \xi(\sigma_0))$$

and

$$\begin{aligned} d \log M_t &= \phi d \log(p_t) + \delta(\sigma_t) dt + \psi d \xi(\sigma_t) \\ &= [\phi \mu(\sigma_t) + \delta(\sigma_t) + \psi m(\sigma_t)] dt \\ &\quad + [\phi \sigma_t \rho(\sigma_t) + \psi \Lambda(\sigma_t)] dW_t^1 + \left[\phi \sigma_t \sqrt{1 - \rho^2(\sigma_t)} \right] dW_t^2 \\ &\quad + (\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}) dJ_{r,\sigma} + \phi c_{r,t}^J dJ_r + \psi c_{\sigma,t}^J dJ_\sigma. \end{aligned}$$

Using Ito's lemma and denoting by dM_t^c the continuous part of dM_t , we obtain

$$\begin{aligned} dM_t &= M_t d \log(M_t^c) + \frac{1}{2} M_t [\phi^2 \sigma_t^2 + \psi^2 \Lambda^2(\sigma_t) + 2\phi\psi\rho(\sigma_t)\sigma_t\Lambda(\sigma_t)] dt \\ &\quad + M_t \left(\exp^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1 \right) dJ_{r,\sigma} + M_t \left(\exp^{\phi c_{r,t}^J} - 1 \right) dJ_r + M_t \left(\exp^{\psi c_{\sigma,t}^J} - 1 \right) dJ_\sigma. \end{aligned}$$

Now define $B_t = B_0 e^{rt}$ and $Y_{1,t} = B_t M_t$, so that

$$dY_{1,t} = Y_{1,t} \left(\frac{dM_t}{M_t} + r dt \right).$$

To guarantee absence of arbitrage, $Y_{1,t}$ has to be martingale. To this extent, setting $\frac{1}{dt} E_t[dY_{1,t}] = 0$, we obtain Eq. (18).

Similarly, consider $Y_{2,t} = p_t M_t$. By Ito's Lemma, we get

$$\begin{aligned} \frac{dY_{2,t}}{Y_{2,t}} &= \frac{dM_t^c}{M_t} + \frac{dp_t^c}{p_t} + (\phi \sigma_t^2 + \psi \rho(\sigma_t) \Lambda(\sigma_t) \sigma_t) dt \\ &\quad + \left(\exp^{(\phi+1)c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1 \right) dJ_{r,\sigma} + \left(\exp^{(\phi+1)c_{r,t}^J} - 1 \right) dJ_r + \left(\exp^{\psi c_{\sigma,t}^J} - 1 \right) dJ_\sigma. \end{aligned}$$

Again, $Y_{2,t}$ should be a martingale. Hence, using $E_t \left[\frac{dM_t}{M_t} \right] = -r dt$, we obtain Eq. (19).

Now, consider $Y_{3,t} = M_t \mathcal{D}(t, p_t, \xi(\sigma_t))$, where $\mathcal{D}(t, p_t, \xi(\sigma_t))$ is - as in Christoffersen et al. (2013) -

the value at time t of a traded asset with payoff at $T > t$ equal to $\mathcal{D}(p_T)$. By Ito's Lemma:

$$\begin{aligned} d\mathcal{D}_t = & \frac{\partial \mathcal{D}}{\partial t} dt + \frac{\partial \mathcal{D}}{\partial p} dp^c + \frac{\partial \mathcal{D}}{\partial \xi(\sigma)} d\xi^c(\sigma_t) + \frac{\partial^2 \mathcal{D}}{\partial p \partial \xi(\sigma)} \sigma_t \Lambda(\sigma_t) \rho(\sigma_t) dt + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial p^2} \sigma_t^2 p_t^2 dt + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial \xi(\sigma)^2} \Lambda^2(\sigma_t) dt \\ & + \left[\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ}}, \xi(\sigma_t) + c_{\sigma,t}^{JJ}) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] dJ_{r,\sigma} + \left[\mathcal{D}(t, p_t \exp^{c_{r,t}^J}, \xi(\sigma_t)) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] dJ_r \\ & + \left[\mathcal{D}(t, p_t, \xi(\sigma_t) + c_{\sigma,t}^J) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] dJ_\sigma \end{aligned}$$

and

$$\begin{aligned} \frac{dY_{3,t}}{Y_{3,t}} = & \frac{dM_t^c}{M_t} + \frac{d\mathcal{D}_t^c}{\mathcal{D}_t} + \frac{1}{\mathcal{D}_t} \left(\frac{\partial \mathcal{D}}{\partial p} (\phi \sigma_t^2 + \psi \rho(\sigma_t) \Lambda(\sigma_t) \sigma_t) + \frac{\partial \mathcal{D}}{\partial \xi(\sigma)} (\phi \sigma_t \Lambda(\sigma_t) \rho(\sigma_t) + \psi \Lambda^2(\sigma_t)) \right) dt \\ & + c_{Y_{3,t}}^{r,\sigma} dJ_{r,\sigma} + c_{Y_{3,t}}^r dJ_r + c_{Y_{3,t}}^\sigma dJ_\sigma \end{aligned}$$

with

$$\begin{aligned} c_{Y_{3,t}}^{r,\sigma} &= \frac{\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ}}, \xi(\sigma_t) + c_{\sigma,t}^{JJ}) \exp^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - \mathcal{D}(t, p_t, \xi(\sigma_t))}{\mathcal{D}(t, p_t, \xi(\sigma_t))}, \\ c_{Y_{3,t}}^r &= \frac{\mathcal{D}(t, p_t \exp^{c_{r,t}^J}, \xi(\sigma_t)) \exp^{\phi c_{r,t}^J} - \mathcal{D}(t, p_t, \xi(\sigma_t))}{\mathcal{D}(t, p_t, \xi(\sigma_t))}, \\ c_{Y_{3,t}}^\sigma &= \frac{\mathcal{D}(t, p_t, \xi(\sigma_t) + c_{\sigma,t}^J) \exp^{\psi c_{\sigma,t}^J} - \mathcal{D}(t, p_t, \xi(\sigma_t))}{\mathcal{D}(t, p_t, \xi(\sigma_t))}. \end{aligned}$$

By equating $\frac{1}{dt} E \left[\frac{dY_{3,t}}{Y_{3,t}} \right]$ to zero, we obtain

$$\begin{aligned} & \left(-r - \lambda_{r,\sigma}(\sigma_t) \left[\exp^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - 1 \right] - \lambda_r(\sigma_t) \left[\exp^{\phi c_{r,t}^J} - 1 \right] - \lambda_\sigma(\sigma_t) \left[\exp^{\psi c_{\sigma,t}^J} - 1 \right] \right) \mathcal{D}(t, p_t, \xi(\sigma_t)) \\ & + \frac{\partial \mathcal{D}}{\partial t} + \frac{\partial \mathcal{D}}{\partial p} (r + \tilde{\mu}(\sigma_t^2)) + \frac{\partial \mathcal{D}}{\partial \xi(\sigma)} m(\sigma_t) + \frac{\partial^2 \mathcal{D}}{\partial p \partial \xi(\sigma)} \sigma_t \Lambda(\sigma_t) \rho(\sigma_t) + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial p^2} \sigma_t^2 p_t^2 + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial \xi(\sigma)^2} \Lambda^2(\sigma_t) \\ & + \frac{\partial \mathcal{D}}{\partial p} (\phi \sigma_t^2 + \psi \sigma_t \Lambda(\sigma_t) \rho(\sigma_t)) + \frac{\partial \mathcal{D}}{\partial \xi(\sigma)} (\phi \sigma_t \Lambda(\sigma_t) \rho(\sigma_t) + \psi \Lambda^2(\sigma_t)) \\ & + \lambda_{r,\sigma}(\sigma_t) E \left[\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ}}, \xi(\sigma_t) + c_{\sigma,t}^{JJ}) \exp^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] \\ & + \lambda_r(\sigma_t) E \left[\mathcal{D}(t, p_t \exp^{c_{r,t}^J}, \xi(\sigma_t)) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] \\ & + \lambda_\sigma(\sigma_t) E \left[\mathcal{D}(t, p_t, \xi(\sigma_t) + c_{\sigma,t}^J) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] = 0, \end{aligned}$$

and, simplifying, we have

$$\begin{aligned} & -r \mathcal{D}(t, p_t, \xi(\sigma_t)) + \frac{\partial \mathcal{D}}{\partial t} + \frac{\partial \mathcal{D}}{\partial p} (r + \tilde{\mu}(\sigma_t^2)) + \frac{\partial \mathcal{D}}{\partial \xi(\sigma)} m(\sigma_t) + \frac{\partial^2 \mathcal{D}}{\partial p \partial \xi(\sigma)} \sigma_t \Lambda(\sigma_t) \rho(\sigma_t) + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial p^2} \sigma_t^2 p_t^2 + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial \xi(\sigma)^2} \Lambda^2(\sigma_t) \\ & + \frac{\partial \mathcal{D}}{\partial p} (\phi \sigma_t^2 + \psi \sigma_t \Lambda(\sigma_t) \rho(\sigma_t)) + \frac{\partial \mathcal{D}}{\partial \xi(\sigma)} (\phi \sigma_t \Lambda(\sigma_t) \rho(\sigma_t) + \psi \Lambda^2(\sigma_t)) \\ & + \lambda_{r,\sigma}(\sigma_t) E \left[\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ}}, \xi(\sigma_t) + c_{\sigma,t}^{JJ}) \exp^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] \\ & + \lambda_r(\sigma_t) E \left[\mathcal{D}(t, p_t \exp^{c_{r,t}^J}, \xi(\sigma_t)) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] \\ & + \lambda_\sigma(\sigma_t) E \left[\mathcal{D}(t, p_t, \xi(\sigma_t) + c_{\sigma,t}^J) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right] = 0. \end{aligned} \tag{50}$$

Now, for every traded asset with price $\mathcal{D}(t, p, \xi(\sigma))$, we also have, slightly abusing notation and using (σ_t, p_t) in place of (σ, p) ,

$$\begin{aligned}
& -r\mathcal{D}(t, p_t, \xi(\sigma_t)) + \frac{\partial \mathcal{D}}{\partial t} + \frac{\partial \mathcal{D}}{\partial p} \left(r - \lambda_{r,\sigma}^*(\sigma_t) E(e^{c_{r,t}^{JJ*}} - 1) - \lambda_r^*(\sigma_t) E(e^{c_{r,t}^{J*}} - 1) \right) \\
& + \frac{\partial \mathcal{D}}{\partial \xi(\sigma)} m^*(\sigma_t) + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial p^2} \sigma_t^2 p_t^2 + \frac{1}{2} \frac{\partial^2 \mathcal{D}}{\partial \xi(\sigma)^2} \Lambda^2(\sigma_t) + \frac{\partial^2 \mathcal{D}}{\partial p \partial \xi(\sigma)} \sigma_t \Lambda(\sigma_t) \rho(\sigma_t) \\
& + \lambda_{r,\sigma}^*(\sigma_t) E \left[\left(\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ*}}, \xi(\sigma_t) + c_{\sigma,t}^{JJ*}) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right] \\
& + \lambda_r^*(\sigma_t) E \left[\left(\mathcal{D}(t, p_t \exp^{c_{r,t}^{J*}}, \xi(\sigma_t)) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right] \\
& + \lambda_\sigma^*(\sigma_t) E \left[\left(\mathcal{D}(t, p_t, \xi(\sigma_t) + c_{\sigma,t}^{J*}) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right] = 0.
\end{aligned} \tag{51}$$

If we now equate Eq. (50) with Eq. (51), we obtain

$$\begin{aligned}
& \lambda_{r,\sigma}(\sigma_t) E \left[\exp^{\phi c_{r,t}^{JJ} + \psi c_{\sigma,t}^{JJ}} \left(\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ}}, \xi(\sigma_t) + c_{\sigma,t}^{JJ}) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right] \\
& = \lambda_{r,\sigma}^*(\sigma_t) E \left[\left(\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ*}}, \xi(\sigma_t) + c_{\sigma,t}^{JJ*}) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right],
\end{aligned} \tag{52}$$

$$\begin{aligned}
& \lambda_r(\sigma_t) E \left[\exp^{\phi c_{r,t}^{JJ}} \left(\mathcal{D}(t, p_t \exp^{c_{r,t}^{JJ}}, \xi(\sigma_t)) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right] \\
& = \lambda_r^*(\sigma_t) E \left[\left(\mathcal{D}(t, p_t \exp^{c_{r,t}^{J*}}, \xi(\sigma_t)) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right],
\end{aligned} \tag{53}$$

$$\begin{aligned}
& \lambda_\sigma(\sigma_t) E \left[\exp^{\psi c_{\sigma,t}^{JJ}} \left(\mathcal{D}(t, p_t, \xi(\sigma_t) + c_{\sigma,t}^{JJ}) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right] \\
& = \lambda_\sigma^*(\sigma_t) E \left[\left(\mathcal{D}(t, p_t, \xi(\sigma_t) + c_{\sigma,t}^{J*}) - \mathcal{D}(t, p_t, \xi(\sigma_t)) \right) \right],
\end{aligned} \tag{54}$$

as well as Eq. (19) and Eq. (20).

Lemma B.1. Assume c_r and c_σ are defined as in Subsection 8.2. Then,

$$E[e^{ac_r + bc_\sigma}] = e^{a\mu_r + \frac{1}{2}a^2\sigma_r^2} \frac{1}{1 - \mu_\sigma(b + a\rho_J)},$$

for all constants a and b such that $\mu_\sigma > \frac{1}{b + a\rho_J}$.

Proof.

$$\begin{aligned}
E[e^{ac_r + bc_\sigma}] &= E \left[E[e^{ac_r + bc_\sigma} | c_\sigma] \right] \\
&= E[e^{bc_\sigma} e^{a\mu_r + a\rho_J c_\sigma + \frac{1}{2}a^2\sigma_r^2}] \\
&= e^{a\mu_r + \frac{1}{2}a^2\sigma_r^2} E[e^{(b + a\rho_J)c_\sigma}] \\
&= e^{a\mu_r + \frac{1}{2}a^2\sigma_r^2} \frac{1}{1 - \mu_\sigma(b + a\rho_J)}.
\end{aligned}$$

C Finite Sample Issues

C.1 Measuring spot volatility with microstructure noise and jumps

The spot variance measures are obtained as follows. Each day, we split the day into 6 consecutive slots of 65 minutes, starting at 9.30am. Denote by $\sigma_{t,i}^2$ the spot variance estimator corresponding to day t , hour i , $i = 1, \dots, 6$. To estimate $\sigma_{t,i}^2$, we first compute logarithmic prices $\log p_{t,i,k}$ every minute, with $k = 0, \dots, 65$, by pre-averaging all observed transaction prices inside each minute. Doing so softens the impact of microstructure noise by reducing the contaminations handled by pre-averaging (see Jacod et al., 2009, for further discussions). Next, we compute one-minute logarithmic returns $r_{t,i,k} = \log p_{t,i,k} - \log p_{t,i,k-1}$ for $k = 1, \dots, 65$. The spot variance estimates are obtained by applying the jump-robust threshold bipower variation estimator in Eq. (13). These estimates are then scaled with a constant scale factor so as to guarantee that the average of the hourly spot variance estimates equals the threshold bipower variation estimated on daily close-to-close returns. Only variance estimates with a sufficiently high number of transactions inside an hour are retained. The exact filter is: the number of transactions in an hour should be no less than one third of the average number of transactions in the last month. We also eliminate spot variance estimates lower than 0.01. In our sample, this simple filter retains 98.88% of the estimated volatilities. Finally, we also record the last price $\log p_{t,i}$ in each hour.

C.2 Infinitesimal cross-moment estimation with measured volatilities

To estimate the moments, we first compute an exponential moving average of estimated volatilities $\tilde{\sigma}_{t,i}^2$ with 20 lags. These smoother volatilities are only used inside the kernels. The exact moment expressions are:

$$\begin{aligned}\tilde{\vartheta}_{p_1, p_2}(\sigma) &= \frac{\sum_{t=1}^{T-1} \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right) (\log p_{t+1,i} - \log p_{t,i})^{p_1} (\log \hat{\sigma}_{t+1,i}^2 - \log \hat{\sigma}_{t,i}^2)^{p_2}}{\Delta \sum_{t=1}^T \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right)}, \quad p_2 = 0, 1 \\ \tilde{\vartheta}_{p_1, 2}(\sigma) &= \frac{\sum_{t=1}^{T-1} \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right) (\log p_{t+1,i} - \log p_{t,i})^{p_1} \left[(\log \hat{\sigma}_{t+1,i}^2 - \log \hat{\sigma}_{t,i}^2)^2 - 2c \right]}{\Delta \sum_{t=1}^T \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right)}, \\ \tilde{\vartheta}_{p_1, 3}(\sigma) &= \frac{\sum_{t=1}^{T-1} \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right) (\log p_{t+1,i} - \log p_{t,i})^{p_1} \left[(\log \hat{\sigma}_{t+1,i}^2 - \log \hat{\sigma}_{t,i}^2)^3 - 6c (\log \hat{\sigma}_{t+1,i}^2 - \log \hat{\sigma}_{t,i}^2) \right]}{\Delta \sum_{t=1}^T \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right)},\end{aligned}$$

$$\tilde{\vartheta}_{p_1,4}(\sigma) = \frac{\sum_{t=1}^{T-1} \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right) (\log p_{t+1,i} - \log p_{t,i})^{p_1} \left[(\log \hat{\sigma}_{t+1,i}^2 - \log \hat{\sigma}_{t,i}^2)^4 - 12c (\log \hat{\sigma}_{t+1,i}^2 - \log \hat{\sigma}_{t,i}^2)^2 - 12c^2 \right]}{\Delta \sum_{t=1}^T \sum_{i=1}^N \mathbf{K} \left(\frac{\tilde{\sigma}_{t,i} - \sigma}{h} \right)},$$

with $N = 6$ and $c = 2.61/59$ in our case. The finite-sample adjustments for $p_2 = 2, 3, 4$ are designed to correct for measurement error in the spot variance estimates. The adjustments are computed using the asymptotic results which hold for our assumed spot variance estimates, namely

$$\hat{\sigma}_{t,i}^2 = \sigma_{t,i}^2 + \varepsilon_{t,i},$$

where $\varepsilon_{t,i}$ is normally distributed with mean zero and variance $c\sigma_{t,i}^4$ (Bandi and Renò, 2009, and Corsi et al., 2010). To account for the log-transformation, the delta method has been employed. This is accurate since, in our sample, $\sqrt{c} \ll 1$.

C.3 Infinitesimal cross-moment estimation for a fixed Δ

The infinitesimal nature of the moments is so that finite sample contaminations may remain in the estimates if Δ is not small enough. To this extent, we implement exact first-order corrections (in Δ) for the estimates. These corrections are immaterial asymptotically and useful in finite samples. The recursive formula to compute them is:

$$\tilde{\vartheta}_{p_1,p_2}^{\Delta} = \tilde{\vartheta}_{p_1,p_2} - \frac{\Delta}{2} \sum_{j_1=0}^{p_1} \sum_{j_2=0}^{p_2} \binom{p_1}{j_1} \binom{p_2}{j_2} \tilde{\vartheta}_{j_1,j_2} \tilde{\vartheta}_{p_1-j_1,p_2-j_2}$$

with $\hat{\vartheta}_{0,0} = 0$.

C.4 Simulations

The reliability of the estimated infinitesimal cross-moments, and their finite-sample adjustments, are evaluated by simulation. We first simulate the bivariate system in Eq. (15), described in Section 7, with the parameters in Column 5, Table 2. We use a first-order Euler scheme. Initially, σ_t^2 does not contain any estimation error. The estimation error is added using the (asymptotic) formula $\hat{\sigma}_t^2 = \sigma_t^2 + \sqrt{2.61/59} \sigma_t^2 \varepsilon_t$, where ε_t is a standard normal draw and $\sqrt{2.61/59} \sigma_t^2$ is the (asymptotic) standard deviation of threshold bipower variation estimator computed with 60 one-minute returns. With the resulting time series, we estimate the infinitesimal cross-moments using the small-sample corrections described above. The choice of the conditioning points, kernel function, and bandwidth(s) are the same as those used to compute the moments for parametric and nonparametric estimation in the main text. Specifically, we use a Gaussian kernel, the set

$$\sigma^2 = \{0.1311, 0.2227, 0.3041, 0.3913, 0.4922, 0.6169, 0.7829, 1.0328, 1.5018, 3.0984\}$$

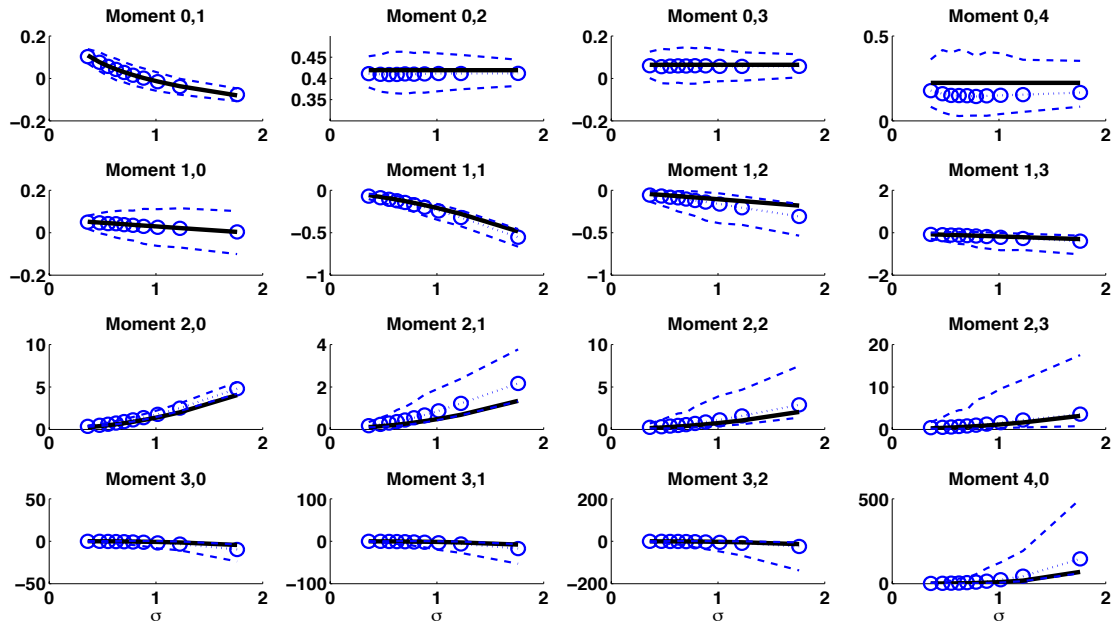


Figure 14: Simulation results for the estimated moments. Dotted lines with circles: median estimates. Dashed lines: 95% confidence bands. Solid lines: generated moments.

for the conditioning points (these are the midpoints of bins which contain the same number of estimated volatilities), and the bandwidth vector (corresponding to each conditioning point)

$$h = \{0.4165, 0.1811, 0.1374, 0.1173, 0.1126, 0.1131, 0.1239, 0.1496, 0.2134, 0.4397\}.$$

Consistent with the asymptotics, h smoothes more in regions with sparse observations near the boundaries of the range of the estimated volatility series. We implement 1,500 simulations. Figure 14 presents the generated moments, together with the estimated median and 95% confidence bands on the replications. The figure shows that our small-sample procedures are able to deliver accurate estimates of the infinitesimal cross-moments. Figure 15 displays the estimated parameters on simulations of the parametric model in Section 7 with the parameters in Table 2, Column 5. Estimation of the parameters via GMM is satisfactory and not affected by meaningful biases.

References

- Aït-Sahalia, Y., J. Fan, and Y. Li (2013). The leverage effect puzzle: Disentangling sources of bias at high frequency. *Journal of Financial Economics* 109, 224–249.
- Alizadeh, S., M. Brandt, and F. Diebold (2002). High and low frequency exchange rate volatility dynamics: Range-based estimation of stochastic volatility models. *Journal of Finance* 57, 1047–1092.

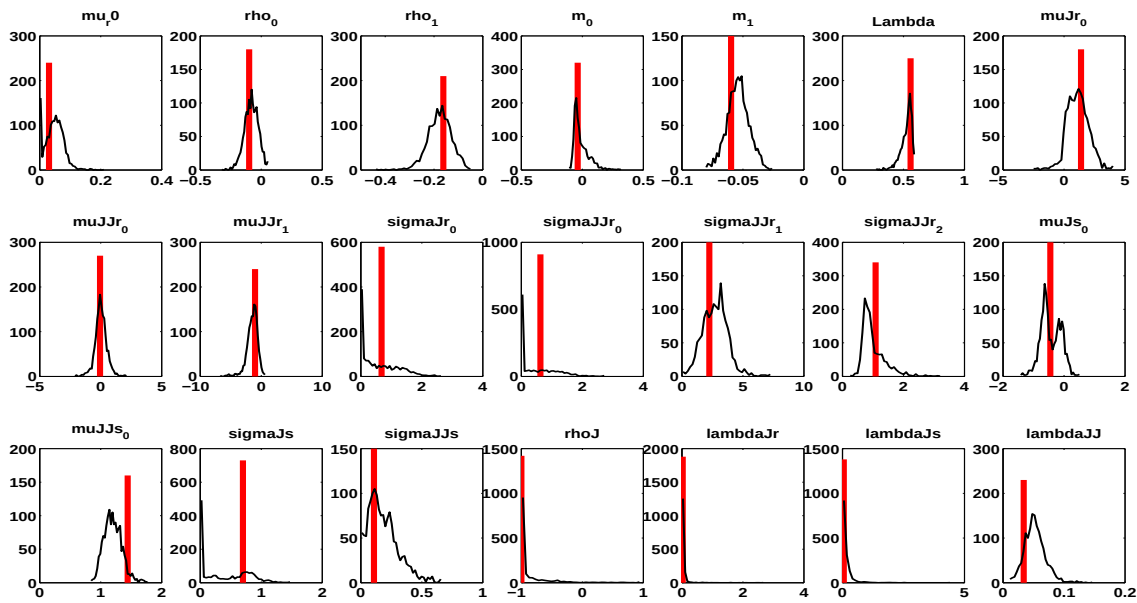


Figure 15: Simulation results for the parametric model. Vertical bars: Generated parameters. Solid lines: Histograms of the estimated parameters on simulations.

- Andersen, T. and L. Benzoni (2011). Stochastic volatility. In *Complex Systems in Finance and Econometrics*, pp. 694–726. Springer.
- Andersen, T., L. Benzoni, and J. Lund (2002). An empirical investigation of continuous-time equity return models. *Journal of Finance* 57, 1239–1284.
- Bakshi, G., C. Cao, and Z. Chen (1997). Empirical performance of alternative option pricing models. *Journal of Finance* 52, 2003–2049.
- Bandi, F. and R. Renò (2009). Nonparametric stochastic volatility. Working Paper.
- Bandi, F. and R. Renò (2012). Time-varying leverage effects. *Journal of Econometrics* 169(1), 94–113.
- Barndorff-Nielsen, O. E. and N. Shephard (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics* 2, 1–48.
- Bates, D. (2000). Post-'87 crash fears in the S&P 500 futures option market. *Journal of Econometrics* 94, 181–238.
- Broadie, M., M. Chernov, and M. Johannes (2007). Model Specification and Risk Premia: Evidence from Futures Options. *The Journal of Finance* 62(3), 1453–1490.
- Chacko, G. and L. Viceira (2003). Spectral GMM estimation of continuous-time processes. *Journal of Econometrics* 116(1-2), 259–292.
- Chernov, M., R. Gallant, E. Ghysels, and G. Tauchen (2003). Alternative models for stock price dynamics. *Journal of Econometrics* 116(1), 225–258.
- Chernov, M. and E. Ghysels (2000). A study towards a unified approach to the joint estimation of objective and risk neutral measures for the purpose of options valuation. *Journal of Financial Economics* 56, 407–458.

- Chib, S., F. Nardari, and N. Shephard (2002). Markov chain Monte Carlo methods for stochastic volatility models. *Journal of Econometrics* 108(2), 281–316.
- Christoffersen, P., S. Heston, and K. Jacobs (2013). Capturing option anomalies with a variance-dependent pricing kernel. *Review of Financial Studies* 16(8).
- Cont, R. and T. Kokholm (2011). A consistent pricing model for index options and volatility derivatives. *Mathematical Finance* 29, 1–27.
- Corsi, F., D. Pirino, and R. Renò (2010). Threshold bipower variation and the impact of jumps on volatility forecasting. *Journal of Econometrics* 159, 276–288.
- Duffie, D., J. Pan, and K. Singleton (2000). Transform analysis and asset pricing for affine jump-diffusions. *Econometrica* 68, 1343–1376.
- Eraker, B. (2004). Do Stock Prices and Volatility Jump? Reconciling Evidence from Spot and Option Prices. *The Journal of Finance* 59(3), 1367–1404.
- Eraker, B., M. Johannes, and N. Polson (2003). The impact of jumps in volatility and returns. *Journal of Finance* 58, 1269–1300.
- Hansen, L. (1982). Large sample properties of generalized method of moments estimators. *Econometrica* 59(4), 1029–1054.
- Harvey, A. and N. Shephard (1996). Estimation of an asymmetric stochastic volatility model for asset returns. *Journal of Business & Economic Statistics* 14(4), 429–434.
- Heston, S. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of financial studies* 6, 327–343.
- Jacod, J., C. Kluppelberg, and G. Muller (2012a). Functional relationships between price and volatility jumps and their consequences for discretely observed data. *Journal of Applied Probability* 49, 901–914.
- Jacod, J., C. Kluppelberg, and G. Muller (2012b). Testing for non-correlation between price and volatility jumps. Working paper.
- Jacod, J., Y. Li, P. Mykland, M. Podolskij, and M. Vetter (2009). Microstructure noise in the continuous case: the pre-averaging approach. *Stochastic Processes and their Applications* 119(7), 2249–2276.
- Jacod, J. and A. N. Shiryaev (2003). *Limit Theorems for Stochastic Processes*. Springer.
- Jacod, J. and V. Todorov (2009). Testing for common arrivals of jumps for discretely observed multidimensional processes. *Annals of Statistics* 37, 1792–1838.
- Jacod, J. and V. Todorov (2010). Do price and volatility jump together? *Annals of Applied Probability* 20, 1425–1469.
- Jacquier, E., N. Polson, and P. Rossi (1994). Bayesian analysis of stochastic volatility models. *Journal of Business and Economic Statistics* 12, 371–389.
- Jacquier, E., N. Polson, and P. Rossi (2004). Bayesian analysis of stochastic volatility models with fat-tails and correlated errors. *Journal of Econometrics* 122(1), 185–212.
- Jones, C. (2003). Nonlinear mean reversion in the short-term interest rate. *Review of Financial Studies* 16, 765–791.
- Lee, S. and P. Mykland (2008). Jumps in financial markets: A new nonparametric test and jump dynamics. *Review of Financial studies* 21(6), 2535.
- Mancini, C. (2009). Non-parametric threshold estimation for models with stochastic diffusion coefficient and jumps. *Scandinavian Journal of Statistics* 36(2), 270–296.
- Pan, J. (2002). The jump-risk premia implicit in options: Evidence from an integrated time series study. *Journal of Financial Economics* 63, 3–50.
- Piazzesi, M. (2010). Affine term structure models. In L. P. Hansen and Y. Ait-Sahalia (Eds.),

- Handbook of Financial Econometrics*. Amsterdam: North-Holland.
- Revuz, D. and M. Yor (1994). *Continuous Martingales and Brownian Motion*. Springer.
- Todorov, V. and G. Tauchen (2010). Activity signature functions for high-frequency data analysis. *Journal of Econometrics* 54, 125–138.
- Todorov, V. and G. Tauchen (2011). Volatility jumps. *Journal of Business and Economic Statistics* 29, 356–371.
- van Zanten, H. (2000). A multivariate central limit theorem for continuous local martingales. *Statistics & Probability Letters* 50(3), 229–235.
- Yu, J. (2005). On leverage in a stochastic volatility model. *Journal of Econometrics* 127(2), 165–178.
- Yu, J. (2012). A semiparametric stochastic volatility model. *Journal of Econometrics* 167, 473–482.